Year 12 Mathematics Solutions

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Learning Strategies

Mathematics is often the most challenging subject for students. Much of the trouble comes from the fact that mathematics is about logical thinking, not memorizing rules or remembering formulas. It requires a different style of thinking than other subjects. The students who seem to be "naturally" good at math just happen to adopt the correct strategies of thinking that math requires – often they don't even realise it. We have isolated several key learning strategies used by successful maths students and have made icons to represent them. These icons are distributed throughout the book in order to remind students to adopt these necessary learning strategies:



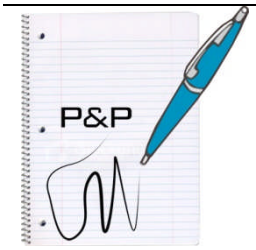
Talk Aloud Many students sit and try to do a problem in complete silence inside their heads. They think that solutions just pop into the heads of 'smart' people. You absolutely must learn to talk aloud and listen to yourself, literally to talk yourself through a problem. Successful students do this without realising. It helps to structure your thoughts while helping your tutor understand the way you think.



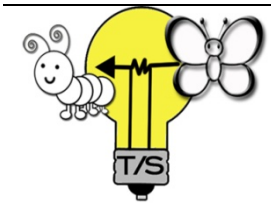
BackChecking This means that you will be doing every step of the question twice, as you work your way through the question to ensure no silly mistakes. For example with this question: $3 \times 2 - 5 \times 7$ you would do "3 times 2 is 5 ... let me check – no 3×2 is 6 ... minus 5 times 7 is minus 35 ... let me check ... minus 5×7 is minus 35. Initially, this may seem time-consuming, but once it is automatic, a great deal of time and marks will be saved.



Avoid Cosmetic Surgery Do not write over old answers since this often results in repeated mistakes or actually erasing the correct answer. When you make mistakes just put one line through the mistake rather than scribbling it out. This helps reduce silly mistakes and makes your work look cleaner and easier to backcheck.



Pen to Paper It is always wise to write things down as you work your way through a problem, in order to keep track of good ideas and to see concepts on paper instead of in your head. This makes it easier to work out the next step in the problem. Harder maths problems cannot be solved in your head alone – put your ideas on paper as soon as you have them – always!



Transfer Skills This strategy is more advanced. It is the skill of making up a simpler question and then transferring those ideas to a more complex question with which you are having difficulty.

For example if you can't remember how to do long addition because you can't recall exactly how to carry the one: $\begin{array}{r} 5889 \\ +\,4587 \\ \hline \end{array}$ then you may want to try adding numbers which you do know how to calculate that also involve carrying the one: $\begin{array}{r} 5 \\ +\,9 \\ \hline \end{array}$

This skill is particularly useful when you can't remember a basic arithmetic or algebraic rule, most of the time you should be able to work it out by creating a simpler version of the question.

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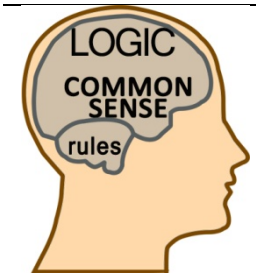
Format Skills These are the skills that keep a question together as an organized whole in terms of your working out on paper. An example of this is using the "=" sign correctly to keep a question lined up properly. In numerical calculations format skills help you to align the numbers correctly.

This skill is important because the correct working out will help you avoid careless mistakes. When your work is jumbled up all over the page it is hard for you to make sense of what belongs with what. Your "silly" mistakes would increase. Format skills also make it a lot easier for you to check over your work and to notice/correct any mistakes.

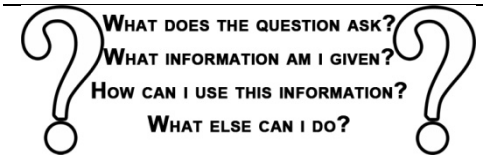
Every topic in math has a way of being written with correct formatting. You will be surprised how much smoother mathematics will be once you learn this skill. Whenever you are unsure you should always ask your tutor or teacher.



Its Ok To Be Wrong Mathematics is in many ways more of a skill than just knowledge. The main skill is problem solving and the only way this can be learned is by thinking hard and making mistakes on the way. As you gain confidence you will naturally worry less about making the mistakes and more about learning from them. Risk trying to solve problems that you are unsure of, this will improve your skill more than anything else. It's ok to be wrong – it is NOT ok to not try.



Avoid Rule Dependency Rules are secondary tools; common sense and logic are primary tools for problem solving and mathematics in general. Ultimately you must understand Why rules work the way they do. Without this you are likely to struggle with tricky problem solving and worded questions. Always rely on your logic and common sense first and on rules second, always ask Why?



Self Questioning This is what strong problem solvers do naturally when they get stuck on a problem or don't know what to do. Ask yourself these questions. They will help to jolt your thinking process; consider just one question at a time and Talk Aloud while putting Pen To Paper.

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Year 12 Mathematics

Series & Sequences

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Exercise 1

Arithmetic Sequences

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1) Calculate the value of d in the following sequences

a) 2, 4, 6, 8, 10, ...

$4 - 2 = 6 - 4 = 2$

$d = 2$

b) 1, 4, 7, 10, ...

$4 - 1 = 7 - 4 = 3$

$d = 3$

c) 4, ____, 16, _____, 28

$2d = 16 - 4, d = 6$

d) 16, 12, 8, 4, ...

$12 - 16 = 8 - 12 = -4$

$d = -4$

e) 64, ____, _____, 28, ...

$3d = 28 - 64, d = 12$

2) Calculate the value of a in the following sequences

a) ____, 6, 10, 14, ...

$d = 4, a = 2$

b) ____, ____, 15, 18, ...

$d = 3, a = 15 - 2d = 9$

c) ____, ____, 22, ____, ____, 43

$3d = 43 - 22 = 21, d = 7$

$a = 22 - 2d = 8$

d) ____, ____, ____76, 68, ...

$d = 68 - 76 = -8$

$a = 76 + 3d = 100$

e) ____, ____, ____, 7, ____, ____, ____, 3

$4d = 3 - 7, d = -1$

$a = 7 + 3d = 10$

3) Find the 5th term of the sequence with first term 4 and a common difference of 3

$a_n = a_1 + (n - 1)d$

$a_5 = 4 + (5 - 1)3 = 16$

4) Find the 25th term of the sequence with first term 6 and a common difference of 7

$a_n = a_1 + (n - 1)d$

$a_{25} = 6 + (25 - 1)7 = 174$

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5) Find the common difference of the sequence with a first term of 5 and a twentieth term of 195

$a_n = a_1 + (n-1)d$

$a_{20} = 5 + (20-1)d = 195$

$5 + 19d = 195$

$19d = 190$

$d = 10$

6) Find the first term of the arithmetic sequence whose tenth term is 14 and whose twentieth term is 62

$a_{10} = a_1 + 9d = 14$

$a_{20} = a_1 + 19d = 62$

$a_1 = 14 - 9d$

$a_1 = 62 - 19d$

$14 - 9d = 62 - 19d$

$10d = 48$

$d = 4.8$

7) An arithmetic sequence has a third term of x, and a fifteenth term of $3x - 2$.

a) What are the values of a and d?

$a_3 = a_1 + 2d = x$

$a_{15} = a_1 + 14d = 3x - 2$

$a_1 = x - 2d$

$a_1 = 3x - 2 - 14d$

$x - 2d = 3x - 2 - 14d$

$2 - 2x = -12d$

$d = \frac{x-1}{6}$

$a_1 = x - 2\left(\frac{x-1}{6}\right)$

$a_1 = x - \left(\frac{x-1}{3}\right)$

$a_1 = \frac{2x+1}{3}$

b) List the first three terms of the sequence when $x = 4$

$a_1 = 3, d = \frac{1}{2}$

$3, \frac{7}{2}, 4$

c) List the first 3 terms of the sequence when $x = -1$

$a_1 = -\frac{1}{3}$

$d = -\frac{1}{3}$

$-\frac{1}{3}, -\frac{2}{3} - 1$

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8) An arithmetic sequence has a common difference of 4 and a twentieth term of 102. What is the ninth term of this sequence?

$a_{20} = a_1 + (19)4 = 102$

$a_1 = 102 - 76 = 26$

$a_9 = 26 + (8)4 = 58$

9) There are two arithmetic sequences: A and B. $A_{10} = B_{28} = 40$, whilst the value of their first term is the same. If the common difference of sequence A is 3, list the first 4 terms of each sequence

For A:

$a_{10} = a_1 + 9 \times 3 = 40$

$a_1 = 13$

For B:

$a_{28} = 13 + 27d = 40$

$d = 1$

$A = 13,16,19$

$B = 13,14,15$

10) Arithmetic sequence A has a first term of (-20) and a twentieth term of 56. Arithmetic sequence B has a first term of 40 and a 5th term of 8. Which term number gives the same value for both sequences, and what is this value?

For A:

$a_{20} = -20 + 19d = 56$

$d = 4$

For B:

$a_5 = 40 + 4d = 8$

$d = -8$

We need

$a_n \, for \, A = \, a_n \, for \, B$

$-20 + (n-1)4 = 40 + (n-1)(-8)$

$-20 + 4n - 4 = 40 - 8n + 8$

$12n = 72$

$n = 6$

Term 6 with a value of 0 will be the same

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Exercise 2

Geometric Progressions

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1) Calculate the value of r in the following sequences

a) 2, 4, 8, 16, ...

$$\frac{4}{2} = \frac{8}{4} = 2$$

b) 3, 4.5, 6.75, ...

$$\frac{4.5}{3} = \frac{6.75}{4.5} = 1.5$$

c) 20, 10, 5, ...

$$\frac{10}{20} = \frac{5}{10} = \frac{1}{2}$$

d) 1000, 200, 40, ...

$$\frac{200}{1000} = \frac{40}{200} = 0.2$$

e) ___, 12, ___, 27, ...

$$r^2 = \frac{27}{12} = \frac{9}{4}$$

$$r = \frac{3}{2}$$

f) ___, ___, 100, ___, 9, ...

$$r^2 = \frac{9}{100}$$

$$r = \frac{3}{10}$$

2) Calculate the value of a in the following sequences

a) ___, ___, 8, 16, 32

$r = 2$

$ar^2 = 8$

$a = 2$

b) ___, ___, 9, ____, 20.25, ...

$9r^2 = 20.25$

$r = 1.5$

$ar^2 = 9$

$r = 4$

c) ____, ____, 25, ____, 6.25

$25r^2 = 6.25$

$r = \frac{1}{2}$

$ar^2 = 25$

$a = 100$

d) ____, ____, 100, ____, 6.25

$100r^2 = 6.25$

$r = \frac{1}{4}$

$ar^2 = 100$

$a = 1600$

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3) Find the 5th term of the sequence with a first term 2 and a common ratio of 3

$a_n = ar^{n-1}$

$a_5 = 2 \times 3^4$

$a = 162$

4) Find the 20th term of the sequence with first term 0.5 and common ratio 4

$a_n = ar^{n-1}$

$a_{20} = 0.5 \times 4^{19}$

$a = 137438953472$

5) What is the value of the first term of the sequence with an 8th term of 874.8 and a common ratio of 3?

$a_n = ar^{n-1}$

$874.8 = a \times 3^7$

$a = 2.5$

6) A geometric sequence has a first term of -2 and a 10th term of 1024. What is the value of the common ratio?

$a_n = ar^{n-1}$

$1024 = (-2)r^9$

$r = -2$

7) The 2nd term of a geometric sequence is 96 and the 5th term is 1.5. What are the common ratio and the first term?

$a_2 = ar = 96$

$a_5 = ar^4 = 1.5$

$ar \times r^3 = 1.5$

$96r^3 = 1.5$

$r = \frac{1}{4}$

8) A geometric sequence has a first term of x and an eleventh term of $59049x$. What is the common ratio of the sequence?

$a_{11} = xr^{10} = 59049x$

$r^{10} = 59049$

$r = 3$

9) The fifth term of a geometric sequence is 48, and the third term is 108. What is the first term and the sixth term?

$a_5 = ar^4 = 48$

$a_3 = ar^2 = 108$

$\frac{r^4}{48} = \frac{r^2}{108}$

$\frac{r^4}{r^2} = \frac{48}{108}$

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$$r^2 = \frac{4}{9}$$

$$r = \frac{2}{3}$$

10) A geometric sequence has a first term of $(x + 3)$, and a third term of $\frac{9x+27}{4}$. In terms of x, what is the fifth term?

$$a_3 = (x + 3)r^2 = \frac{9x + 27}{4}$$

$$r^2 = \frac{9(x + 3)}{4(x + 3)} = \frac{9}{4}$$

$$r = \frac{3}{2}$$

$$a_5 = (x + 3)\left(\frac{3}{2}\right)^4 = (x + 3)\frac{81}{16}$$

11) The fifth term of geometric sequence A is 4, and its ninth term is $\frac{1}{4}$. The second term of geometric sequence B is $\frac{1}{16}$, and its fifth term is (-4). Which term number will give the same value for each sequence, and what will this value be?

For A:

$$a_5 = ar^4 = 4$$

$$a_9 = ar^8 = \frac{1}{4}$$

$$\frac{ar^8}{ar^4} = \frac{\frac{1}{4}}{4}$$

$$r^4 = \frac{1}{16}$$

$$r = \frac{1}{2}$$

$$a = 64$$

For B:

$$a_2 = ar = \frac{1}{16}$$

$$a_5 = ar^4 = (-4)$$

$$\frac{ar^4}{ar} = \frac{(-4)}{\frac{1}{16}}$$

$$r^3 = -64$$

$$r = -4$$

$$a = -\frac{1}{64}$$

Need the value of n for which

$$64\left(\frac{1}{2}\right)^{n-1} = -\frac{1}{64}(-4)^{n-1}$$

$$4096 = (-8)^{n-1}$$

$$n - 1 = 4$$

$$n = 5$$

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Exercise 3

Arithmetic Series

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1) Find the sum of the first 20 terms of the following arithmetic series

a) 7, 11, 15, ...

$$s_n = \frac{n}{2}(a_1 + a_n)$$

$$s_{20} = \frac{20}{2}(7 + a_{20})$$

$$a_{20} = 7 + (19)4 = 83$$

$$s_{20} = 10 \times 90 = 900$$

b) 10, 12, 14, 16, ...

$$s_n = \frac{n}{2}(a_1 + a_n)$$

$$s_{20} = \frac{20}{2}(10 + a_{20})$$

$$a_{20} = 10 + (19)2 = 48$$

$$s_{20} = 10 \times 58 = 580$$

c) -8, -5, -2, ...

$$s_n = \frac{n}{2}(a_1 + a_n)$$

$$s_{20} = \frac{20}{2}(-8 + a_{20})$$

$$a_{20} = -8 + (19)3 = 49$$

$$s_{20} = 10 \times 41 = 410$$

d) ___, ___, 12, ___, ___, 21, .

$$a_1 = 6, d = 3$$

$$s_n = \frac{n}{2}(a_1 + a_n)$$

$$s_{20} = \frac{20}{2}(6 + a_{20})$$

$$a_{20} = 6 + (19)3 = 63$$

$$s_{20} = 10 \times 69 = 690$$

e) ___, ___, ___, -12, -8, ...

$$a = -24, d = 4$$

$$s_n = \frac{n}{2}(a_1 + a_n)$$

$$s_{20} = \frac{20}{2}(-24 + a_{20})$$

$$a_{20} = -24 + (19)4 = 52$$

$$s_{20} = 10 \times 28 = 280$$

2) What is the first term of an arithmetic series with a common difference of 8 and a sum to 30 terms of 4500?

$$s_{30} = \frac{30}{2}(a_1 + a_{31}) = 4500$$

$$a_{30} = a_1 + 29 \times 8 = a_1 + 232$$

$$\frac{30}{2}(a_1 + a_1 + 232) = 4500$$

$$2a_1 + 232 = 4500 \times \frac{2}{30}$$

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$$2a_1 = 300 - 232$$

$$a_1 = 34$$

3) The 15th term of an arithmetic series is 92. If the first term is 64, what is the sum of the first 25 terms?

$$s_{25} = \frac{25}{2}(64 + a_{25})$$

$$a_{15} = 64 + 14 \times d = 92$$

$$d = 2$$

$$a_{25} = 64 + 24 \times 2 = 112$$

$$s_{25} = \frac{25}{2}(64 + 112) = 2200$$

4) The first term of an arithmetic series is (-10), and the sum of the first 10 terms is 35. What is the common difference?

$$s_{10} = \frac{10}{2}(-10 + a_{10}) = 35$$

$$-10 + a_{10} = 7$$

$$a_{10} = 17$$

$$17 = -10 + 9d$$

$$9d = 27$$

$$d = 3$$

5) The sum of the first twelve terms of an arithmetic series is 348, and the sum of the first 30 terms of the same series is 1950. Write the first four terms of the series

$$s_{12} = \frac{12}{2}(a_1 + a_{12}) = 348$$

$$a_1 + a_{12} = 58$$

$$a_1 + a_1 + 11d = 58$$

$$2a_1 = 58 - 11d$$

$$s_{30} = \frac{30}{2}(a_1 + a_{30}) = 1950$$

$$a_1 + a_{30} = 130$$

$$a_1 + a_1 + 29d = 130$$

$$2a_1 = 130 - 29d$$

$$58 - 11d = 130 - 29d$$

$$d = 4$$

$$a_1 = 7$$

6) The sum of the first 8 terms of an arithmetic series is 36, and the sum of the first 100 terms is 14250. What is the sum of the first 50 terms?

$$s_8 = \frac{8}{2}(a_1 + a_8) = 36$$

$$a_1 + a_8 = 9$$

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$a_1 + a_1 + 7d = 9$

$2a_1 = 9 - 7d$

$s_{100} = \frac{100}{2}(a_1 + a_{100}) = 14250$

$a_1 + a_{100} = 285$

$a_1 + a_1 + 99d = 285$

$2a_1 = 285 - 99d$

$285 - 99d = 9 - 7d$

$d = 3$

$a_1 = -6$

$s_{50} = \frac{50}{2}(a_1 + a_{50}) = 25(-6 + a_{50})$

$a_{50} = -6 + 49 \times 3 = 141$

7) What is the sum of the first ten terms of an arithmetic series with first term x, and a common difference of $(2x - 1)$?

$s_{10} = \frac{10}{2}(x + 2x - 1) = 15x - 5$

8) The first term of an arithmetic series is x^2, and the sum of the first six terms is $2x^2 - 4x - 3$. If the sixth term is equal to zero, what are the possible values of x?

$s_6 = \frac{6}{2}(x^2 + a_6) = 2x^2 - 4x - 3$

$3x^2 + 3a_6 = 2x^2 - 4x - 3$

$a_6 = 0$

So: $x^2 + 4x + 3 = 0$

$x = -3 \; or \; x = -1$

9) Calculate the sum of the series $1 + 5 + 9 + \cdots + 49 + 53$

$a_n = a + (n - 1)d$

$a_n = 1 + (n - 1)4 = 53$

$4n - 4 = 52$

$n = 14$

$s_{14} = \frac{14}{2}(1 + 53) = 378$

10) If the sum of the first n terms of an arithmetic series is 98, the common difference is 4 and the first term is 2, what is the value of n?

$s_n = \frac{n}{2}(2 + a_n) = 98$

$a_n = 2 + (n - 1)4$

$a_n = 2 + 4n - 4 = 4n - 2$

$98 = \frac{n}{2}(2 + 4n - 2)$

$2n^2 = 98$

$n = 7$

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Exercise 4

Geometric Series

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1) Calculate the sum of the first 8 terms of the following geometric series

a) 1, 2, 4, ...

$$s_n = a\frac{1-r^n}{1-r}$$

$$s_8 = 1 \times \frac{1-2^8}{1-2} = \frac{-255}{-1} = 255$$

b) 2, 3, 4.5, ...

$$s_n = a\frac{1-r^n}{1-r}$$

$$s_8 = 2 \times \frac{1-1.5^8}{1-1.5} = 2 \times \frac{-25.6289}{-0.5} \cong 102.52$$

c) ____, 5, ____, 20, ...

$$r = 2, a = 2.5$$

$$s_n = a\frac{1-r^n}{1-r}$$

$$s_8 = 2.5 \times \frac{1-2^8}{1-2} = 2.5 \times \frac{-255}{-1} = 637.5$$

d) 675, ____, ____, 200, ...

$$a = 675, r = \frac{2}{3}$$

$$s_n = a\frac{1-r^n}{1-r}$$

$$s_8 = 675 \times \frac{1-(\frac{2}{3})^8}{1-\frac{2}{3}} = 675 \times \frac{1-\frac{256}{6561}}{1-\frac{2}{3}} \cong 1946$$

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e) $____, -500, ____, -125, ..$

$r = \frac{1}{2}, a = -1000$

$s_n = a\frac{1-r^n}{1-r}$

$s_8 = -1000 \times \frac{1-\left(\frac{1}{2}\right)^8}{1-\frac{1}{2}} = -1000 \times \frac{\frac{255}{256}}{\frac{1}{2}} \cong 1992$

2) The sum of the first 4 terms of a geometric series is 90 and the sum of the first two terms is 18. Write the first 4 terms of the series

$s_4 = a\frac{1-r^4}{1-r} = 90$

$s_2 = a\frac{1-r^2}{1-r} = 18$

$5a\frac{1-r^2}{1-r} = a\frac{1-r^4}{1-r}$

$5 - 5r^2 = 1 - r^4$

$r^4 - 5r^2 + 4 = 0$

$r^2 = 4 \; or \; r^2 = 1$

$r = \pm 1$ gives unreal solutions for a, so $r = \pm 2$

For $r = 2, a = 6$

For $r = -2, a = -18$

Series is $6 + 12 + 24 + 48$

Or

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$-18 + 36 - 72 + 144$

3) Calculate the sum of the first ten terms of the geometric series

$40, -20, 10, -5, \ldots \ldots$

$a = 40, r = -\frac{1}{2}$

$$s_{10} = 40 \times \frac{1 - \left(-\frac{1}{2}\right)^{10}}{1 - \left(-\frac{1}{2}\right)} = 40 \times \frac{1 - \frac{1}{1024}}{\frac{3}{2}} = 40 \times \frac{2}{3} \times \frac{1023}{1024} \cong 26.64$$

4) The sixth term of a geometric series is 40, and $r^5 = 20$, what is the value of the first term?

$a_6 = ar^5 = 40$

$20a = 40$

$a = \frac{1}{2}$

5) The sum of the first 4 terms of a geometric series is 30, and $r = 2$. What is the value of the first term of the series?

$$s_4 = a \times \frac{1 - 2^4}{1 - 2} = 30$$

$$a = \frac{-30}{-15} = 2$$

6) The sum of the first 4 terms of a geometric series is 800, and the first term is 20. What is the value of r?

$$s_4 = a \times \frac{1 - r^4}{1 - r} = 800$$

$$800 = 20 \times \frac{1 - r^4}{1 - r}$$

$40(1 - r) = 1 - r^4$

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$r^4 - 39r + 40 = 0$

By trial and error, graphing or logarithms

$r = 3$

7) The sum of the first nine terms of a geometric series is 1, and the sum of the first ten terms is 0. What is the value of the first term, and the value of r?

$s_9 = a \times \frac{1 - r^9}{1 - r} = 1$

$a(1 - r^9) = 1 - r$

$s_{10} = a \times \frac{1 - r^{10}}{1 - r} = 0$

$a(1 - r^{10}) = 0$

$1 - r^{10} = 0$

$r = \pm 1$

$r = 1, a = 0$ which does not lead to a geometric series

$r = -1, a = 1$

8) If the fourth term of a geometric series is $\frac{5}{64}$, and the common ratio is $\frac{1}{2}$, what is the sum of the first six terms?

$a_4 = ar^3 = \frac{5}{64}$

$\frac{1}{8}a = \frac{5}{64}$

$a = \frac{5}{64} \times 8 = \frac{5}{8}$

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$$s_6 = \frac{5}{8} \times \frac{1 - \left(\frac{1}{2}\right)^6}{1 - \frac{1}{2}} = \frac{5}{8} \times \frac{1 - \frac{1}{64}}{\frac{1}{2}} = \frac{5}{8} \times 2 \times \frac{63}{64} = \frac{630}{512} = \frac{315}{256}$$

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Exercise 5

Series Notation; Convergence & Divergence

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1) Determine if the following sequences converge, diverge or oscillate. If the sequence converges, state the limiting value

a) $a_n = \frac{1}{n}$

$1, \frac{1}{2}, \frac{1}{4}, \ldots$

$r = \frac{1}{2}$

Geometric sequence with $r < 1$, so sequence converges

b) $a_n = (-1)^n$

$-1, 1, -1, 1, \ldots\ldots$

Sequence oscillates

c) $a_n = \frac{n}{n+1}$

$\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots\ldots$

$$\lim_{n\to\infty} \frac{n}{n+1} = \lim_{n\to\infty} \frac{1}{1+\frac{1}{n}} = 1$$

Sequence diverges

d) $a_n = 2n + 3$

$$\lim_{n\to\infty} 2n + 3 = \lim_{n\to\infty} 2 + \frac{3}{n} = 2$$

Sequence diverges

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2) Calculate the sum of the first ten terms of the geometric series

$$\sum_{k=1}^{10} a^k$$

for the given value of a, and determine if the series converges, diverges or neither

a) $a = \frac{1}{2}$

$$s_{10} = \frac{1}{2} \times \frac{1 - \left(\frac{1}{2}\right)^{10}}{1 - \frac{1}{2}} = \frac{1}{2} \times \frac{1 - \frac{1}{1024}}{\frac{1}{2}} = \frac{1023}{1024}$$

Series converges

b) $a = 2$

$$s_{10} = 2 \times \frac{1 - 2^{10}}{1 - 2} = 2 \times 1023 = 2046$$

Series diverges

c) $a = -1$

$$s_{10} = -1 + 1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 + 1 = 0$$

Series oscillates

d) $a = -\frac{1}{4}$

$$s_{10} = -\frac{1}{4} \times \frac{1 - \left(-\frac{1}{4}\right)^{10}}{1 - -\frac{1}{4}} = \frac{1}{3}\left(1 - \frac{1}{1048576}\right) = \frac{1048575}{3145728} \cong \frac{1}{3}$$

Series converges

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e) $a = 1$

$s_{10} = 10$

Neither

3) From your answers to question 2, for what value(s) of r does a geometric series converge?

$|r| < 1$

4) Write the following series in summation notation

a) $1 + 3 + 5 + 7 + \cdots . + 33$

Series is arithmetic with $a = 1, d = 2$

$a_n = 1 + (n - 1)2 = 33$

$1 + 2n - 2 = 33$

$n = 17$

$$\sum_{k=1}^{17} 2k - 1$$

b) $2 + 4 + 8 + \cdots + 256$

Series is geometric with $a = 2, r = 2$

$a_n = 2 \times 2^{n-1} = 256$

$2^n = 256$

$n = 8$

$$\sum_{k=1}^{8} 2^k$$

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c) $\frac{1}{3}+\frac{1}{5}+\frac{1}{7}+\cdots+\frac{1}{21}$

Series is neither geometric nor arithmetic

By inspection, $\frac{1}{21}$ represents the tenth term

$$\sum_{k=1}^{10}\frac{1}{2k+1}$$

d) $-3+6-9+12\ldots.60$

Series is neither geometric nor arithmetic

By inspection, 60 represents the 20th term

$$\sum_{k=1}^{20}(-1)^k 3k$$

5) List the first four terms of the following series

a) $\sum_{k=1}^{N}(k^2)$

$1, 4, 8, 16$

b) $\sum_{k=1}^{N}(2k+2)$

$4, 6, 8, 10$

c) $\sum_{k=0}^{N}(k-2)$

$-2, -1, 0, 1$

d) $\sum_{k=-2}^{N}|k|$

$2, 1, 0, 1$

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6) Calculate the sum of the first 5 terms of the series generated by the notation

$$\sum_{k=0}^{N} 4^{-k}$$

$$1+\frac{1}{4}+\frac{1}{16}+\frac{1}{64}+\frac{1}{256}=\frac{341}{256}$$

7) A Find the 40th term of the series generated by the notation

$$\sum_{k=1}^{N} 2k+2$$

Series is $4, 6, 8, \ldots$

Which is an arithmetic series with $a = 4, d = 2$

$$s_{40} = \frac{40}{2}(4 + a_{40})$$

$$a_{40} = 82$$

$$s_{40} = 20 \times 86 = 1720$$

8) Calculate the sum of the first ten terms of the series generated by the notation

$$\sum_{k=1}^{N} 2k-1$$

Series is $1, 3, 5, \ldots$

Series is arithmetic with $a = 1, d = 2$

Term 10 is 19

$$s_{10} = \frac{10}{2}(1 + 19) = 100$$

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9) Determine the terms of the following series, and express the sum in terms of n

$$\sum_{k=0}^{n} k^2 - (k+1)^2$$

Series is $-1, -3, -5, -7, \ldots$

$$\sum_{n=1}^{N} 1 - 2n$$

10) Determine the terms of the following series, and express the sum in terms of n

$$\sum_{k=1}^{n} \left(\frac{1}{k+1} - \frac{1}{k}\right)$$

Series is $-\frac{1}{2}, -\frac{1}{6}, -\frac{1}{12}, \ldots \ldots$

$$\sum_{n=1}^{N} \frac{-1}{n(n+1)}$$

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Exercise 6

Sum to Infinity

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1) Calculate the sum to infinity of the following sequences

a) $8, 4, 2, 1, \ldots$

$$s_\infty = \frac{a}{1-r} = \frac{8}{1-\frac{1}{2}} = 16$$

b) $10, 1, 0.1, 0.01, \ldots.$

$$s_\infty = \frac{a}{1-r} = \frac{10}{1-\frac{1}{10}} = \frac{100}{9}$$

c) 12, 3, 0.75, ….

$$s_\infty = \frac{a}{1-r} = \frac{12}{1-\frac{1}{4}} = 16$$

d) $\frac{5}{8}, \frac{5}{16}, \frac{5}{32}, \ldots.$

$$s_\infty = \frac{a}{1-r} = \frac{\frac{5}{8}}{1-\frac{1}{2}} = \frac{5}{4}$$

e) 6.4, 0.8, 0.1, ….

$$s_\infty = \frac{a}{1-r} = \frac{6.4}{1-\frac{1}{8}} = \frac{512}{70}$$

f) $2, 4, 8, 16, \ldots \ldots$

Sum to infinity cannot be calculated as $r > 1$

2) Calculate the following

$$\sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k$$

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$$s_\infty = \frac{a}{1-r} = \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1$$

3) The sum to infinity of a geometric series is 18. If the common ratio is $\frac{2}{3}$, what is the first term of the series?

$$s_\infty = \frac{a}{1-r} = \frac{a}{1-\frac{2}{3}} = 18$$

$$a = 6$$

4) The first term of a geometric series is 21, and its sum to infinity is 28. What is the common ratio?

$$s_\infty = \frac{a}{1-r} = \frac{21}{1-r} = 28$$

$$1 - r = \frac{21}{28}$$

$$r = \frac{7}{28} = \frac{1}{4}$$

5) Prove with the use of a geometric series that $0.\dot{9} = 1$

$0.\dot{9} = 0.9 + 0.09 + 0.009 + \cdots.$

Which is a geometric series with $a = 0.9 = \frac{9}{10}$ and $r = \frac{1}{10}$

$$s_\infty = \frac{a}{1-r} = \frac{\frac{9}{10}}{1-\frac{1}{10}} = \frac{\frac{9}{10}}{\frac{9}{10}} = 1$$

6) Which scenario would get you more money?

- \$10 on day 1 and $\frac{3}{4}$ of what you received the day before from then on
- \$20 on day 1 and $\frac{1}{2}$ of what you received the day before from then on

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Series 1

$$s_\infty = \frac{10}{1-\frac{3}{4}} = \$40$$

Series 2

$$s_\infty = \frac{20}{1-\frac{1}{2}} = \$40$$

You would receive the same amount

7) A form of Zeno's paradox (Zeno was a contemporary of Socrates) postulates that one can never walk across a room, since first one must cover half the distance of the room, then half the remaining distance, then half the remaining distance and so on. Since there will always be a fraction of a distance to cover, the total journey is impossible. Reconcile this paradox with the use of a geometric series

The travel is a sum to infinity with $a = \frac{1}{2}\ and\ r = \frac{1}{2}$

$$s_\infty = \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1$$

The sum of the halves is the total distance

8) A person weighing 210 kg plans to lose 10 kg in the first month of their diet, then 8 kg in the second month, 6.4 kg in the third month, and so on repeating the pattern of weight loss. Their goal is to eventually reach 150 kg. Will they be successful with this strategy? Explain your answer.

The weight to lose is a geometric series with $a = 10, and\ r = \frac{8}{10}$

$$s_\infty = \frac{10}{1-\frac{8}{10}} = 50$$

The maximum weight loss is 50 kg, so the goal weight will never be reached

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9) If the person from question 8 wanted to achieve their goal weight, but maintaining the same pattern of weight loss, how much weight would they have to lose in the first month?

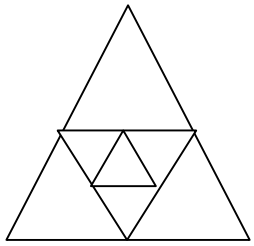
Needs to lose 60 kg

$$s_\infty = \frac{a}{1 - \frac{8}{10}} = 60$$

$$a = 12$$

Needs to lose 12 kg in the first month, and 20% thereafter

10) An equilateral triangle has a side length of x cm. Another equilateral triangle is inscribed inside the first one such that the vertices of the second triangle sit at the midpoint of the sides of the larger triangle. (See diagram). This process is repeated infinitely. What is the sum of the perimeters of the triangles?



x

Perimeter of triangle 1 is $3x$

Perimeter of next smallest triangle is $\frac{3x}{2}$

Similarly the next size triangle has a perimeter of $\frac{3x}{4}$,

This gives a geometric series with $a = 3x \; and \; r = \frac{1}{2}$

$$s_\infty = \frac{3x}{1 - \frac{1}{2}} = 6x$$

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Exercise 7

Arithmetic & Geometric Mean

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1) Calculate the arithmetic mean of the first 8 terms of the following sequences

a) 2, 4, 6, ...,

$a_8 = 2 + 7 \times 2 = 16$

Mean $\frac{s_8}{8} = \frac{4(2+16)}{8} = 9$

b) 3, 9, 15, 21, ...

$a_8 = 3 + 7 \times 6 = 45$

Mean $\frac{s_8}{8} = \frac{4(3+45)}{8} = 24$

c) $-6, -2, 2, \ldots..$

$a_8 = -6 + 7 \times 4 = 22$

Mean $\frac{s_8}{8} = \frac{4(-6+22)}{8} = 8$

d) 1.5, 2.75, 4, ...

$a_8 = 1.5 + 7 \times 2.25 = 19.25$

Mean $\frac{s_8}{8} = \frac{4(1.5+19.25)}{8} = 10.375$

e) $\frac{5}{8}, \frac{31}{24}, \frac{47}{24}, \ldots\ldots,$

$a_8 = \frac{5}{8} + 7 \times \frac{2}{3} = \frac{127}{24}$

Mean $\frac{s_8}{8} = \frac{4\left(\frac{5}{8}+\frac{127}{24}\right)}{8} = \frac{142}{48} = \frac{71}{24}$

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2) Calculate the arithmetic mean of the series denoted by

$$\sum_{n=1}^{10} 10n - 13$$

Series is $-3, 7, 17, \ldots$.

$a_{10} = -3 + 9 \times 10 = 87$

Mean $= \frac{s_{10}}{10} = \frac{5(-3+87)}{10} = 42$

3) What is the arithmetic mean of the set of multiples of 6 between 18 and 96 inclusive?

Series is $18, 24, 30, \ldots., 96$

$a_{10} = 18 + (n-1)6 = 96$

$6n - 6 = 78$

$n = 12$

Mean $= \frac{s_{12}}{12} = \frac{6(18+96)}{12} = 57$

4) An arithmetic series has the following terms

$2x + 1, 2x + 3, \ldots \ldots . 2x + 31$

If the arithmetic mean of the series is 40, calculate the value of x

$a_n = (2x + 1) + (n-1)2 = 2x + 31$

$2x + 1 + 2n - 2 = 2x + 31$

$2n = 32$

$n = 16$

$s_n = 8(2x + 1 + 2x + 31) = 8(4x + 32)$

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Mean $= \frac{8(4x+32)}{16} = 2x + 16$

5) Calculate the geometric mean of the sequence $\frac{1}{4}, 1, 4$

Mean $= \sqrt[3]{4 \times 1 \times \frac{1}{4}} = 1$

6) Calculate the geometric mean of the set of numbers 22, 20, 12, 4, 0

Product of terms is zero, so mean is zero

7) Calculate the geometric mean of the set of numbers 3.6, 5, 6.9, 10

Mean $= \sqrt[4]{3.6 \times 5 \times 6.9 \times 10} = \sqrt[4]{1242} \cong 5.94$

8) Insert two geometric means in each of the following geometric series

a) 1, ____, ____, 27

$3, 9$

b) $\frac{3}{8}$, ____, _____, $\frac{1}{9}$

$r = \frac{2}{3}$

$\frac{1}{4}, \frac{1}{6}$

c) $\frac{2}{5}$, ___, ____, $\frac{1}{160}$

$r = \frac{1}{4}$

$\frac{1}{10}, \frac{1}{40}$

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d) $\sqrt{6}, ____, ____, 2\sqrt{6}$

$r = \sqrt[3]{2}$

$\sqrt{6}\sqrt[3]{2}, \sqrt{6}\sqrt[3]{4}$

e) $\frac{1}{8}, ____, ____, \sqrt{8}$

$r = \sqrt{8}$

$\frac{1}{\sqrt{8}}, 1$

9) Prove with two examples if the geometric mean is always, sometimes or never larger than the arithmetic mean for the same data set

For $1, 3, 10, 22$

Arithmetic mean $=9$

Geometric mean $= \sqrt[4]{880} \cong 5.45$

For $4, 100, 1000$

Arithmetic mean $= 368$

Geometric mean $= \sqrt[3]{400000} \cong 73.68$

From the examples, the geometric mean is smaller than the arithmetic mean

10) Write a set of data for which the arithmetic and geometric means are the same

Any set of data with the same value for all terms (e.g. 2, 2, 2,2)

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Exercise 8

Applications of Series

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1) Convert the following recurring decimals to fractions

a) $0.\dot{4}\dot{3}$

$$0.\dot{4}\dot{3} = \frac{43}{100} + \frac{43}{10000} + \cdots$$

Is a geometric series with:

$$a = \frac{43}{100}, r = \frac{1}{100}$$

$$s_\infty = \frac{\frac{43}{100}}{1 - \frac{1}{100}} = \frac{43}{99}$$

b) $0.\dot{2}\dot{6}$

Similarly $= \frac{26}{99}$

c) $0.\dot{5}$

Similarly $= \frac{5}{9}$

d) $0.\dot{1}\dot{4}$

Similarly $= \frac{14}{99}$

e) $0.\dot{1}$

Similarly $= \frac{1}{9}$

2) A man has \$20 in his bank as at July 1st, and each day thereafter puts \$4 more into it than his balance

a) How much money will he have in the piggy bank on July 4th?

Series is arithmetic with $a = 20, d = 4$

For $d = 4$

$$s_3 = \frac{4}{2}(20 + 32) = \$104$$

b) How much money will he have in his piggy bank on August 6th?

For $d = 37$

$$s_{37} = \frac{37}{2}(20 + a_{37})$$

$$a_{37} = 20 + 36 \times 4 = 164$$

$$s_{37} = \frac{37}{2} \times 184 = \$3404$$

c) He is saving to buy a suit that costs \$800. When will he be able to buy the suit?

$$s_n = \frac{n}{2}(20 + a_n) = 800$$

$$a_n = 20 + (n-1)4$$
$$= 16 + 4n$$

$$800 = \frac{n}{2}(20 + 16 + 4n)$$

$$1600 = 36n + 4n^2$$

$$n^2 + 9n - 400 = 0$$

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$n = 16\ or\ -25$

Therefore can afford suit on July 16th

3) A set of rocks is stacked in rows with 30 on the bottom and 2 less on each subsequent row

a) How many rocks are in the 5th row?

$a = 30, d = -2$

$a_n = 30 + 4 \times (-2) = 22$

b) In which row are there 8 rocks?

$a_n = 30 + (n-1)(-2) = 8$

$8 = 32 - 2n$

$n = 12$

c) How many rows do you need to have a total of 168 rocks?

$s_n = \frac{n}{2}(30 + a_n) = 168$

$a_n = 30 + (n-1)(-2)$

$a_n = 32 - 2n$

$168 = \frac{n}{2}(30 + 32 - 2n)$

$336 = 62n - 2n^2$

Solving for n gives

$n = 7\ or\ n = 24$

$n = 24$ is not a realistic solution in this context, since it would give rows of negative blocks

Therefore there are 7 rows containing 168 rocks

d) How many rocks in the entire set?

The last row contains 2 blocks

$a_n = 30 + (n-1)(-2) = 2$

$30 - 2n + 2 = 2$

$n = 15$

$s_{15} = \frac{15}{2}(30 + 2) = 240$

4) A company pays off a loan by paying $200 in the first week and increases their repayments by $30 each subsequent week

a) How much will they repay in week 10?

$a = 200, d = 30$

$a_{10} = 200 + 9 \times 30 = \$470$

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b) How much will they have repaid by week 20?

$$s_{20} = \frac{20}{2}(200 + a_{20})$$

$$a_{20} = 200 + 19 \times 30$$

$$a_{20} = 770$$

$$s_{20} = 10(200 + 770)$$
$$= \$9700$$

c) The loan requires 50 payments; how much will the last payment be?

$$a_{50} = 200 + 49 \times 30$$
$$= \$1670$$

d) How much will the company repay in this time?

$$s_{50} = \frac{50}{2}(200 + 1670)$$

$$s_{50} = \$93500$$

5) Find the sum of

a) The first 40 multiples of 9

$$a = 9, d = 9$$

$$a_{40} = 9 + 39 \times 9 = 360$$

$$s_{40} = \frac{40}{2}(9 + 360)$$
$$= 7380$$

b) The multiples of 8 between 0 and 1000

$$a = 8, d = 8$$

$$a_n = 8 + (n - 1)8$$
$$= 1000$$

$$8n = 1000$$

$$n = 125$$

$$s_{125} = \frac{125}{2}(8 + 1000)$$
$$= 63000$$

c) The multiples of 7 between 500 and 1000

$$a = 504, d = 7$$

$$a_n = 7 + (n - 1)7$$
$$= 1000$$

$$7n = 1000$$

$$n = 142.9$$

Term 72 is first multiple of 7 over 500

Term 142 is last multiple of 7 under 1000 =994

There are 71 terms in the series

$$Sum = \frac{71}{2}(504 + 994)$$
$$= 53179$$

Type equation here.

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6) A line is cut into 5 sections which form a geometric sequence. The shortest length is 4cm and the longest is 64cm. What is the length of the line?

$a = 4, n = 8, a_5 = 64$

$s_5 = a\dfrac{1 - r^5}{1 - r}$

$ar^4 = 64$

$r = 2$

$s_5 = 4 \times \dfrac{1 - 32}{1 - 2} = 124\ cm$

7) A ball is dropped from 2 metres, and each time it bounces it reaches 70% of its previous height.

a) How high will it bounce after its fourth bounce?

Downward travel forms a geometric series with

$a = 2, r = 0.7$

Upward travel forms a geometric series with

$a = 1.4. r = 0.7$

After 4th upward bounce

$a_4 = 1.4 \times 0.7^3 \cong 0.48\ m$

b) What distance will it have travelled when it hits the ground for the 6th time?

Addition of 2 series

Downward travel

$s_6 = 2 \times \dfrac{1 - 0.7^6}{1 - 0.7}$
$\cong 5.88\ m$

Upward travel

$s_5 = 1.4 \times \dfrac{1 - 0.7^5}{1 - 0.7}$
$\cong 3.88\ m$

Total distance $= 9.76\ m$

c) How far does it travel before coming to rest?

For downward

$s_\infty = \dfrac{2}{1 - 0.7} \cong 6.67$

For downward

$s_\infty = \dfrac{1.4}{1 - 0.7} \cong 4.67\ m$

Total distance $= 11.34\ m$

8) You are given a new job with a choice of pay method

a) $1000 on your first day and a pay increase of $200 per

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day (that is on day two you earn another $1200 etc)

Arithmetic series

$a = 1000, d = 200$

For 30 days

$a_{30} = 1000 + 29 \times 200$

$a_{30} = \$6800$

$$s_{30} = \frac{30}{2}(1000 + 6800) = \$117000$$

b) 1 cent on your first day, and double the previous day's pay each day (that is on day two you earn another 2 cents, on day 3 you earn another 4 cents etc)

Geometric series

$a = 0.01, r = 2$

$$s = 0.01\frac{1 - 2^{30}}{1 - 2}$$

$s_{30} \cong 10737418$

Which pay option would you choose?

Option 2 pays much more

9) Each hour a bell rings the number of times corresponding to the time of day (for example, at 4 o'clock it rings 4 times). How many times does the bell ring per day? (Assume not a 24 hour clock)

Arithmetic series with $a = 1, d = 1$

From 1 to 12

$$s_{12} = \frac{12}{2}(1 + 12) = 78$$

This happens twice a day

Bell rings 156 times

10) At the end of the year 2002, world oil reserves were about 950 billion barrels.
During 2003, about 30 billion barrels of oil were consumed. Over the past decade, oil consumption has been increasing at about 1% a year. Assuming oil consumption increases at this rate in the future, how long will reserves last?

Geometric series

$a = 30, r = 1.01$

For what value of n does:

$s_n = 950$

$$s_n = 30 \times \frac{1 - 1.01^n}{1 - 1.01} = 950$$

$$1 - 1.01^n = \frac{950 \times (-0.01)}{30} = -0.317$$

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$1.01^n = 1.317$

Solving for n $\cong 30\ years$

11) Every day person consumes 10 micrograms of a toxin which leaves the body at a rate of 3% per day. How much toxin is accumulated in the body in the long run?

Geometric series

$a = 10, r = 0.97$

$$s_\infty = \frac{10}{1 - 0.97} \cong 333\ \mu g$$

12) A plant is eaten by a caterpillar, the caterpillar by a fish, the fish by a bigger fish, and the large fish eaten by a man. If only 20% of the energy is transformed from one stage to the next, how many calories must be supplied by plant food to provide the man with 2,000 calories from the large fish?

Geometric series

$r = 0.2, n = 5$

Need $a_5 = 2000$

$$a_5 = a \times \frac{1 - 0.2^5}{1 - 0.2} = 2000$$

$$a = \frac{2000 \times 0.8}{0.99968} \cong 1600$$

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Exercise 9

Financial Applications

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1) The value of a computer depreciates at the rate of 12.5% per year. If it originally costs \$5000, how much will it be worth after 5 years?

$V = 5000 \times (0.875)^4 \cong \$2931$

2) Calculate the value of an investment of \$4000 compounded at 5% annually after 6 years

$V = 4000 \times 1.05^5 \cong \$5105$

3) An investment fund returns 7.5% interest annually. Each year a man puts \$2000 into the fund. How much will he have in his fund after 20 years?

After one year he will have $\$2000 \times 1.075$

After 2 years he will have $2000 \times 1.075^2 + 2000 \times 1.075$

After 3 years he will have $2000 \times 1.075^3 + 2000 \times 1.075^2 + 2000 \times 1.075$

Etc

The series can be simplified for 20 years to

Amount $= 2000(1.075 + 1.075^2 + 1.075^3 + \cdots \ldots + 1.075^{20})$

The bracketed sum is a geometric series with $a = 1.075, and\ r = 1.075$

$$s_{20} = 1.075 \times \frac{1 - 1.075^{20}}{1 - 1.075} = 1.075 \times \frac{-3.25}{-0.075} = 46.58$$

Amount $= 46.58 \times 2000 \cong \$93166.67$

4) A man borrows \$10,000 at 2% per month reducible interest, and makes repayments each month. What should his repayments be to have the loan paid off after 5 years?

Let repayment amount be P

After 1 month, amount owing $= (10000 \times 1.02) - P$

After 2 months, amount owing $= \big((10000 \times 1.02 - P) \times 1.02\big) - P$

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$= 10000 \times 1.02^2 - 1.02P - P$

After 3 months, amount owing $= 10000 \times 1.02^3 - 1.02^2P - 1.02P - P$

After 60 months, amount owing

$= 10000 \times 1.02^{60} - (1.02^{59}P + 1.02^{58}P + 1.02^{57}P + \cdots .. 1.02^2P + 1.02P + P)$

The bracketed quantity is a geometric series with first term P and a common ratio of 1.02

$$s_{60} = P\frac{1 - 1.02^{60}}{1 - 1.02} = 114P$$

Require amount owing to be zero

Therefore: $10000 \times 1.02^{60} - 114P = 0$

$$P = \frac{10000 \times 1.02^{60}}{114} \cong \$288$$

5) A government uses proceeds from a federal grant to provide a tax rebate for land owners. Suppose an individual receives a $600 rebate and spends 90% of this, and each of the recipients of the money spent by this individual also spends 90% of what he or she receives, and this process continues without end. According to the multiplier effect theory in economics, the effect of the original $600 tax rebate on the economy is multiplied many times. What is the total amount spent if the process continues as indicated?

The multiplier is a geometric series with a=600 and r=0.9

$$s_\infty = \frac{a}{1 - r} = \frac{600}{1 - 0.9} = \$6000$$

6) A sweepstakes has $4,000,000 in prizes. The first ticket drawn wins $15, the second ticket drawn wins $45; the third ticket drawn wins $135, and so on.

a) How many tickets can be drawn without giving away more than the allotted prize money?

The prize allocation is a geometric series with a=15 and r=3

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$$s_n = 15\frac{1-3^n}{1-3} = -7.5 \times (1-3^n) = 4000000$$

When $1-3^n = -533{,}333.33$

$3^n = 533{,}334.333$

$n \log 3 = \log 533{,}334.33$

$n = 12$

b) How much is the last prize?

$a_{12} = ar^{11} = 15 \times 3^{11} = \$2{,}657{,}205$

7) After how many months will an investment of $15000 be worth more than $18000 given that the interest rate is 10% per annum calculated monthly?

Value $= 15000 \times \left(1+\frac{0.1}{12}\right)^n = 15000 \times 1.00833^n = 18000$

When $1.00833^n = \frac{18000}{15000} = 1.2$

$$n = \frac{\log 1.2}{\log 1.00833}$$

$n = 22$

Investment will pass $18000 after 22 months

8) There are two investment schemes available. Scheme A pays simple interest of 8% paid yearly, while scheme B pays 6.5% interest compounded annually. Which investment will give the greater return?

Let amount invested be Q

Under simple interest, I=PRT

$I = Q \times 0.08 \times t$

Total amount $= I + Q$

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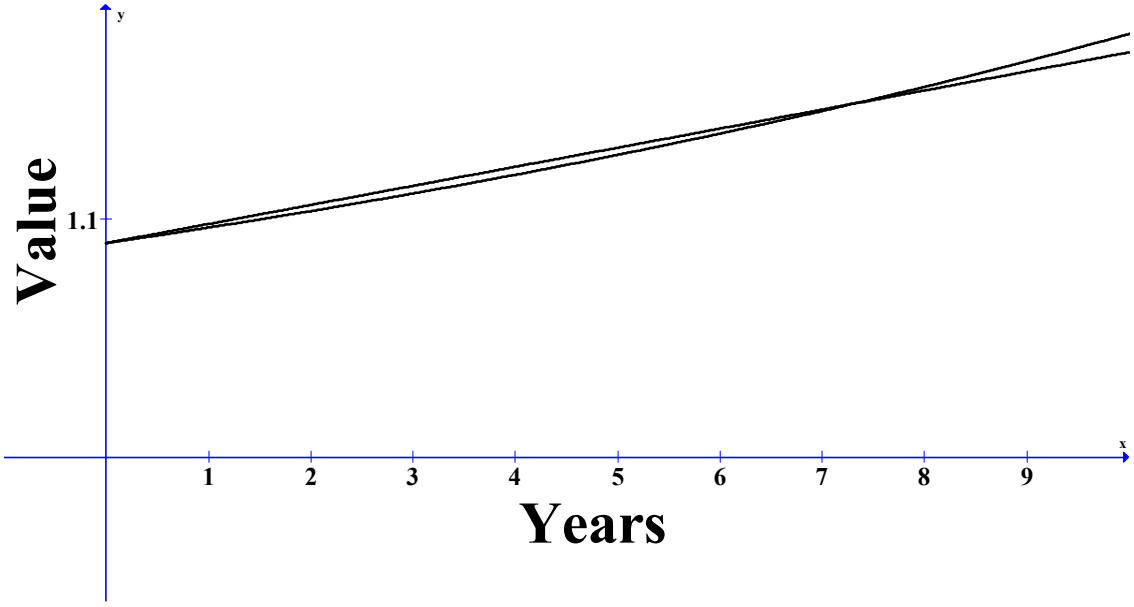
Under compound interest
$A = Q(1 + 0.065)^t$

When does $Q + (Q \times 0.08t) = Q(1.065)^t$

Removing Q from both sides

$1 + 0.08t = 1.065^t$

Solving graphically it can be seen that simple interest is worth more until year 7, and compound interest is worth more from year 8



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Year 12 Mathematics

Chance

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Exercise 1

Probability

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1) A six sided die is thrown. Find the probability that

a) A six is thrown

$\frac{1}{6}$

b) An even number is thrown

$\frac{3}{6} = \frac{1}{2}$

c) A number greater than 2 is thrown

$\frac{4}{6} = \frac{2}{3}$

d) An odd number less than 5 is thrown

$\frac{2}{6} = \frac{1}{3}$

e) An even number greater than 5 is thrown

$\frac{1}{6}$

2) A die has 12 sides, numbered from 1 to 12. Find the probability that when it is thrown

a) The number is 5

$\frac{1}{12}$

b) The number is less than 8

$\frac{7}{12}$

c) The number is an even number less than 8

$\frac{3}{12}$

d) The number is an odd number greater than 11

0

3) Two six sided dice are thrown. What is the probability that

a) The total of the two dice is 8

$\frac{5}{36}$

b) The total of the two dice is less than 10

$\frac{30}{36} = \frac{5}{6}$

c) Both numbers are even

$\frac{9}{36} = \frac{1}{4}$

d) One number is greater than 2 and the other is less than 4

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For dice 1 >2, and dice 2 <4 there are 12 possibilities

For dice 1<4 and dice 2>2 there are 12 possibilities

However the combination 3, 3 only occurs once but is in both sets

Therefore probability $= \frac{23}{36}$

e) Both numbers are a 3

$$\frac{1}{36}$$

4) There are 80 tickets in a raffle. How many tickets must someone buy to have a better than 50% chance of winning?

41

5) A card is drawn from a standard deck of 52. What is the probability that

a) It is a king

$$\frac{4}{52} = \frac{1}{13}$$

b) It is a number less than 5

$$\frac{16}{52} = \frac{4}{13}$$

c) It is a king or a number less than 5

$$\frac{20}{52} = \frac{5}{13}$$

d) It is a picture card

$$\frac{12}{52} = \frac{3}{13}$$

e) It is a six

$$\frac{4}{52} = \frac{1}{13}$$

f) It is black

$$\frac{26}{52} = \frac{1}{2}$$

g) It is a black 6

$$\frac{2}{52} = \frac{1}{26}$$

6) Are the events drawing king and drawing a number less than 5 mutually exclusive? Explain

Yes, the sum of the probabilities equals the probability of the events together

7) There are 10 cards in a pack, 5 red and 5 black, each numbered from 1 to 5. Find the probability that a card drawn at random

a) Is a 5

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$$\frac{2}{10} = \frac{1}{5}$$

b) Is red

$$\frac{5}{10} = \frac{1}{2}$$

c) Is a red 5

$$\frac{1}{10}$$

d) Is a red or a 5

$$\frac{6}{10} = \frac{3}{5}$$

e) Are the events "drawing a 5" and "drawing a red card" mutually exclusive? Explain

No, there is a red 5, so the sum of the probabilities is not equal to the probability of the two events

8) Two coins are tossed. Find the probability that

a) The first coin shows a head

$$\frac{1}{2}$$

b) The second coin shows a tail

$$\frac{1}{2}$$

c) The first coin shows a head and the second shows a tail

$$\frac{1}{4}$$

d) The first shows a head or the second shows a tail

$$\frac{3}{4}$$

e) Are the events mutually exclusive? Explain

Yes, the sum of the probabilities equals the probability of the two events; one event does not affect the other

9) A man is driving around some square city blocks. At each intersection he either turns left, right, or goes straight ahead. If he goes through three intersections, what is the probability that he ends up back where he started from?

At each intersection probability of going in a certain direction is $\frac{1}{3}$

There are 27 possibilities
There are 4 ways to return to his starting point

Probability $= \frac{4}{27}$

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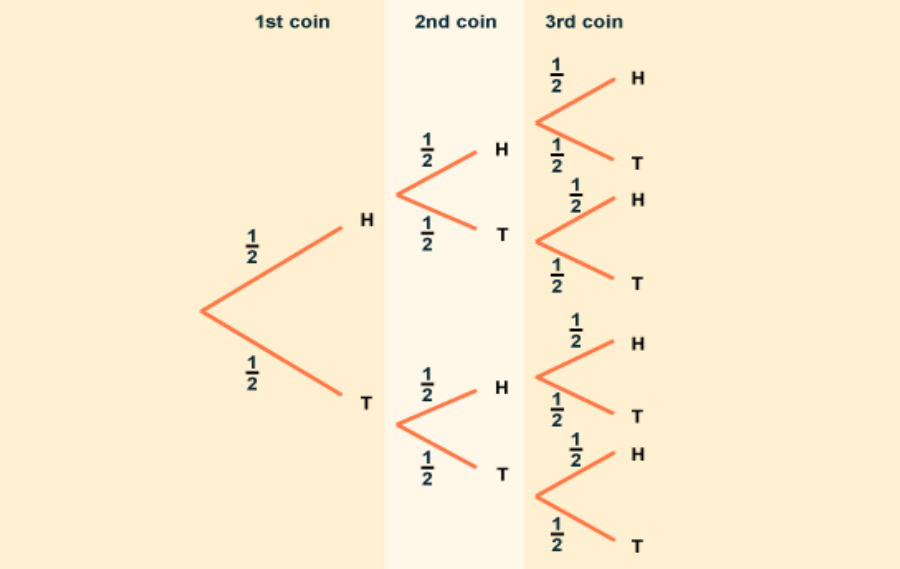
Exercise 2

Compound Probability

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1) A coin is tossed three times. Draw a tree diagram that shows all possible outcomes and from it calculate the probability of

a) Three heads



$\frac{1}{8}$

b) Two heads and one tail

$\frac{3}{8}$

c) At least two tails

$\frac{4}{8} = \frac{1}{2}$

d) One or two heads

$\frac{7}{8}$

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2) There are 5 black and 5 white shirts in a draw. Three are taken out without replacement. Draw a tree diagram and from it calculate the probability of

a) Three black shirts being taken

$$\frac{1}{2}\times\frac{4}{9}\times\frac{3}{8}=\frac{1}{12}$$

b) Two white and one black shirt being taken

$$\frac{1}{2}\times\frac{5}{9}\times\frac{1}{2}=\frac{5}{36}$$

$$+$$

$$\frac{1}{2}\times\frac{4}{9}\times\frac{5}{8}=\frac{5}{36}$$

$$+$$

$$\frac{1}{2}\times\frac{5}{9}\times\frac{1}{2}=\frac{5}{36}$$

$$=\frac{15}{36}=\frac{5}{12}$$

c) At least one white shirt being taken

= 1- probability of three whites being taken

$$=1-\frac{1}{2}\times\frac{4}{9}\times\frac{3}{8}=\frac{11}{12}$$

d) All three shirts being the same colour

$$\frac{1}{12}+\frac{1}{12}=\frac{1}{6}$$

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3) The probability of rain on any particular day in May is 70%. If three days from the month are chosen, use a tree diagram to calculate the probability that

a) All three are rainy

$0.7 \times 0.7 \times 0.7 = 0.343$

b) Two of the three days are dry

$= 3 \times (0.3 \times 0.3 \times 0.7) = 0.189$

c) It rained on at least one day

=1 – probability rained on all three days

$1 - 0.343 = 0.657$

d) The last day of the three is wet

$$(0.7 \times 0.7 \times 0.7) + (0.7 \times 0.3 \times 0.7) + (0.3 \times 0.7 \times 0.7) + (0.3 \times 0.3 \times 0.7) = 0.7$$

4) There are forty balls in a bag. Two of the balls have a star on them. If a man draws 5 balls from the bag, what is the probability that at least one has a star?

Probability = 1-probability that no starred balls are drawn

$$= 1 - \left(\frac{38}{40} \times \frac{37}{39} \times \frac{36}{38} \times \frac{35}{37} \times \frac{34}{36}\right) = \frac{37}{156}$$

5) There are 5 red, 3 green and 2 blue blocks in a box. Three are drawn out without replacement. What is the probability that

a) All are blue

Cannot draw all blue as there are only 2 blue socks to begin with

b) They are all different colours

There are 6 combinations of different colours

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Each has a probability of $\frac{1}{24}$

Total probability $= \frac{1}{4}$

c) All three are red

$$\frac{5}{10} \times \frac{4}{9} \times \frac{3}{8} = \frac{1}{12}$$

d) At least 2 are green

= probability all three are green + probability of 2 being green

$$= \left(\frac{1}{120}\right) + \left(\frac{7}{40}\right) = \frac{11}{60}$$

e) No red blocks are drawn

Possibilities are:

BGG $\left(\frac{12}{720}\right)$

BBG $\left(\frac{6}{720}\right)$

GBG $\left(\frac{12}{720}\right)$

GBB $\left(\frac{6}{720}\right)$

GGG $\left(\frac{6}{720}\right)$

Total probability $= \frac{42}{720} = \frac{7}{120}$

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6) When John plays striker in his soccer team he scores a goal 2 games out of 5 on average. His chance of playing striker is 25%. What is the probability that John scores a goal two games in a row?

Probability of scoring in one game $= 0.25 \times 0.4 = 0.1$

Probability of scoring two games in a row $= 0.1 \times 0.1 = 0.01$

7) Three cards are drawn from a standard deck of 52 with no replacement. What is the probability of drawing the king of spades, followed by the 7 of clubs then the 2 of hearts?

$$\frac{1}{52} \times \frac{1}{51} \times \frac{1}{50} = \frac{1}{132600}$$

8) Four numbers are drawn out of a barrel of fifty numbers with no replacement. The first number is 5. What is the probability that the next three numbers are **NOT** 6, 7, and 8 in that order

Probability = 1-probability of 6 then 7 then 8

$$= 1 - \left(\frac{1}{49} \times \frac{1}{48} \times \frac{1}{47}\right) = \frac{1}{110544}$$

9) There are 45 numbers in a lotto draw. Jim has one ticket with 6 numbers on it. What is the probability that he does not win first prize (all 6 numbers drawn)

Probability = 1-probability his 6 numbers drawn

$$= 1 - \left(\frac{6}{45} \times \frac{5}{44} \times \frac{4}{43} \times \frac{3}{42} \times \frac{2}{41} \times \frac{1}{40}\right) = 1 - \frac{1}{8145060}$$
$$= \frac{8145059}{8145060} \ (\cong 0.999999877)$$

10) When Ben sits a test he has a 70% chance of getting a question correct. If there are 10 questions on a test what is the probability of Ben getting at least one question correct?

Probability =1-probability of getting all wrong

$$1 - (0.3^{10}) = 1 - 0.0000059 = 0.9999941$$

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Year 12 Mathematics

Geometric Applications of Differentiation

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Exercise 1

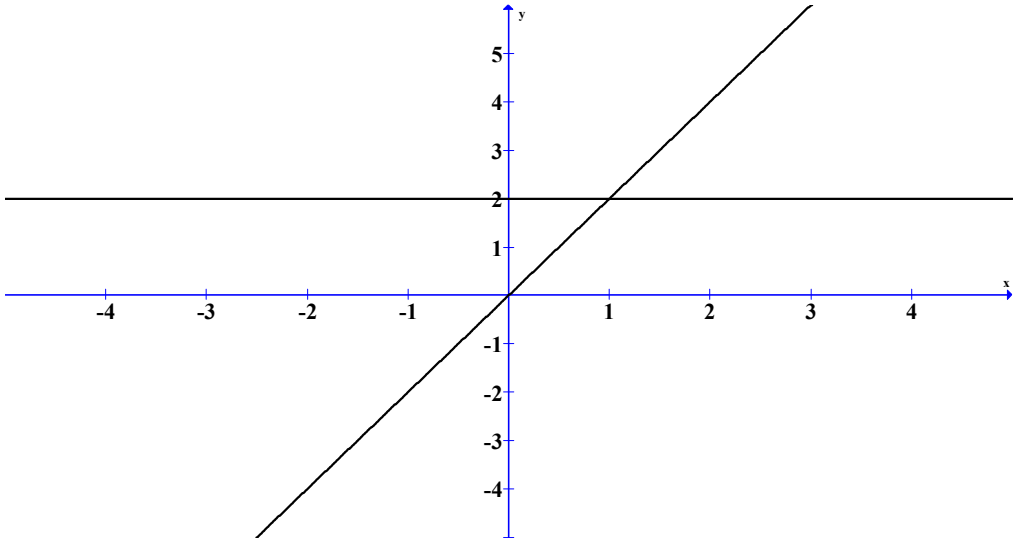
Critical Points of Functions

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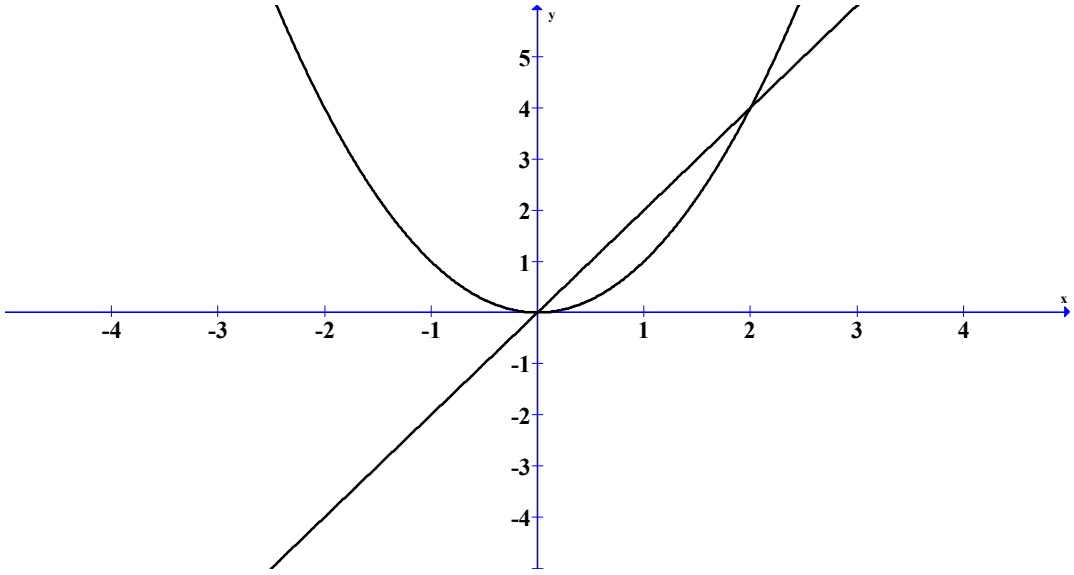
1) For each of the following functions:

- Graph the function in the domain $-4 \leq x \leq 4$
- Using the same domain and scale, graph the derivative of the function
- Complete the table

a) $y = 2x$

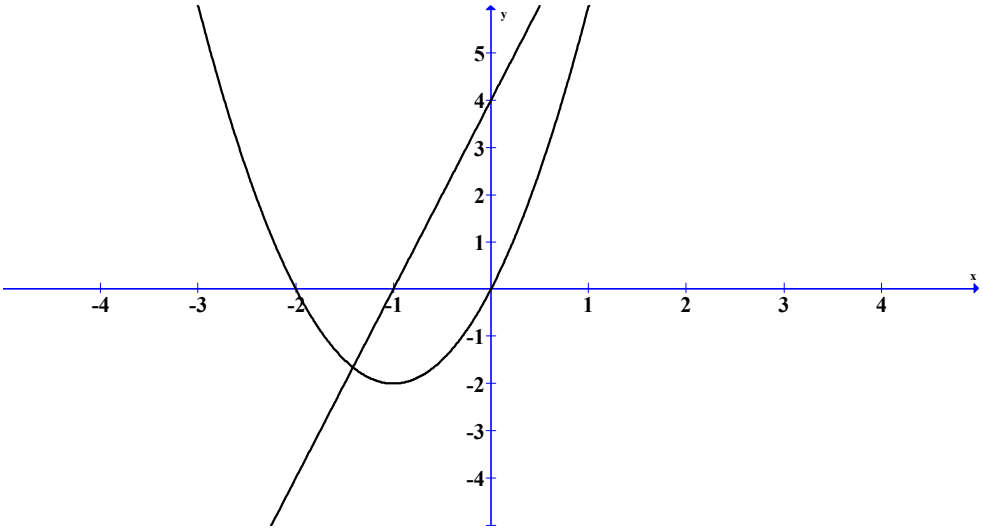


b) $y = x^2$

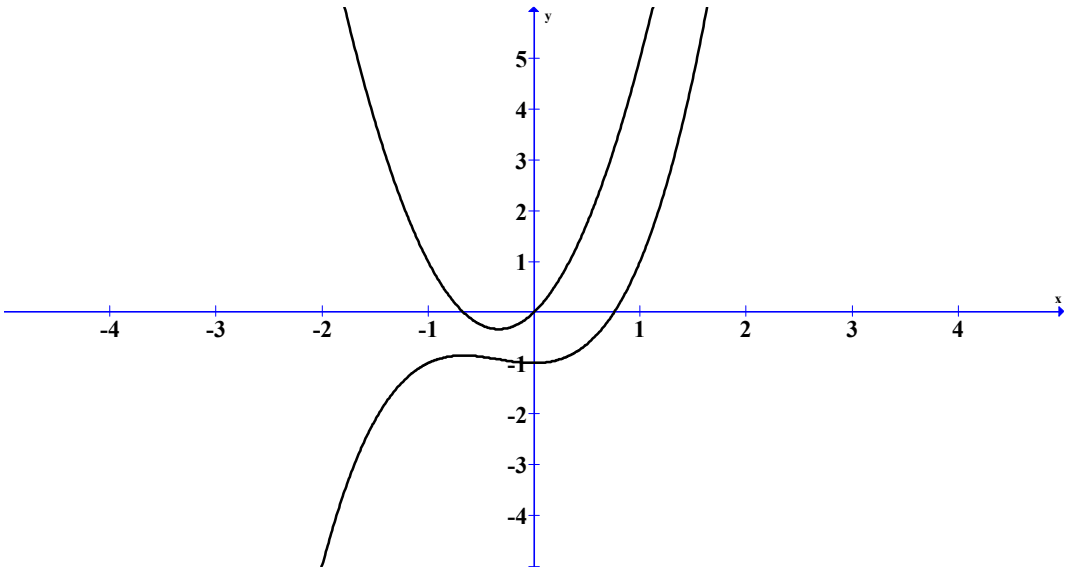


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c) $y = 2x^2 + 4x$

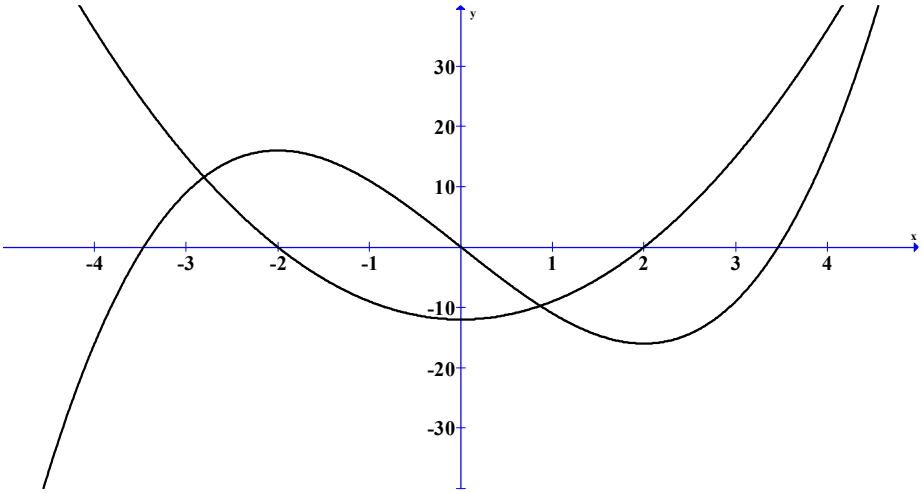


d) $y = x^3 + x^2 - 1$

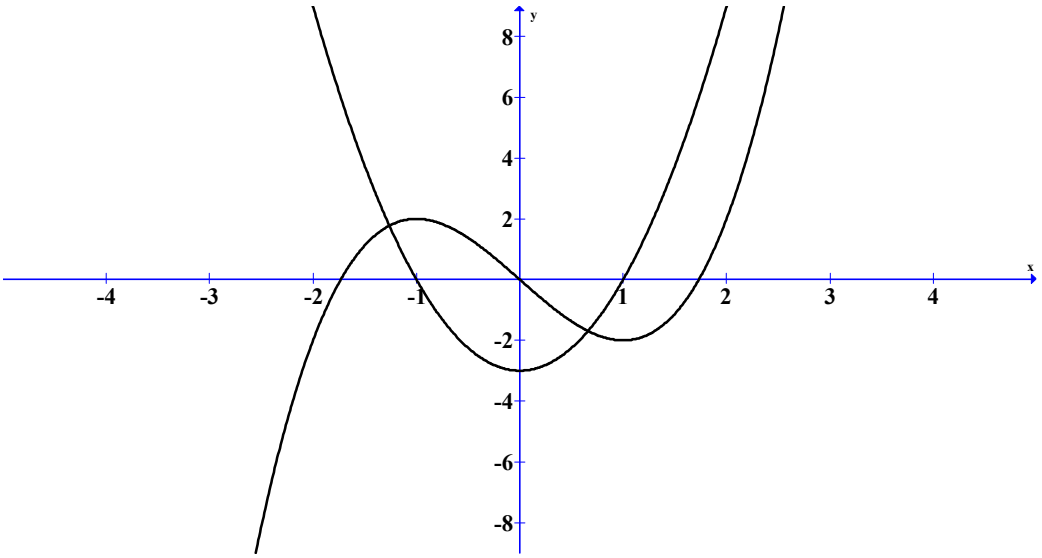


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e) $y = x^3 - 12x$



f) $y = x^3 - 3x$



Function	$\frac{dy}{dx} =$	Turning point(s) of y	Point(s) where $\frac{dy}{dx} = 0$ (Critical point)	Values of x where function is increasing	Values of x where function is decreasing
$y = 2x$	2	None	None	All	None
$y = x^2$	$2x$	$x = 0$	$x = 0$	$x > 0$	$x < 0$
$y = 2x^2 + 4x$	$4x + 4$	$x = -1$	$x = -1$	$x > -1$	$x < -1$
$y = x^3 + x^2 - 1$	$3x^2 + 2x$	$x = 0 \text{ or } -\frac{2}{3}$	$x = 0 \text{ or } -\frac{2}{3}$	$x < -\frac{2}{3} \text{ or } x > 0$	$-\frac{2}{3} < x < 0$

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$x^3 - 12x$	$3x^2 - 12$	$x = 2 \text{ or } x = -2$	$x = 2 \text{ or } x = -2$	$x < -2 \text{ or} x > 2$	$-2 < x < 2$
$x^3 - 3x$	$3x^2 - 3$	$x = 1 \text{ or } x = -1$	$x = 1 \text{ or } x = -1$	$x < -1 \text{ or } x > 1$	$-1 < x < 1$

2) What happens to a function at a critical point?

The slope of the function is zero

3) What happens to the value of the derivative function at a critical point?

The value of the derivative is zero

4) What relationship is there between a function and its derivative at each critical point?

The value of the derivative is equal to the gradient of the function at that point

5) For each of the functions in question 1, identify all local maxima and minima (use your graphs), and the global maxima and minima over the domain graphed

a) No local max or min

Global max at (4, 8); global min at (-4,-8)

b) No local max; local min at (0, 0)

Global max at (4, 16) and (-4, 16); global min at (0, 0)

c) No local max; local min at (-1, -2)

Global max at (4, 48); global min at (-1, -2)

d) Local max at $\left(-\frac{2}{3}, -\frac{23}{27}\right)$; local min at (0, -1)

Global max at (4, 79); global min at (-4, -47)

e) Local max at (-2, 16); local min at (2, -16)

Global max at (4, 112); global min at (-4, -16)

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f) Local max at (-1, 2); local min at (1, -2)

Global max at (4, 52); global min at (-4, -52)

6) For each function in question 1, calculate the second derivative

a) 0

b) 2

c) 4

d) $6x + 2$

e) $6x$

f) $6x$

7) Calculate the value of the second derivative at each critical point.

a) 0

b) 2

c) 4

d) For $x = 0, second\ derivative = 2$; for $x = -\frac{2}{3}, second\ derivative = -2$

e) For $x = 2, second\ derivative = 12$; for $x = -2, second\ derivative = -12$

f) For $x = 1, second\ derivative = 6$; for $x = -1, second\ derivative = -6$

8) What is the relationship between the value of the second derivative at a critical point and the nature of the original function?

If critical point is a minimum, second derivative is positive; if critical point is a maximum, second derivative is negative

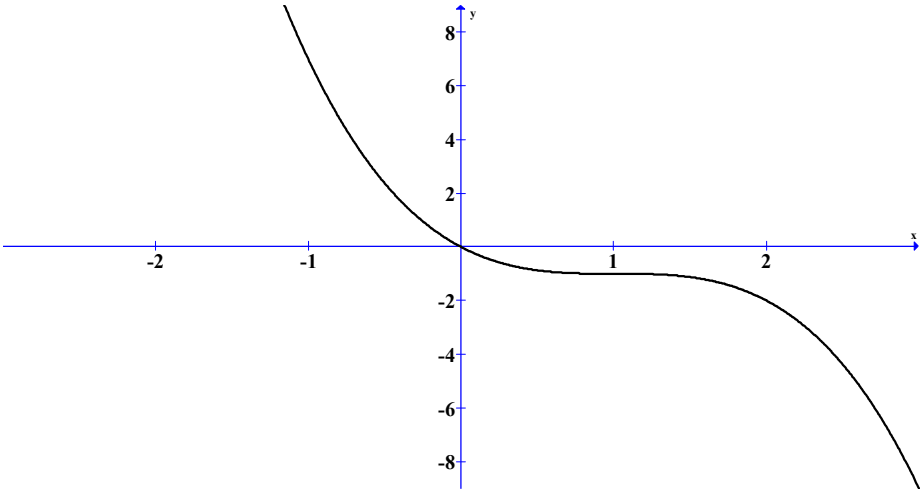
9)

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a) Graph the function$(y = -x^3 + 3x^2 - 3x)$, and calculate the first and second derivatives.

$$\frac{dy}{dx} = -3x^2 + 6x - 3$$

$$\frac{d^2x}{dy^2} = -6x + 6$$



b) From previous work, calculate the co-ordinate(s) of the critical point(s), and the value of the second derivative at that point

$$\frac{dy}{dx} = 0 \; when - 3x^2 + 6x - 3 = 0$$

$$x^2 - 2x + 1 = 0$$

$$x = 1$$

(1, -1) is a critical point

$$at \; x = 1$$
$$\frac{d^2x}{dy^2} = 0$$

c) How is the behaviour of this function at the critical point similar to previous functions in this exercise, and how is it different?

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The function is differentiable twice and has a critical point where the derivative is zero. However the second derivative test does not indicate if the critical point is a max or a min

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Exercise 2

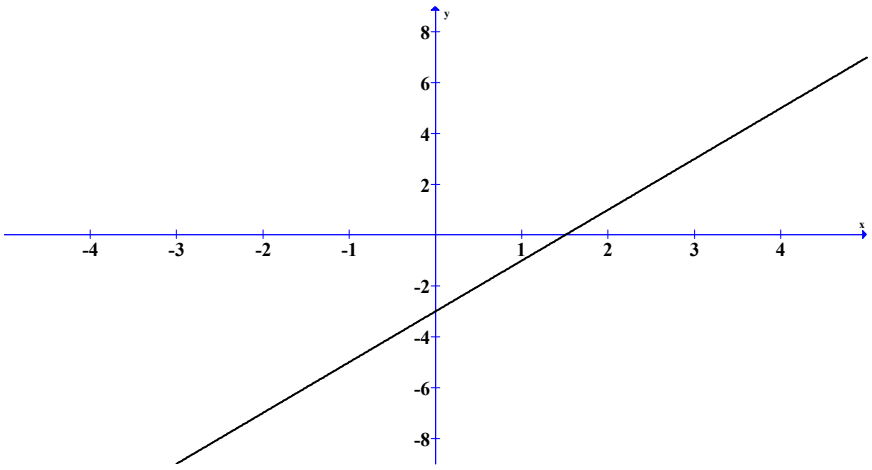
Graphing Functions Using Key Points

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SKETCH each of the functions below by considering and drawing the following key points

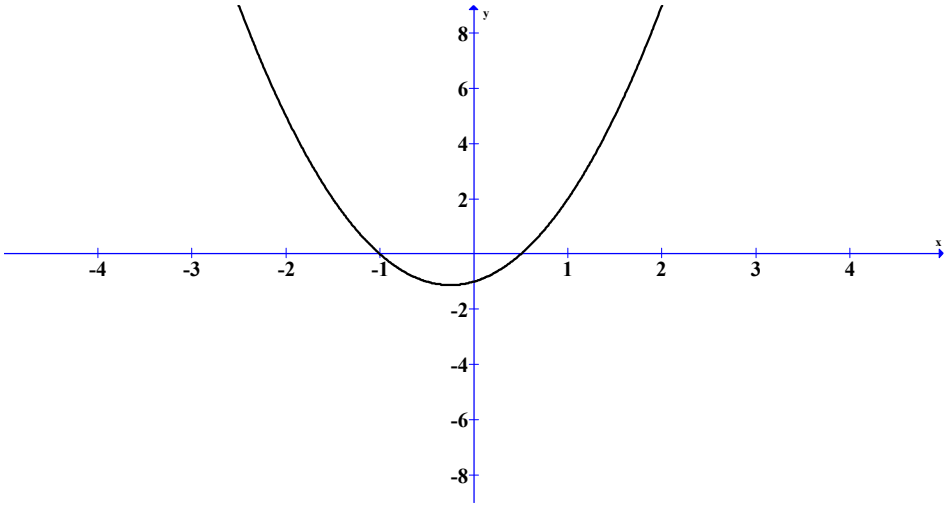
- The values of the function when $x = 0$
- The roots of the equation
- The co-ordinates of the critical point(s)
- The nature of the critical point(s)
- The identification of local maxima and minima and global maxima and minima
- The line of symmetry (if symmetrical)
- Any points where the function is not defined

1) $y = 2x - 3$

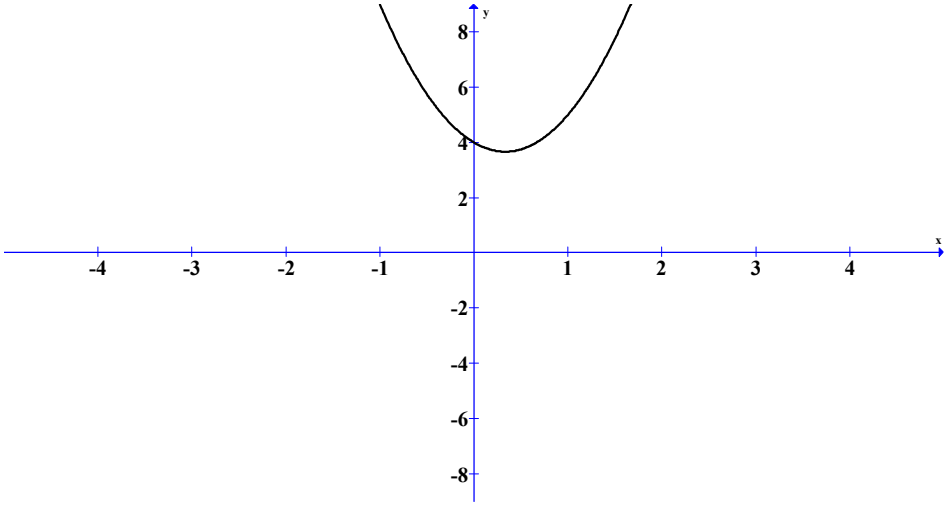


2) $y = 2x^2 + x - 1$

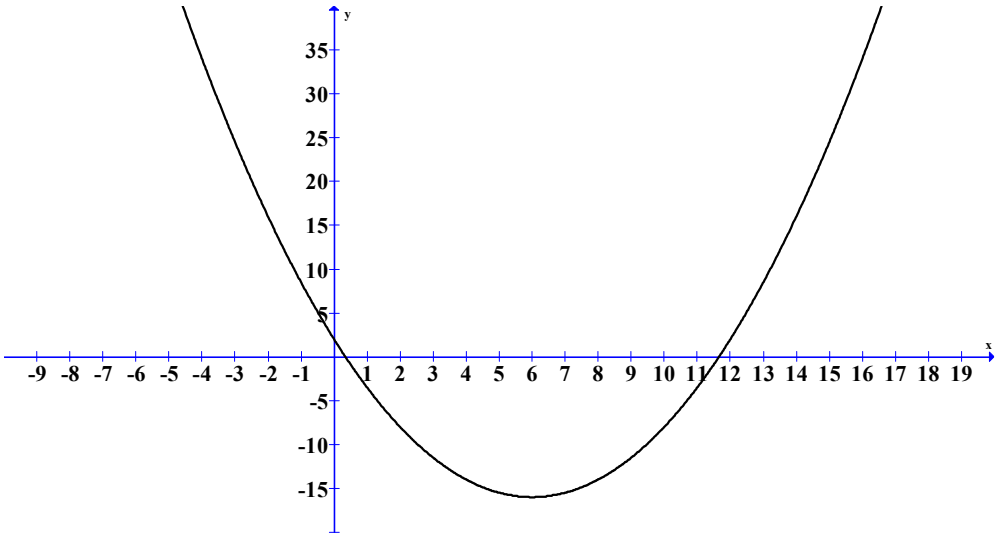
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3) $y = 3x^2 - 2x + 4$

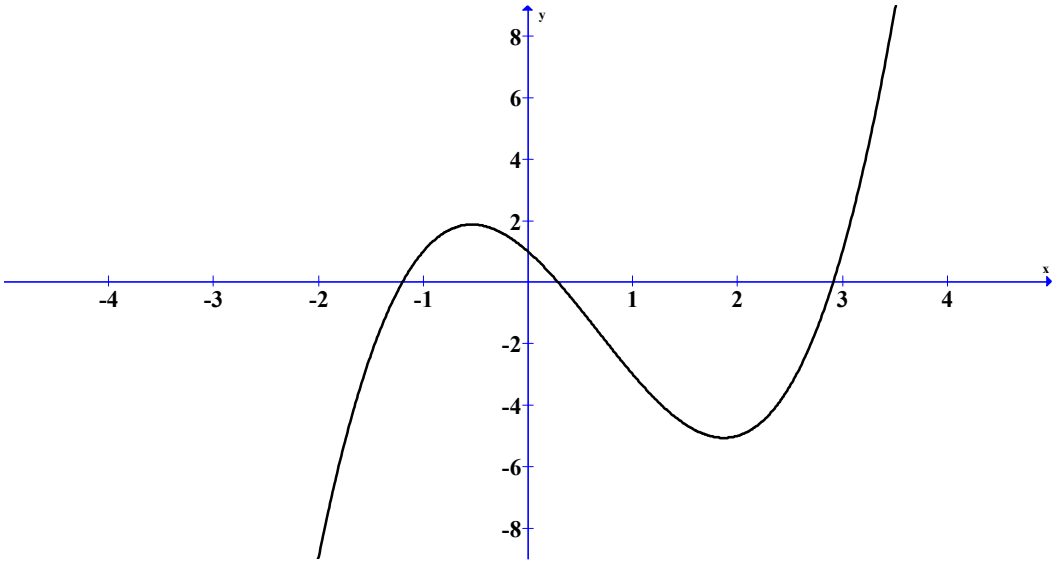


4) $y = \frac{1}{2}x^2 - 6x + 2$

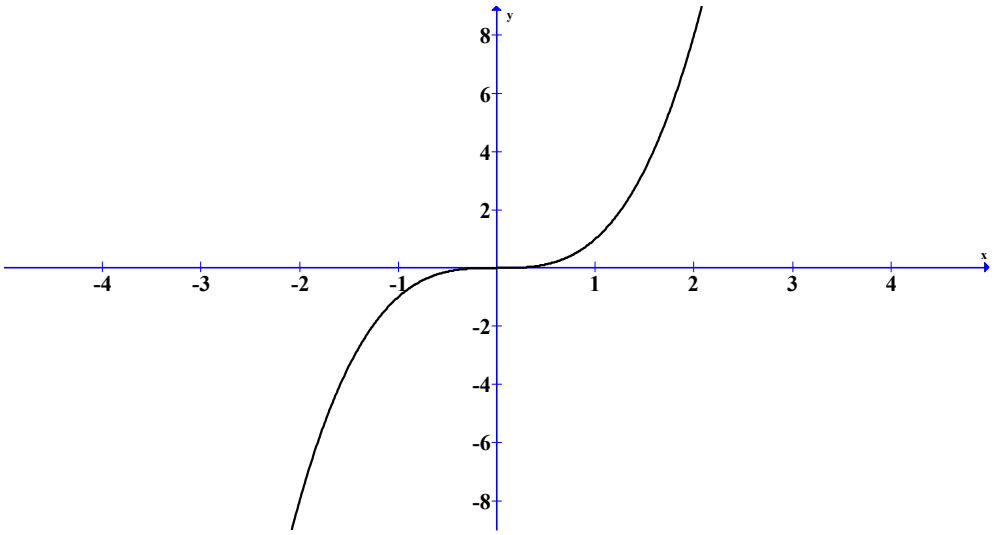


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5) $y = x^3 - 2x^2 - 3x + 1$

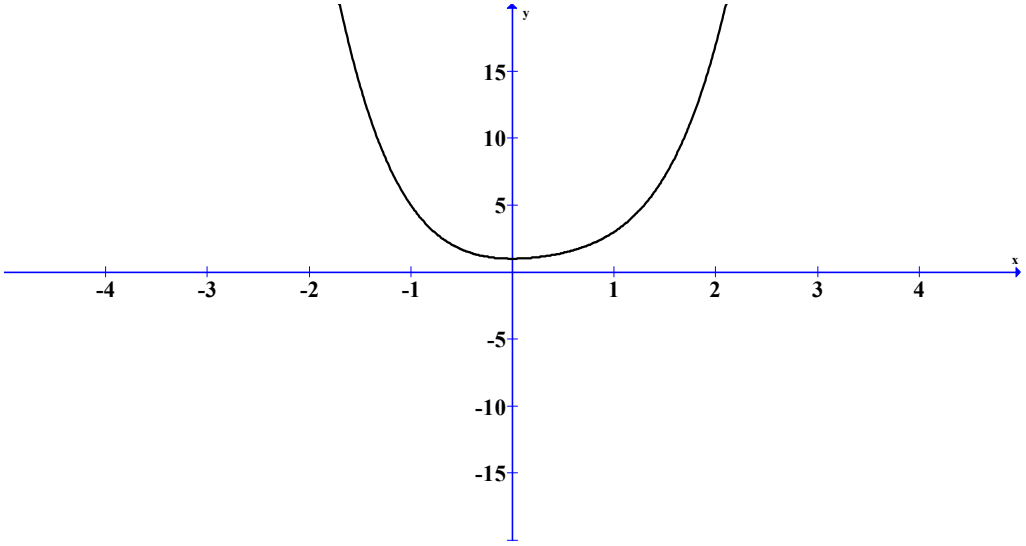


6) $y = x^3$

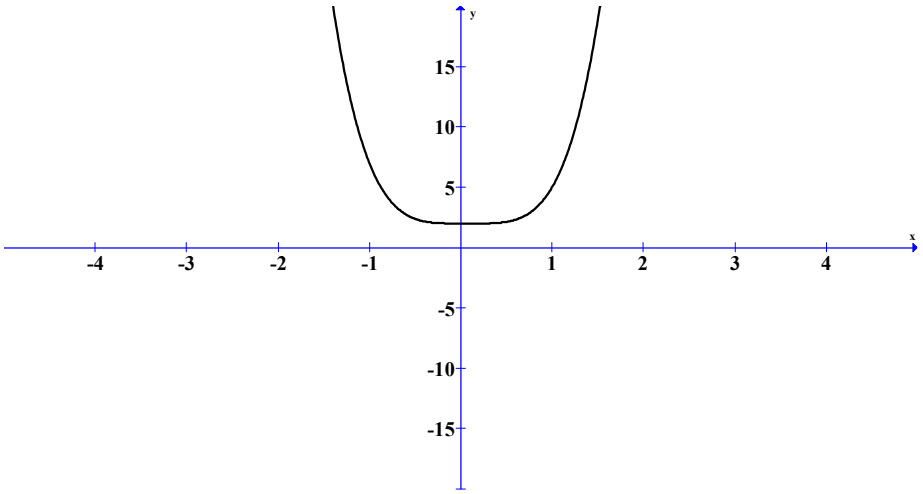


7) $y = x^4 - x^3 + 2x^2 + 1$

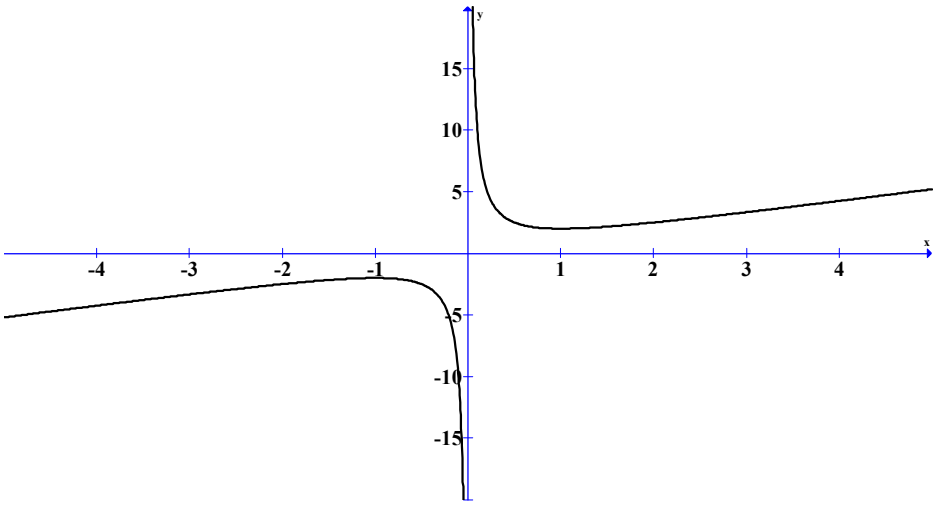
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8) $y = 4x^4 - x^3 + 2$

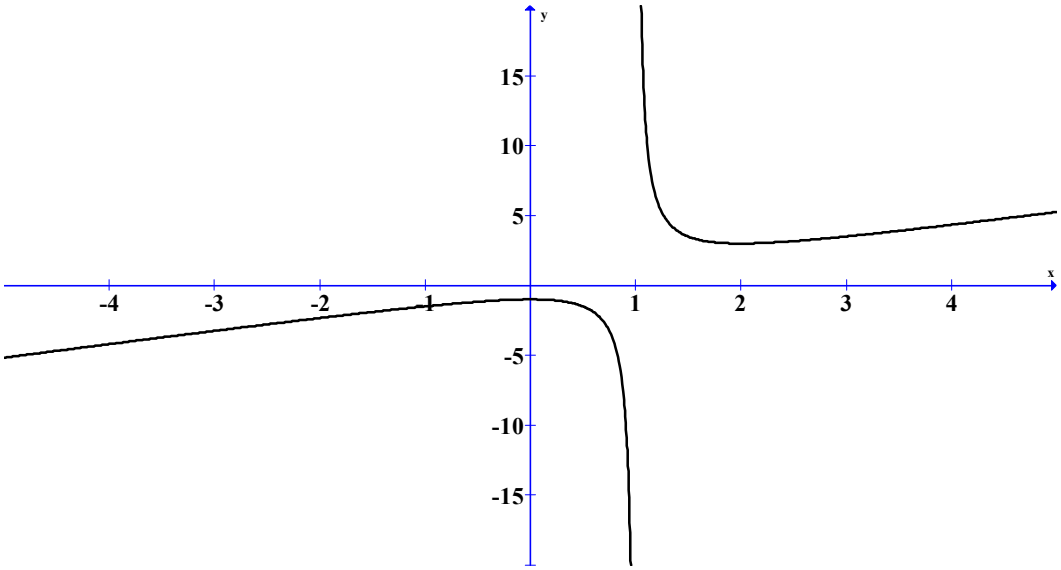


9) $y = x + \frac{1}{x}$



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10) $y = \frac{1}{x-1} + x$



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Exercise 3

Word Problems

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1) Using differentiation, find the local and global maxima and minima of the following functions over the domain$(-5 \leq x \leq 5)$. Indicate if any critical points are a maximum, minimum or neither

a) $y = -3x^2 - 2x + 4$

$\frac{dy}{dx} = -6x - 2$

$= 0 \; when \; x = -\frac{1}{3}$

$\frac{d^2x}{dy^2} = -6$

Point $\left(-\frac{1}{3}, \frac{13}{3}\right)$ is a maximum

b) $y = x^3 + 6x^2 + 9x + 2$

$\frac{dy}{dx} = 3x^2 + 12x + 9$

$= 0 \; when \; x = -1 \; or \; x = -3$

$\frac{d^2x}{dy^2} = 6x + 12$

Is positive for $x = -1$

Is negative for $x = -3$

Point $(-1, -2)$ is a minimum

Point (-3, 2) is a maximum

c) $y = x^4 - 4x$

$\frac{dy}{dx} = 4x^3 - 4$

$= 0 \; when \; x = 1$

$\frac{d^2x}{dy^2} = 12x^2$

Is positive for $x = 1$

Point $(1, -3)$ is a minimum

d) $y = -\frac{1}{2}x^3 - \frac{5}{2}x^2 + 4x$

$\frac{dy}{dx} = -\frac{3}{2}x^2 - 5x + 4$

$= 0 \; when \; x = \frac{2}{3} \; or \; x = -4$

$\frac{d^2x}{dy^2} = -3x - 5$

Is negative for $x = \frac{2}{3}$

Is positive for $x = -4$

Point $\left(\frac{2}{3}, -\frac{38}{27}\right)$ is a maximum

Point (-4, -24) is a minimum

e) $y = x^2 - x^3 + x$

$\frac{dy}{dx} = -3x^2 + 2x + 1$

$= 0 \; when \; x = 1 \; or \; x = -\frac{1}{3}$

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$$\frac{d^2x}{dy^2} = -6x + 2$$

Is negative for $x = 1$

Is positive for $x = -\frac{1}{3}$

Point $(1, 1)$ is a maximum

Point $(-\frac{1}{3}, -\frac{5}{27})$ is a minimum

2) A man is standing on a platform k metres above the ground. He throws a ball upward which then falls to the ground. The height of the ball can be described by the equation $h = -t^2 + 4t + 12$, where t is in seconds $(t > 0)$

a) What is the value of k?

At $t = 0, height\ of\ ball = 12\ m$

Therefore $k = 12$

b) What is the highest point the ball reaches above the ground and how many seconds after it is thrown does it reach this height?

$\frac{dh}{dt} = -2t + 4 = 0\ when\ t = 2$

After 2 seconds, $h = 16\ m$

c) After how many seconds does it hit the ground?

$h = 0\ when - t^2 + 4t + 12 = 0$

$t = 6\ or\ t = -2$

The ball hits the ground after 6 seconds

3) A balloon is blown up then left to deflate. The equation of the volume of the balloon is given by the equation $V = -t^2 + 8t$, where t is in seconds $(t > 0)$, and V is in cubic centimetres

a) What is the maximum volume the balloon reaches?

$\frac{dV}{dt} = -2t + 8 = 0\ when\ t = 4$

At $t = 4, V = 16\ cm^3$

b) After how many seconds does it reach its maximum volume?

From part a, volume is a maximum at t=2 seconds

c) When is its volume 7 cm^3?

$-t^2 + 8t = 7$

When $t^2 - 8t + 7 = 0$

$t = 1\ or\ t = 7$

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4) In a factory with 20 men, each man can produce 200 units of a product per day. For each additional man hired, output drops by 5 units per man. How many men should be employed to maximize production?

(Hint: the total output is equal to the number of men times the amount produced per man; the number of men at any time is $(20 + x)$, and the amount produced per man is $(200 - 5x)$

Output $= (20 + x)(200 - 5x)$

$= 4000 + 100x - 5x^2$

$\frac{do}{dx} = 100 - 10x = 0 \; when \; x = 10$

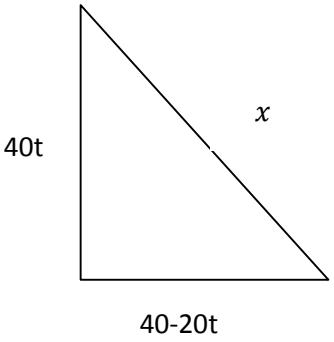
If $x = 10$, number of men to hire is 30 for an output of 150 units per man

5) A car is located 40km east of a truck. At the same time the car starts moving west at a speed of 20 km per hour, and the truck starts moving north at a speed of 40 km per hour. When will they be at the minimum distance from each other, and what will this distance be? (Draw diagram and use Pythagoras' Theorem)

The distance the car is from the trucks starting point is 40-20t

The distance the truck is from its starting point is 40t

Where t is in hours



$x = \sqrt{(40t)^2 + (40 - 20t)^2}$

x

$= \sqrt{1600t^2 + 400t^2 - 1600t + 1600}$

$= \sqrt{2000t^2 - 1600t + 1600}$

$\frac{dx}{dt} = \frac{1}{2} \times \frac{4000t - 1600}{\sqrt{2000t^2 - 1600t + 1600}} = 0$

When $2000t - 800 = 0$

$t = \frac{2}{5}$ of an hour (24 minutes)

At $t = \frac{2}{5}$

$x = \sqrt{320 - 640 + 1600}$

$\cong 35.8 \; km$

6) Calculate the shortest distance from the curve $y = \sqrt{x}$ to the point $(3, 0)$

Distance given by $\sqrt{(x - 3)^2 + y^2}$

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$= \sqrt{x^2 - 6x + 9 + x}$

$D = \sqrt{x^2 - 5x + 9}$

$\frac{dD}{dx} = \frac{1}{2} \frac{2x - 5}{\sqrt{x^2 - 5x - 9}}$

$= 0 \; when \; x - \frac{5}{2} = 0$

$x = \frac{5}{2}$

$D = \sqrt{\left(\frac{5}{2}\right)^2 - 5\left(\frac{5}{2}\right) + 9}$

$\cong 1.66$

7) A bridge is in the shape of a parabola that has the equation: $h = -0.0025x^2 + 1.25x$, where h is the height of the bridge above the water and x is the distance along the bridge.

What is the maximum height of the bridge, at what distance along the bridge does it occur, and what is the total length of the bridge?

$\frac{dh}{dx} = -0.005x + 1.25$

$= 0$ when $x = 250 \; m$

$h = -0.0025(250)^2 + 1.25 \times (250)$

$h = 156.25$

The bridge rises 156.25 m from a point 250 m along

$h = 0$

when $0.0025x^2 = 1.25x$

$x = 0, x = 500$

The bridge is 500 m long

8) A rectangular enclosure is to be constructed from 120 metres of wire. The wire only has to be used on three sides as the fourth side of the enclosure will be a barn wall. What will the length and width of the largest possible enclosure, and hence what will be its area?

Let l represent the length and w the width

$l + 2w = 120$

$l = 120 - 2w$

$A = l \times w$

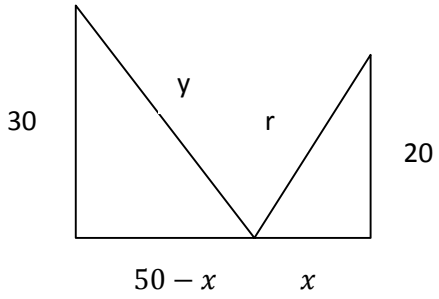
$A = (120 - 2w)w = 120w - 2w^2$

$\frac{dA}{dw} = 120 - 4w = 0 \; when \; w = 30$

The enclosure should be 30 metres wide and 60 metres long

9) Two poles, 30 metres high and 20 metres respectively are 50 metres apart from base to base. A rope is attached to the top of each and secured to a point in the ground between them. Where should the secured point be to minimize the amount of rope used?

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Need to minimize the distance $r + y$, call this distance q

$y^2 = 30^2 + (50 - x)^2$

$r^2 = x^2 + 20^2$

$q = r + y = \sqrt{30^2 + (50 - x)^2} + \sqrt{x^2 + 400}$

$= \sqrt{3400 - 100x + x^2} + \sqrt{x^2 + 400}$

$\frac{dq}{dx} = \frac{1}{2}\frac{(-100 + 2x)}{\sqrt{3400 - 100x + x^2}} + \frac{1}{2}\frac{2x}{\sqrt{x^2 + 400}}$

$= 0 \; when \; \frac{50 - x}{\sqrt{3400 - 100x + x^2}} = \frac{x}{\sqrt{x^2 + 400}}$

Squaring both sides, cross multiplying gives and rationalising like terms gives

$x^2 + 80x - 2000 = 0$

Solving for x gives $x = 20 \; or \; x = -100$

The rope should be placed 20 metres from the base of the small pole and 30 metres from the base of the large pole, forming two isosceles triangles

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Exercise 4

Tangents, Normals & Primitive Functions

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1) Find the equation of the tangent to the curve $y = x^3 + 1$ at the following points

$\frac{dy}{dx} = 3x^2$

a) $(1, 2)$

$m = 3$

$y = 3x + c$

Substituting (1, 2) gives

$c = -1$

Equation is $y = 3x - 1$

b) $(3, 28)$

$m = 27$

$y = 27x + c$

Substituting (3, 28) gives

$c = -25$

Equation is $y = 27x - 25$

c) $(0, 1)$

$m = 0$

$y = 0x + c$

Substituting (0, 1) gives

$c = 2$

Equation is $y = 2$

d) $(2, 9)$

$m = 12$

$y = 12x + c$

Substituting (2, 9) gives

$c = -10$

Equation is $y = 12x - 10$

2) Find the equation of the tangent to the curve $y = x^2 - 2x + 3$ at the following points

$\frac{dy}{dx} = 2x - 2$

a) $(0, 3)$

$m = -2$

$y = -2x + c$

Substituting (0, 3) gives

$c = 3$

Equation is $y = -2x + 3$

b) $(1, 2)$

$m = 0$

$y = 0x + c$

Substituting (1, 2) gives

$c = 2$

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Equation is $y = 2$

c) $(2, 3)$

$m = 3$

$y = 3x + c$

Substituting (2, 3) gives

$c = -3$

Equation is $y = 3x - 3$

d) $(5, 18)$

$m = 8$

$y = 8x + c$

Substituting (5, 18) gives

$c = -22$

Equation is $y = 8x - 22$

3) Find the equation of the tangent to the curve $y = x \sin x$ at the following points

$\frac{dy}{dx} = \sin x + x \cos x$

a) $\left(\frac{3\pi}{2}, -\frac{3\pi}{2}\right)$

$m = -1$

$y = -x + c$

Substituting $(\frac{3\pi}{2}, \frac{3\pi}{2})$ gives

$c = 3\pi$

Equation is $y = -x + 3\pi$

b) $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$

$m = 1$

$y = x + c$

Substituting $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$ gives

$c = 0$

Equation is $y = x$

c) $\left(\frac{\pi}{6}, \frac{\pi}{12}\right)$

$m = 3$

$y = \left(\frac{1}{2} + \frac{\pi}{6}\frac{\sqrt{3}}{2}\right)x + c$

$y = \left(\frac{1}{2} + \frac{\pi}{4\sqrt{3}}\right)x + c$

Let A= $\left(\frac{1}{2} + \frac{\pi}{4\sqrt{3}}\right)$

$y = Ax + c$

Substituting $(\frac{\pi}{6}, \frac{\pi}{6})$ gives

$c = \frac{\pi}{6}(1 - A)$

Equation is $y = Ax + \frac{\pi}{6}(1 - A)$

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d) $(\pi, 0)$

$m = \pi$

$y = \pi x + c$

Substituting $(\pi, 0)$

$c = -\pi^2$

Equation is $y = \pi x - \pi^2$

4) Find the equation of the normal to the curve $y = x^3 + 1$ at the following points

$\frac{dy}{dx} = 3x^2$

a) $(4, 65)$

$m = 48$

Gradient of normal $= -\frac{1}{48}$

$y = -\frac{x}{48} + c$

Substituting $(4, 65)$

$c = \frac{3124}{48}$

Equation is $48y = -x + 3124$

b) $(2, 9)$

$m = 12$

Gradient of normal $= -\frac{1}{12}$

$y = -\frac{x}{12} + c$

Substituting $(2,9)$

$c = \frac{110}{12}$

Equation is $12y = -x + 110$

c) $(1, 2)$

$m = 3$

Gradient of normal $= -\frac{1}{3}$

$y = -\frac{x}{3} + c$

Substituting $(1,2)$

$c = \frac{7}{3}$

Equation is $3y = -x + 7$

d) $(0, 1)$

$m = 0$

Gradient of normal is undefined

Equation is $x = 0$

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5) Find the equation of the normal to the curve $y = x^2 - 2x + 3$ at the following points

$\frac{dy}{dx} = 2x - 2$

a) $(1, 2)$

$m = 0$

Gradient of normal is undefined

Equation is $x = 1$

b) $(4, 11)$

$m = 6$

Gradient of normal $= -\frac{1}{6}$

$y = -\frac{x}{6} + c$

Substituting $(4, 11)$

$c = \frac{70}{6}$

Equation is $6y = -x + 70$

c) $(3, 6)$

$m = 4$

Gradient of normal $= -\frac{1}{4}$

$y = -\frac{x}{4} + c$

Substituting $(3, 6)$

$c = \frac{27}{4}$

Equation is $4y = -x + 27$

d) $(0, 3)$

$m = -2$

Gradient of normal $= \frac{1}{2}$

$y = \frac{x}{2} + c$

Substituting $(0,3)$

$c = 3$

Equation is $y = \frac{1}{2}x + 3$

6) The equation of the tangent to the curve $y = x^2 - 1$ is $y = 4x - 5$.

a) At what point is this the equation of the tangent?

$\frac{dy}{dx} = 2x$

Gradient of tangent is 4, so $x = 2$

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Substituting into either equation gives the point $(2, 3)$

b) What is the equation of the normal at the same point?

Gradient of normal $= -\frac{1}{4}$

$$y = -\frac{1}{4}x + c$$

Substituting $(2, 3)$

$$c = \frac{7}{2}$$

7) The following equations show the second derivative of a function y in terms of x. Find y (the original function) in terms of x

a) $\ddot{y} = 4.$

When $x = 0, \dot{y} = 6$, and when $x = 0, y = 1$

$$\dot{y} = 4x + c$$

Substituting gives $c = 6$

$$\dot{y} = 4x + 6$$

$$y = 2x^2 + 6x + c$$

Substituting gives $c = 1$

$$y = 2x^2 + 6x + 1$$

b) $\frac{d^2y}{dx^2} = 3$

When $x = 0, \frac{dy}{dx} = 5$, and when $x = 0, y = 10$

$$\frac{dy}{dx} = 3x + c$$

Substituting gives $c = 5$

$$\frac{dy}{dx} = 3x + 5$$

$$y = \frac{3}{2}x^2 + 5x + c$$

Substituting gives $c = 10$

$$y = \frac{3}{2}x^2 + 5x + 10$$

c) $\ddot{y} = 2x$

When $x = 1, \dot{y} = 1$, and when $x = 1, y = 1$

$$\dot{y} = x^2 + c$$

Substituting gives $c = 0$

$$\dot{y} = x^2$$

$$y = \frac{x^3}{3} + c$$

Substituting gives $c = \frac{2}{3}$

$$3y = x^3 + 2$$

d) $y'' = 2x - 2.$

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When $x = 1, y' = 0$, and when $x = 3, y = 4$

$y' = x^2 - 2x + c$

Substituting gives $c = 0$

$y' = x^2 - 2x$

$y = \frac{x^3}{3} - x^2 + c$

Substituting gives $c = 4$

$y = \frac{x^3}{3} - x^2 + 4$

8) The gradient function of a curve is $\frac{dy}{dx} = 2x^2 - 2$, and the curve passes through the point $(0, 4)$. Find the equation of the curve

$y = \frac{2}{3}x^3 - 2x + c$

Substituting gives $c = 4$

$y = \frac{2}{3}x^3 - 2x + 4$

9) The gradient function of a curve is $y' = \frac{3}{2}x^{\frac{1}{2}} + 2x$, and the curve passes through the point $(1, 3)$. Find its equation

$y = (x)^{\frac{3}{2}} + x^2 + c$

Substituting gives $c = 1$

$y = (x)^{\frac{3}{2}} + x^2 + 1$

10) The gradient function of a curve is $y' = \sin x$, and the curve passes through the point $(0, 2)$. Find its equation

$y = -\cos x + c$

Substituting gives $c = 3$

$y = -\cos x + 3$

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Year 12 Mathematics

Integration

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Exercise 1

Approximations

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For all approximations in this exercise, the areas discussed are bounded below by the x axis

1) Use the approximation

$$f(x) \cong \frac{A(x)-A(c)}{x-c}$$

to estimate the area under the curve $y = x^2 + 1$ between the points $x = 1$ and $x = 3$

$f(3) = 10$

$f(1) = 2$

$$f(x) \cong \frac{(3 \times 10) - (1 \times 2)}{3 - 1} = 14$$

2) Use the Trapezoidal rule to approximate the area under the curve $y = x^2 + 1$ between the points $x = 1$ and $x = 3$ Use successively smaller subintervals of size 1, 2, and 4

Subinterval 1:

Area $\cong \frac{3-1}{2}\big(f(1) + f(3)\big)$

$= (2 + 10) = 12$

Subinterval 2:

Area $\cong \frac{3-1}{4}\big(f(1) + 2f(2) + f(3)\big)$

$$= \frac{1}{2}(2 + 10 + 10) = 11$$

Subinterval 4:

Area $\cong \frac{1}{4}\big(f(1) + 2f(1.5) +$
$2f2+2f2.5+f3$

$$= \frac{1}{4}(2 + 6.5 + 10 + 14.5 + 10) = 10.75$$

3) Use Simpson's rule to approximate the area under the curve $y = x^2 + 1$ between the points $x = 1$ and $x = 3$

Taking 4 intervals

Area $\cong \frac{1}{6}\big(f(1) + 4f(1.5) +$
$2f2+4f2.5+f3$

$$= \frac{1}{6}(2 + 13 + 10 + 9 + 10) = 10\frac{2}{3}$$

4) Calculate $\int_1^3 x^2 + 1\, dx$, and explain why your answer is exactly equal in this instance to your answer to question 3

$$\int = \frac{x^3}{3} + x = \left(\frac{27}{3} + 3\right) - \left(\frac{1}{3} + 1\right)$$
$$= 10\frac{2}{3}$$

The parabola used to approximate the integral is the same as the function being integrated

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5) Use the approximation used in question 1 to estimate the area under the curve $y = x^3 + 1$ between the points $x = 1$ and $x = 3$

$f(3) = 28$

$f(1) = 2$

$$f(x) \cong \frac{(3 \times 28) - (1 \times 2)}{3 - 1} = 41$$

6) Use the Trapezoidal rule to approximate the area under the curve $y = x^3 + 1$ between the points $x = 1$ and $x = 3$

Subinterval 4:

Area $\cong \frac{1}{4}(f(1) + 2f(1.5) +$ *2f2+2f2.5+f3*

$$= \frac{1}{4}(2 + 8.75 + 18 + 33.25 + 28)$$
$$= 22.5$$

7) Use Simpson's rule to approximate the area under the curve $y = x^3 + 1$ between the points $x = 1$ and $x = 3$

Taking 4 intervals

Area $\cong \frac{1}{6}(f(1) + 4f(1.5) +$ *2f2+4f2.5+f3*

$$= \frac{1}{6}(2 + 17.5 + 18 + 66.5 + 28)$$
$$= 22$$

8) Calculate $\int_1^3 x^3 + 1$

$$\int = \frac{x^4}{4} + x = \left(\frac{81}{4} + 3\right) - \left(\frac{1}{4} + 1\right)$$

$$= 22$$

9) For the function $y = \frac{1}{x+1}$ estimate the area between the points $x = 1$ and $x = 2$, by using first the trapezoidal rule, and then Simpson's rule for the whole interval and then for the two subintervals separated by the point $x = \frac{3}{2}$

Trapezoidal subinterval 4:

Area $\cong \frac{1}{8}(f(1) + 2f(1.5) +$ *2f2+2f2.5+f3*

$$= \frac{1}{8}\left(\frac{1}{2} + \frac{4}{5} + \frac{2}{3} + \frac{4}{7} + \frac{1}{4}\right) \cong 0.35$$

Simpson's rule:

Whole interval

Area $\cong \frac{1}{3}(f(2) + f(1))$

$$= \frac{1}{3}\left(\frac{1}{3} + \frac{1}{2}\right) \cong 0.28$$

Two intervals

Area $\cong \frac{1}{6}(f(2) + 2f(1.5) + f(1))$

$$= \frac{1}{6}\left(\frac{1}{3} + \frac{4}{5} + \frac{1}{2}\right) \cong 0.27$$

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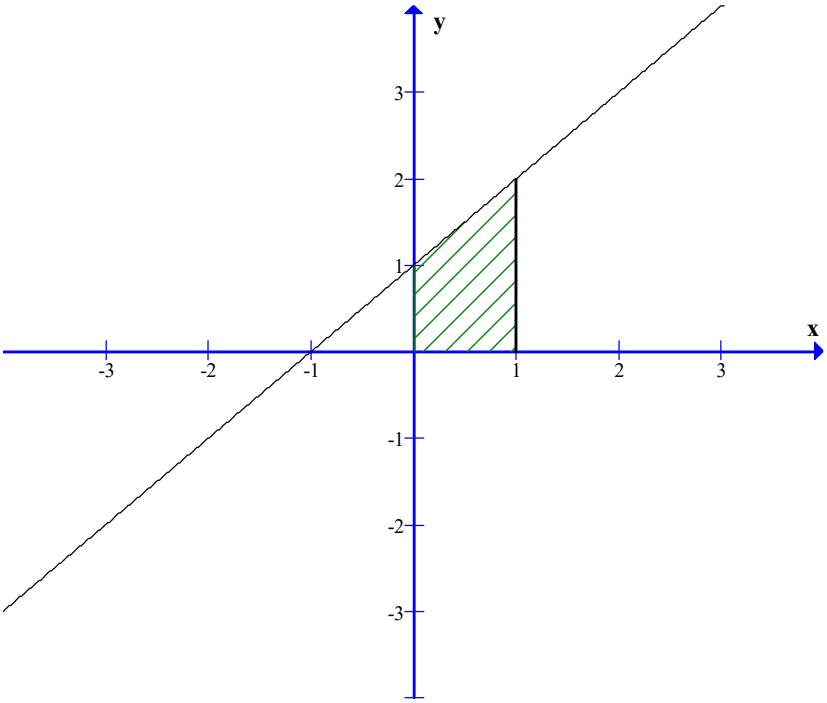
Exercise 2

Calculations &Applications

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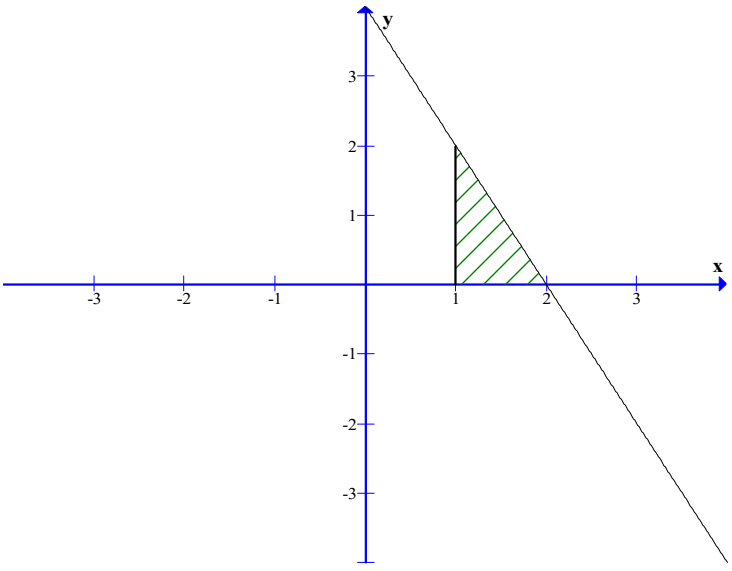
1) Calculate the following definite integrals

a) $\int_0^1 x + 1 \, dx$



$$\int = \frac{x^2}{2} + x = \left(\frac{1}{2} + 1\right) - (0) = \frac{3}{2}$$

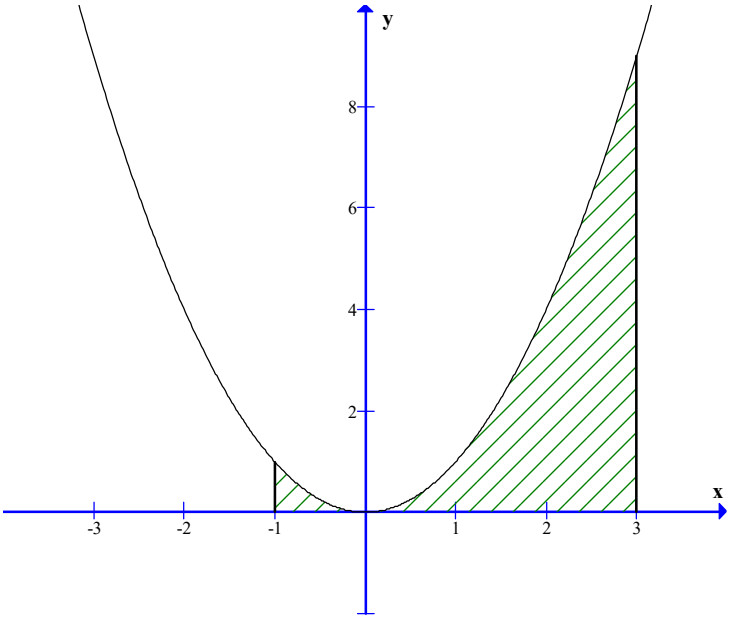
b) $\int_1^2 -2x + 4 \, dx$



$$\int = -x^2 + 4x = (-4 + 8) - (-1 + 4) = 1$$

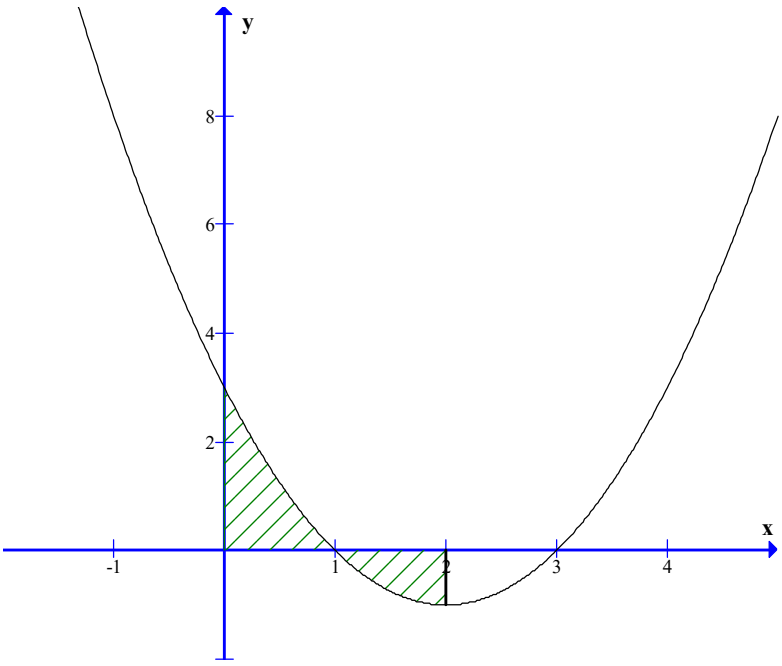
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c) $\int_{-1}^{3} x^2\, dx$



$$\int = \frac{x^3}{3} = (9) - \left(-\frac{1}{3}\right) = \frac{28}{3}$$

d) $\int_{0}^{2} x^2 - 4x + 3\ dx$



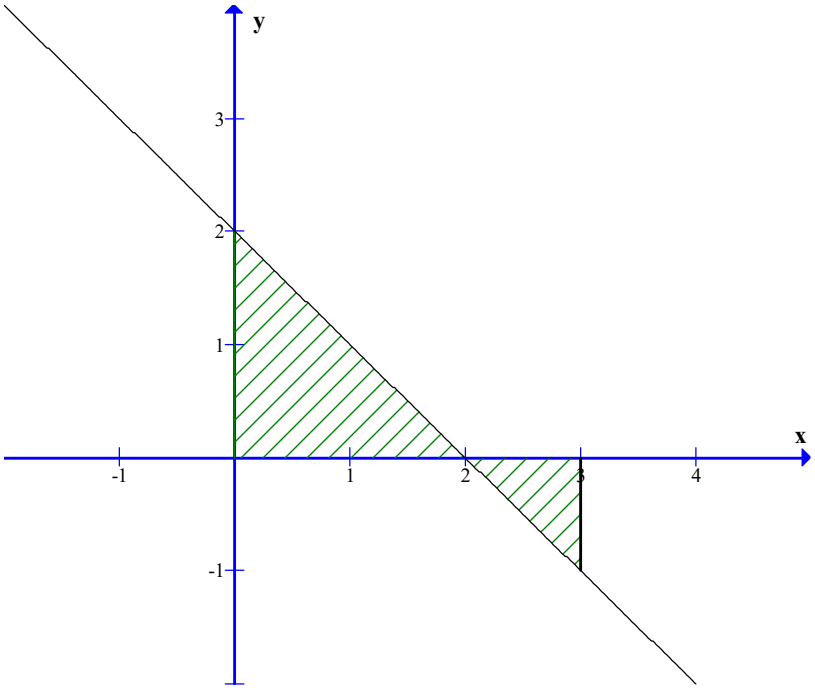
$$\int = \frac{x^3}{3} - 2x^2 + 3x$$

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Area $= \int_0^1 + \int_2^1 = \left(\left(\frac{1}{3} - 2 + 3\right) - (0)\right)\, dx + \left(\left(\frac{1}{3} - 2 + 3\right) - \left(\frac{8}{3} - 8 + 6\right)\right)\, dx$

$= 2$

2) Calculate the area bounded by the y axis, x axis, the line $x = 3$, and the line $y = -x + 2$



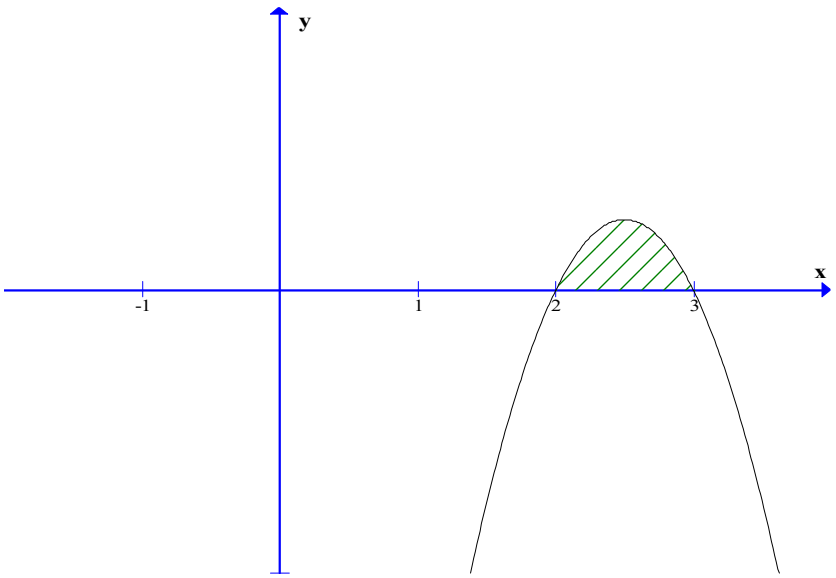
Area $= \left(\int_0^2 -x + 2\right) + \left(\int_3^2 -x + 2\right)\, dx$

$\int = -\frac{x^2}{2} + 2x$ dx

Area $= \left((2) - (0)\right) + \left((2) - \left(\frac{3}{2}\right)\right) = \frac{5}{2}$

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3) Calculate the area between the x axis and the equation $y = -x^2 + 5x - 6$

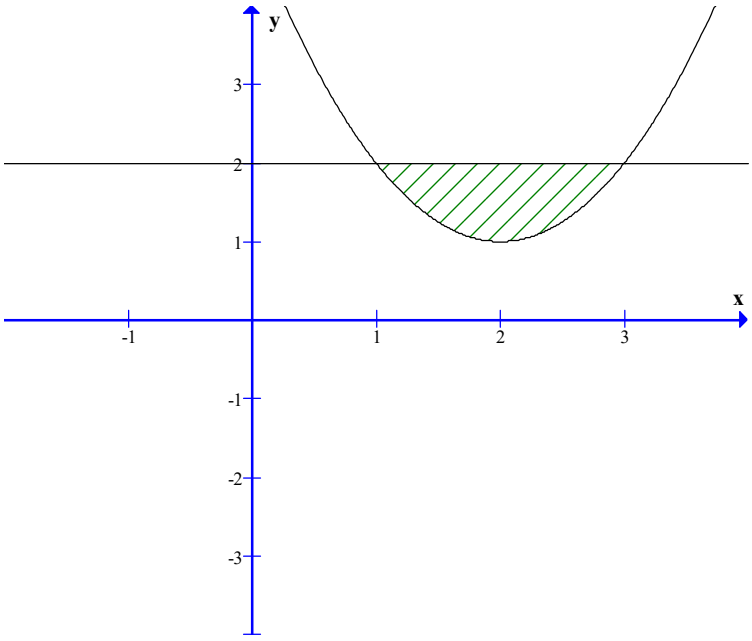


$$\text{Area} = \int_2^3 -x^2 + 5x - 6 \; dx = -\frac{x^3}{3} + \frac{5}{2}x^2 - 6x \; dx$$

$$= \left(-9 + \frac{45}{2} - 18\right) - \left(-\frac{8}{3} + 10 - 12\right)$$

$$= \frac{1}{6}$$

4) Calculate the area bounded by the curve $y = x^2 - 4x + 5$ and the line $y = 2$



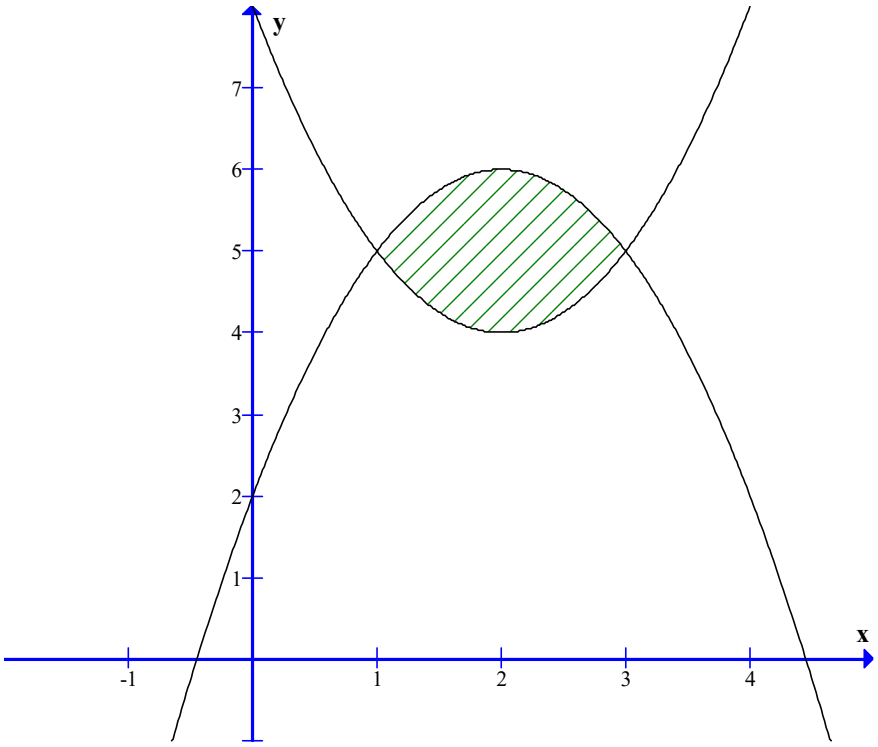
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$$\text{Area} = \int 2 - \int x^2 - 4x + 5\, dx = \int 2 - x^2 + 4x - 5\, dx = \int -x^2 + 4x - 3\, dx$$

$$\int = -\frac{x^3}{3} + 2x^2 - 3x \text{ dx}$$

$$\text{Area} = (-9 + 12 - 3) - \left(-\frac{1}{3} + 2 - 3\right) = \frac{4}{3}$$

5) Calculate the area bounded by the curves $y = x^2 - 4x + 8$ and $y = -x^2 + 4x + 2$

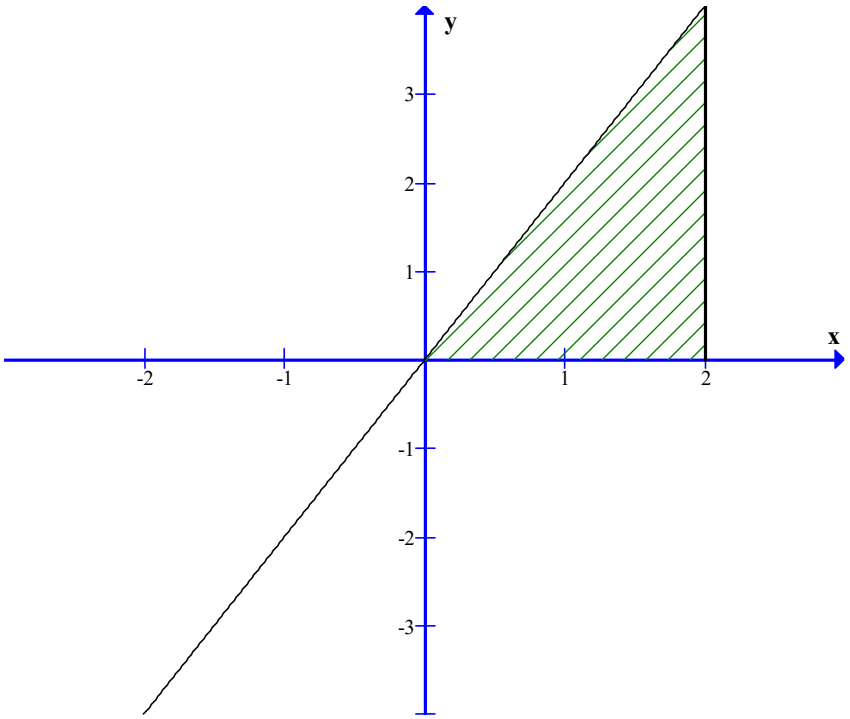


$$\int = \int -x^2 + 4x + 2 - (x^2 - 4x + 8)\, dx = \int -2x^2 + 8x - 6 = -\frac{2}{3}x^3 + 4x^2 - 6x\, dx$$

$$\text{Area} = \left(-\frac{2}{3} \times 27 + 36 - 18\right) - \left(-\frac{2}{3} + 4 - 6\right) = 0 + \frac{8}{3} = \frac{8}{3}$$

6) Calculate the volume generated when the area bounded by the lines $y = 2x$, $x = 2$, and the x axis is rotated about the x axis

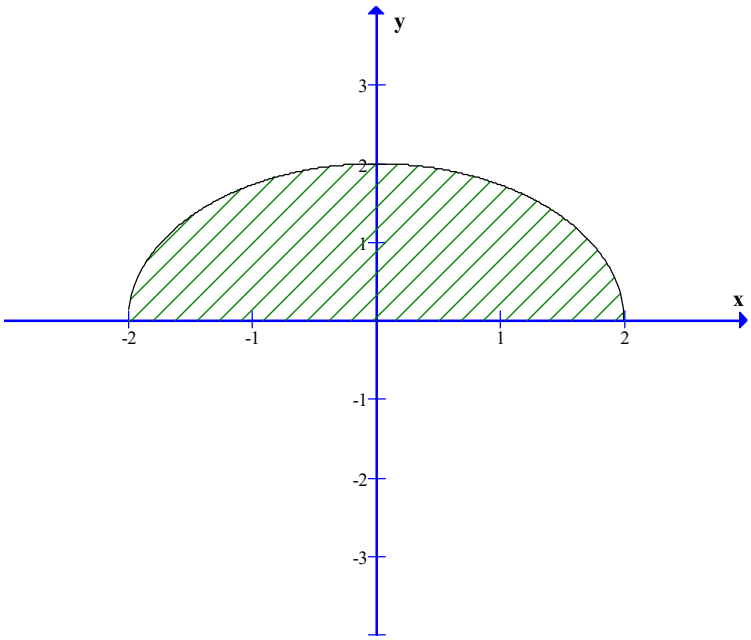
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$$V = \int_0^2 \pi y^2 dx = \int_0^2 \pi(2x)^2 dx = \pi \int_0^2 4x^2 \, dx = \frac{4}{3}\pi x^3$$

Volume $= \frac{4}{3}\pi(2^3 - 0) = \frac{32\pi}{3}$

7) Calculate the volume generated when the area bounded by the semicircle $y = \sqrt{4 - x^2}$ and the x axis is rotated about the x axis

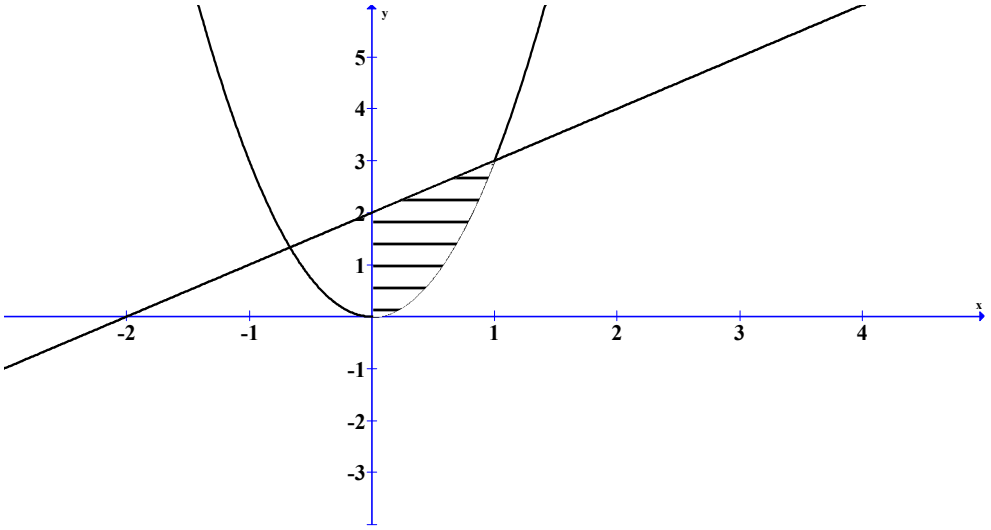


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$$V = \int_{-2}^{2} \pi y^2 \, dx = \pi \int_{-2}^{2} 4 - x^2 \, dx = \pi \left(4x - \frac{x^3}{3} \right)$$

$$V = \pi \left(\left(8 - \frac{8}{3} \right) - \left(-8 - \frac{-8}{3} \right) \right) = \pi \left(16 - \frac{16}{3} \right) = \frac{32\pi}{3}$$

8) Calculate the volume produced by rotating the area between $y = 3x^2$ and $y = x + 2$ with $x \geq 0$ around the x axis



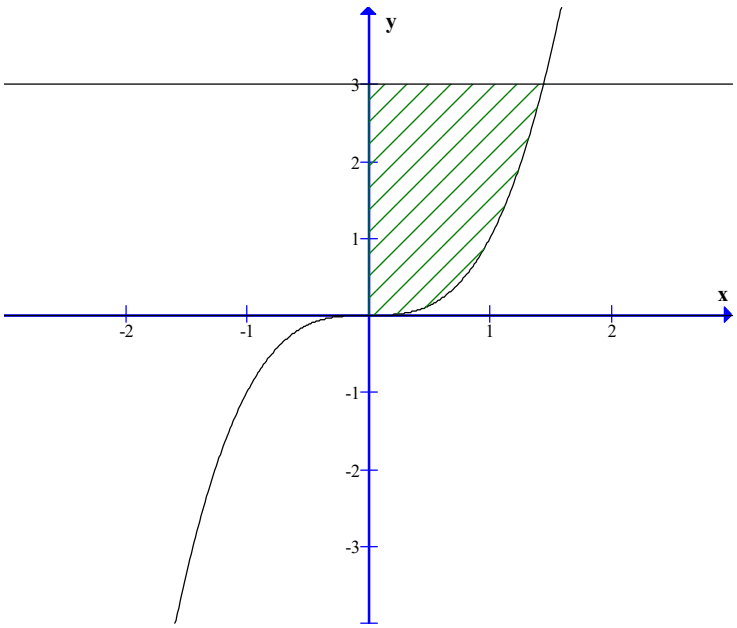
Point of intersection is where $x = 1$

$$V = \pi \int_{0}^{1} (x + 2)^2 - (3x^2)^2 \, dx = \pi \int_{0}^{1} x^2 + 4x + 4 - 9x^4 \, dx = \frac{x^3}{3} + 2x^2 + 4x - \frac{9x^5}{5}$$

$$\text{Volume} = \pi \left(\frac{1}{3} + 2 + 4 - \frac{9}{5} \right) - (0) = \frac{78\pi}{15}$$

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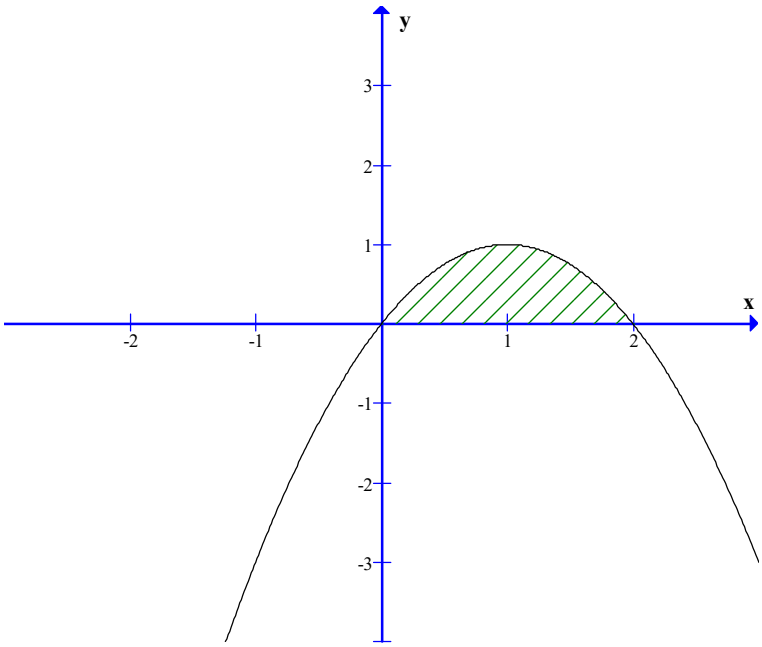
9) Find the volume generated by rotating the curve $y = x^3$ between $y = 0$ and $y = 3$ about the y axis



$$V = \int_0^3 \pi x^2 \, dy = \int_0^3 \pi (y)^{\frac{2}{3}} = \pi \frac{3}{5} (y)^{\frac{5}{3}}$$

$$V = \frac{3}{5}\pi (3)^{\frac{5}{3}} = \frac{\pi}{5} (3)^{\frac{8}{3}}$$

10) Find the volume generated by the curve $y = 2x - x^2$ and $y = 0$ about the x axis



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$$V = \int_0^2 \pi(2x - x^2)^2 \, dx = \pi \int_0^2 4x^2 - 4x^3 + x^4 \, dx = \pi(\frac{4x^3}{3} - x^4 + \frac{x^5}{5})$$

Volume $= \pi\left(\frac{32}{3} - 16 + \frac{32}{5}\right) = \frac{16\pi}{15}$

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Year 12 Mathematics

Applications of Calculus

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Exercise 1

Rates of Change

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1) A tap is slowly opened such that the volume flow rate R varies in time according to the equation $R = kt$, where k is a constant and $t > 0$. Calculate the total volume that flows through the tap in the first 12 seconds if $k = 1.5m^3s^{-2}$

$R = 1.5t$

$V = \int 1.5t\ dt = \int \frac{3}{2}t\ dt = \frac{3}{4}t^2$

Volume in first twelve seconds $= \frac{3}{4} \times 144 - 0 = 108\ m^3$

2) The number of bacteria in a dish after t hours is given by $b = 3t^2 + 5t + 2$. How fast is the population growing after 3 hours?

Rate of growth $= \frac{db}{dt} = 6t + 5$

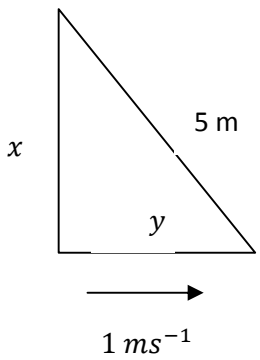
At $t = 3, \frac{db}{dt} = 23\ per\ hour$

3) The rate of change of profit from sales of x beds per week is given by the equation $\frac{dP}{dx} = 50 - 2x$. What is the profit when 20 beds are sold?

Profit $= \int 50 - 2x\ dx = 50x - x^2$

When $x = 20, profit = \$60$

4) A ladder 5 meters long is resting against a wall. If the bottom of the ladder begins sliding away from the wall at the rate of 1 metre per second, how fast is the top of the ladder moving down when the bottom of the ladder is 3 meters from the wall?



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$x^2 + y^2 = 5^2$

$x^2 = 25 - y^2$

$x = \sqrt{25 - y^2}$

$\frac{dx}{dy} = -\frac{1}{2}\frac{2y}{\sqrt{25 - y^2}}$

$\frac{dx}{dt} = \frac{dx}{dy} \times \frac{dy}{dt} = \frac{-y}{\sqrt{25 - y^2}} \times (-1)$

When $y = 3, \frac{dx}{dt} = \frac{3}{4}\ ms^{-1}$

5) In 2005, the population of a town was 1000. Since 2005 the rate of change in the population is modelled by the equation $\frac{dP}{dt} = 4t + 100$, where t is the number of years from 2005. What was the population of the town in 2009?

$P = \int \frac{dP}{dt}\ dt = \int 4t + 100\ dt = 2t^2 + 100t + c$

When $t = 0, P = 1000, so\ c = 1000$

$P = 2t^2 + 100t + 1000$

When $t = 4, P = 1432$

6) A tank is being drained of water at a rate of $r = 1 + 2t - 12t^2$ in litres per minute. After 4 minutes there are 802 litres in the tank. What was the initial volume of the tank, and how much will be left in the tank after 6 minutes?

$V = \int r\ dt = \int 1 + 2t - 12t^2\ dt = t + t^2 - 4t^3 + c$

When $t = 4, V = 764$

$4 + 16 - 4 \times 64 + c = 802$

c =1000

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$V = t + t^2 - 4t^3 + 1000$

$t = 6$

$V = 6 + 36 - (4 \times 216) + 1000 = 178\ litres$

7) The number of fish that a seal can eat per hour (t) is given by $R = 32 - 2t^2$

a) At what rate does the seal initially consume fish?

When $t = 0, R = 32$

b) How many fish did it eat in the second hour? (To nearest whole number)

$$N = \int R\ dt = \int 32 - 2t^2\ dt = 32t - \frac{2t^3}{3}$$

When $t = 1$, (that is during second hour), $N = 32 - \frac{2}{3} \cong 31$

c) When will the seal be full?

When the rate of consumption is 0; that is $32 - 2t^2 = 0, t = 4$ hours

8) In 1970 a rare painting was valued at $50,000. The rate of change in its value is given by the equation: $\frac{dV}{dT} = 200t + 500$.

a) What rate will the value be changing by in the year 2020?

When $t = 50, \frac{dV}{dt} = 200 \times 50 + 500 = \$1050\ per\ year$

b) If an investor purchased the painting in 1970 for $50,000 how much profit will they have made by the year 2020?

$$V = \int \frac{dV}{dt}\ dt = \int 200t + 500\ dt = 100t^2 + 500t + c$$

When $t = 0, V = 50{,}000$

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Therefore $V = 100t^2 + 500t + 50{,}000$

When $t = 50, V = 100 \times (50^2) + 500 \times (50) + 50{,}000 = \$325{,}000$

Profit will be $275,000

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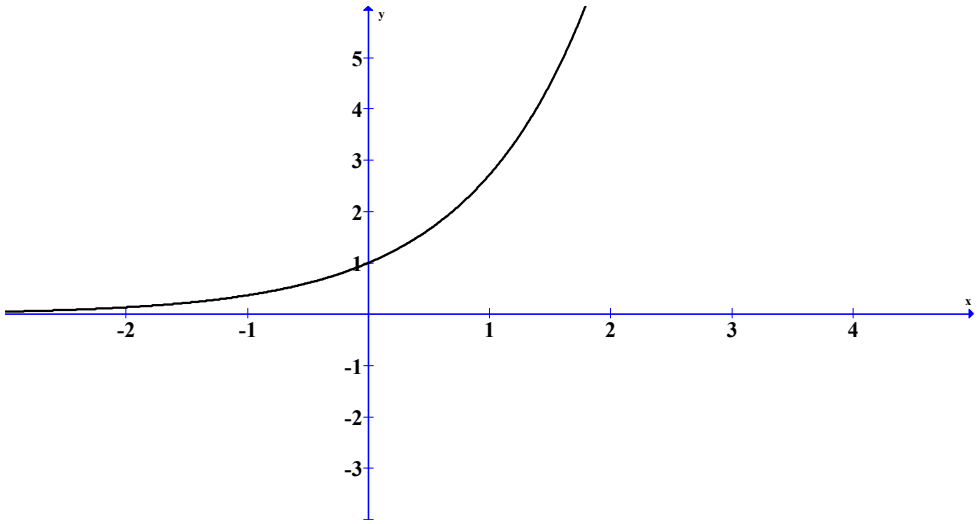
Exercise 2

Exponential Growth & Decay

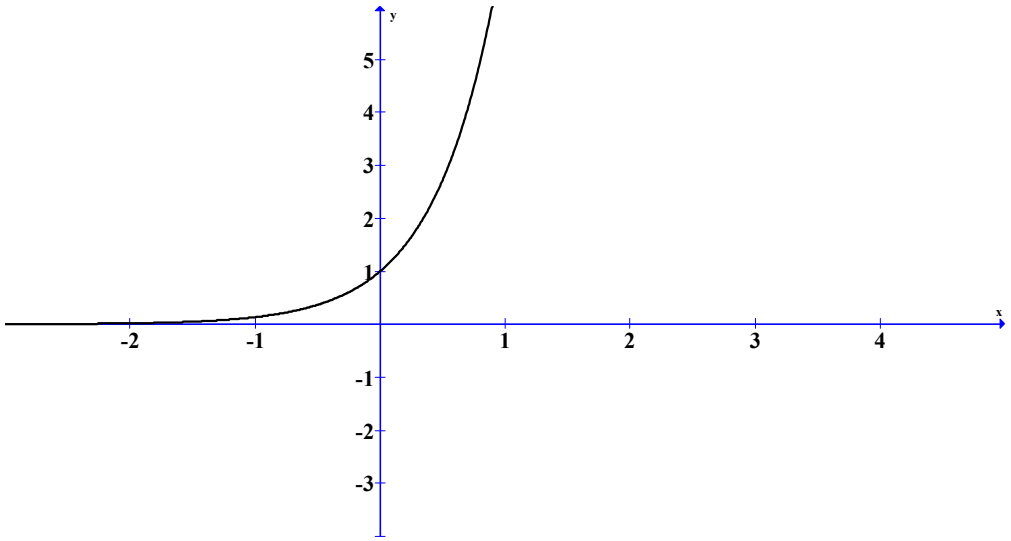
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1) Graph the following over an appropriate range and domain

a) $y = e^x$

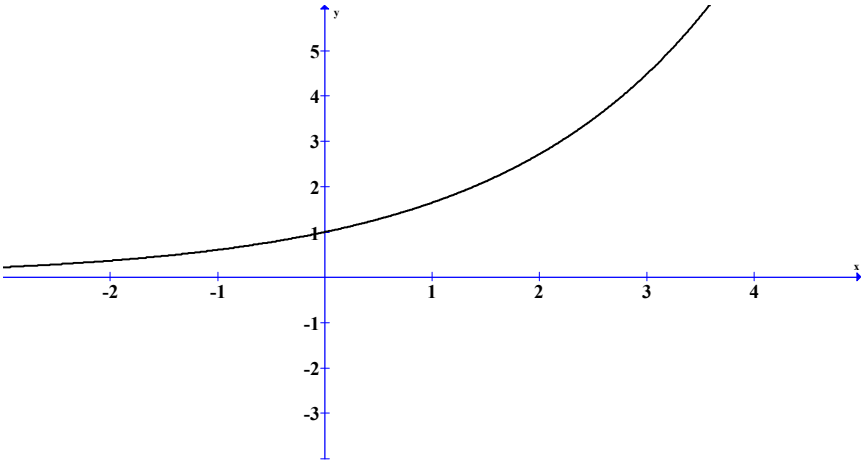


b) $y = e^{2x}$

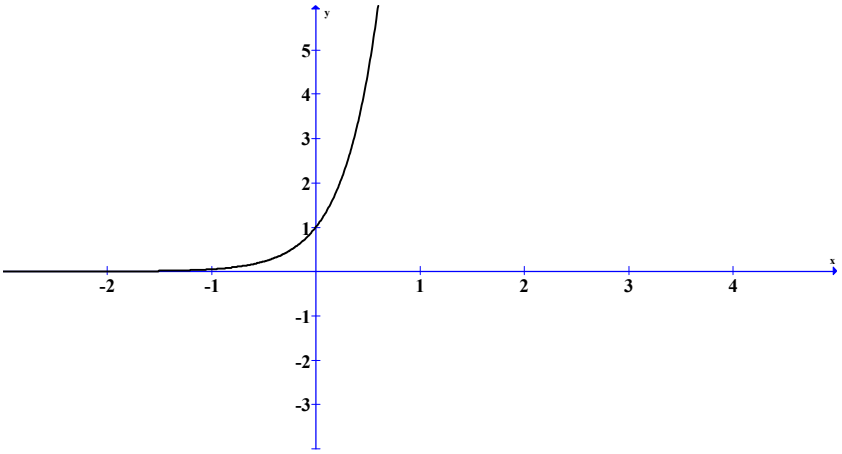


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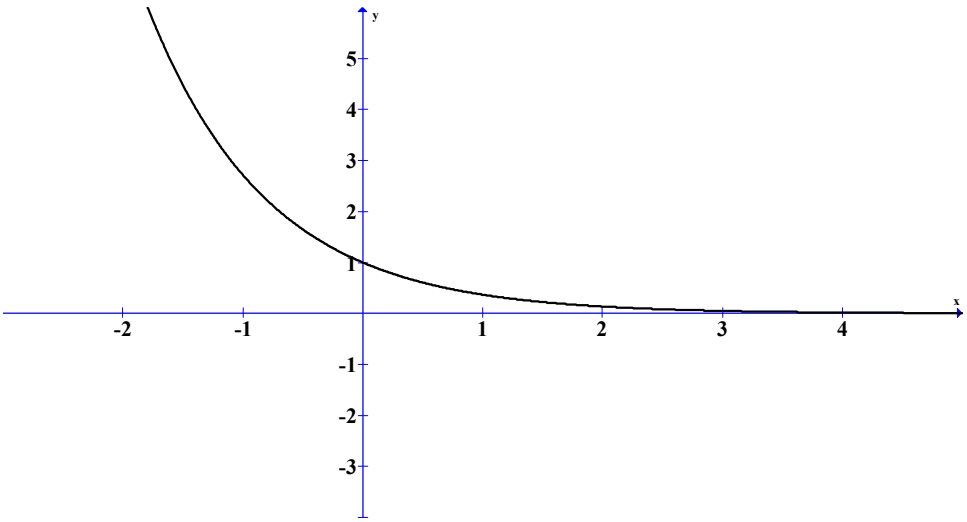
c) $y = e^{\frac{1}{2}x}$



d) $y = e^{3x}$

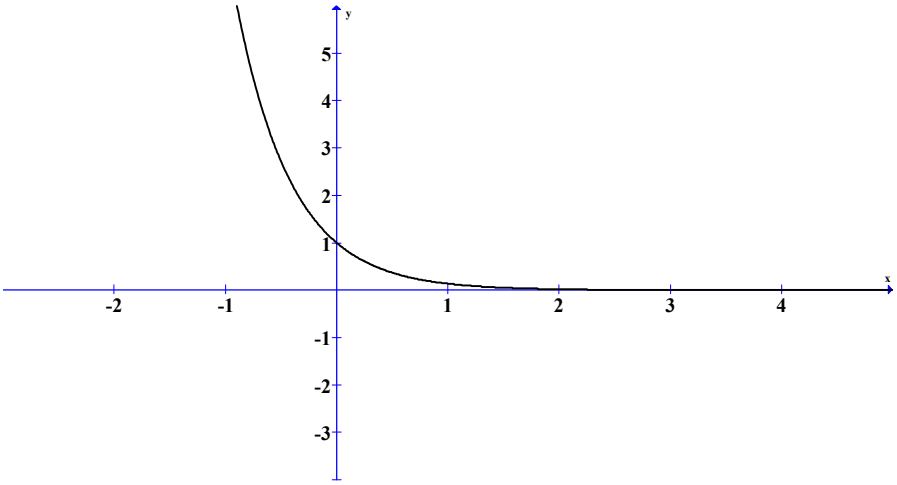


e) $y = e^{-x}$



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f) $y = e^{-2x}$

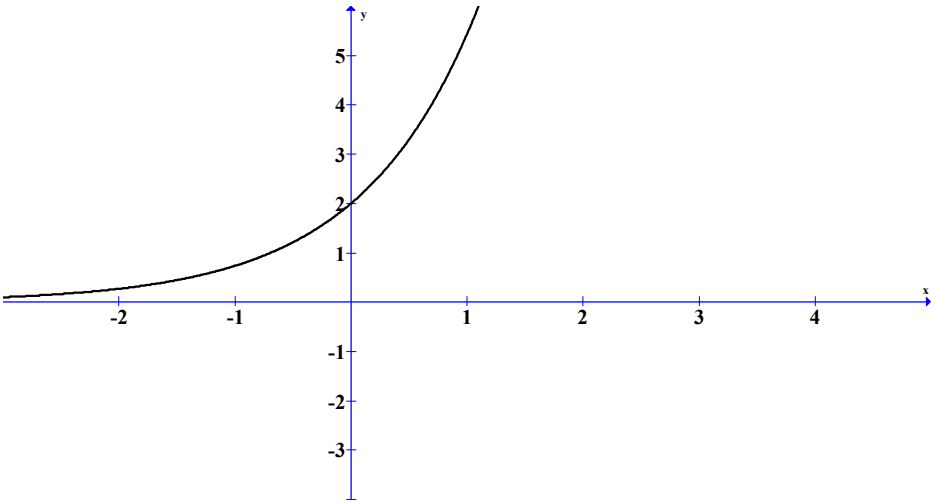


2) What effect does the value of k have on graphs of the form $y = e^{kx}$?

The higher the value of k, the broader the graph; a negative value of k reflects the graph around the y axis

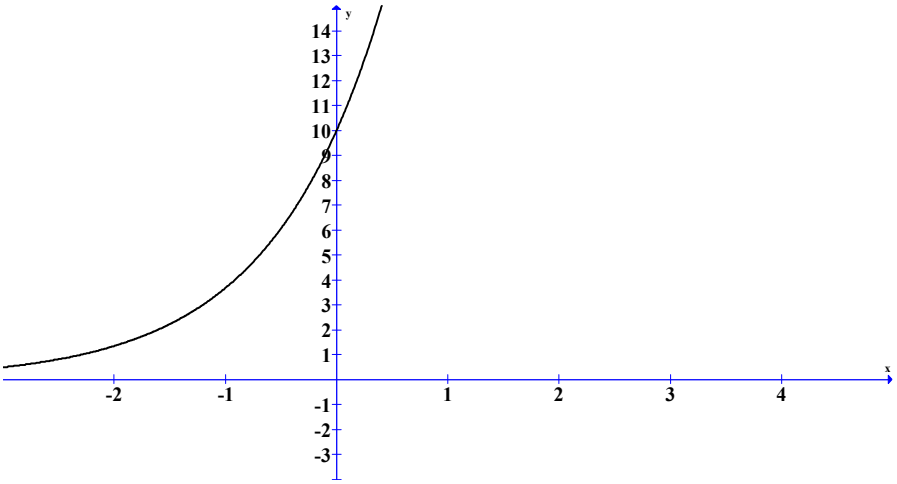
3) Graph the following over an appropriate range and domain

a) $y = 2e^{x}$

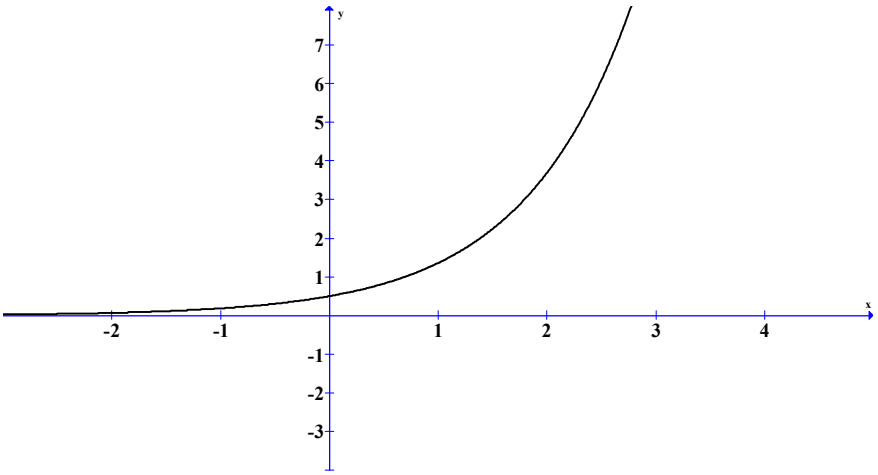


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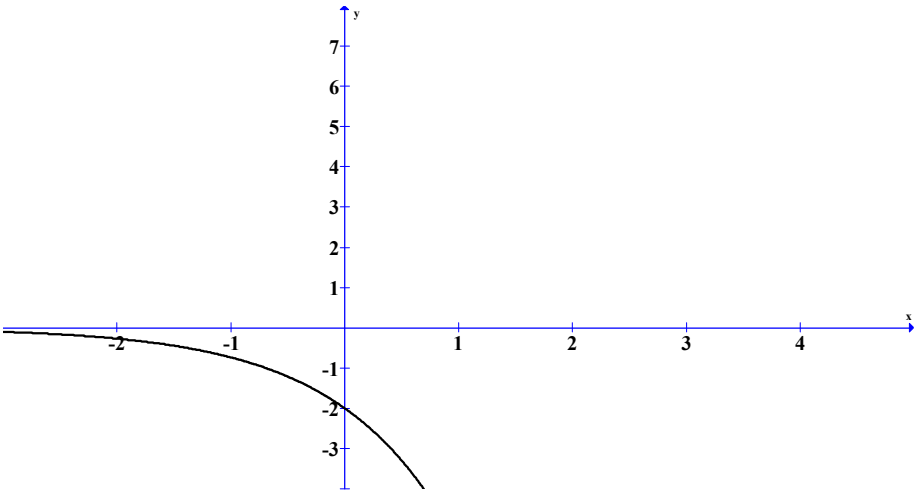
b) $y = 10e^x$



c) $y = 0.5e^x$



d) $y = -2e^x$

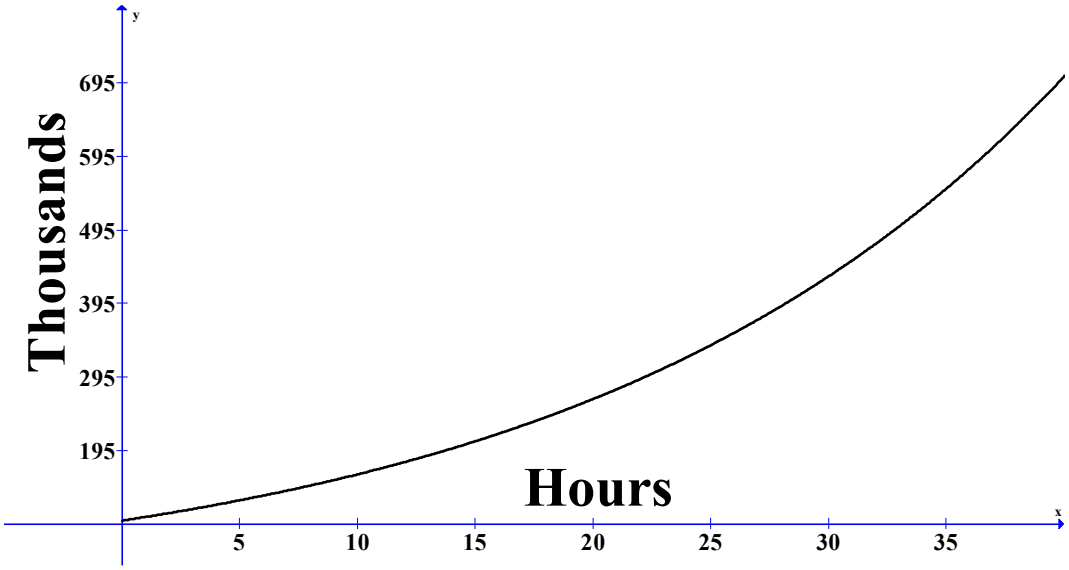


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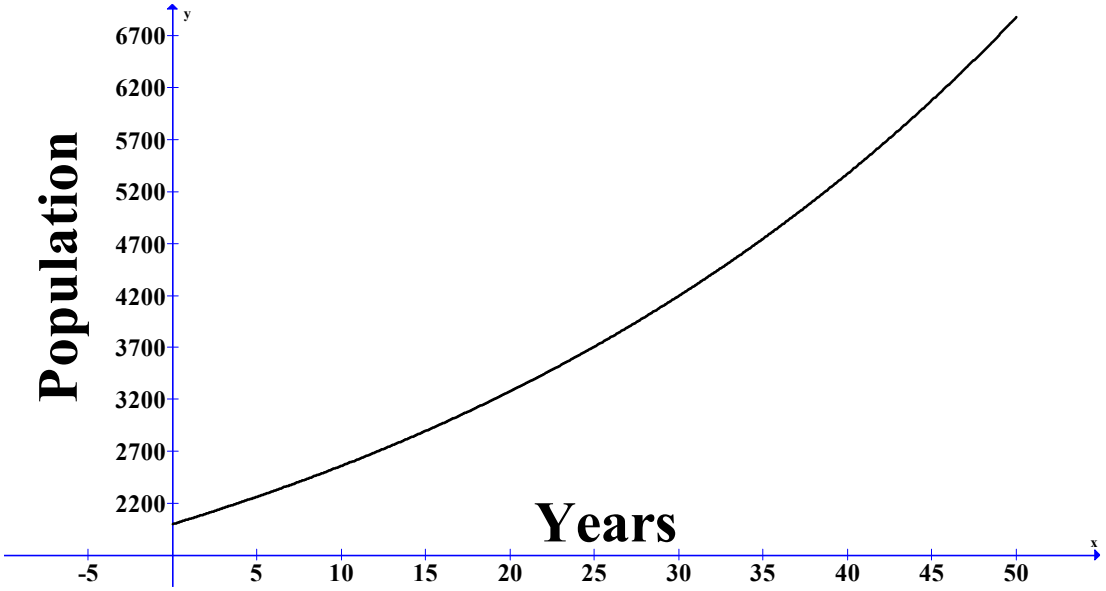
4) What effect does the value of A have on graphs of the form $y = Ae^x$?

The graph is moved up or down the y axis by A units. A negative value of A reflects the graph around the x axis

5) The growth rate per hour of a population of bacteria is 5% of the population. The initial population was 100,000 bacteria. Sketch the curve of the population after 40 hours



6) The initial population of a town is 2000, and it grows at the rate of 2.5% per annum. Graph the curve of the population after 50 years



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7) A mining town is suffering a net population decline due to lack of work. In 1990 the population was 2,000 the decline rate thereafter was 4.5% per annum.

a) What will the population be at the end of 2001?

$P = 2000(0.955^n)$

$n = 11, P = 2000(0.955^{11}) \cong 1205$

b) When will the population drop below 100?

$100 = 2000(0.955^n)$

When $n \log 0.955 = \log \frac{100}{2000}$

$n = 65\ years$

If the population drops at the same rate, it will be less than 100 in 2055

8) The number of mites in a pond was 1500 on January 1st. Each day the size of the colony grows by 8%

a) What will the population be on January 10th?

$P = 1500(1.08^n)$

When $n = 9, P = 1500(1.999) \cong 3000$

b) When will the population reach 5000?

$5000 = 1500(1.08^n)$

When $\log \frac{5000}{1500} = n \log 1.08$

$n = 15.64$

The population will reach 5000 during the 16th day; that is January 17th

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c) The pond can only support 7500 mites. When will this limit be reached?

$7500 = 1500(1.08^n)$

When $\log\frac{7500}{1500} = n\log 1.08$

$n = 20.9$

The maximum will be reached during the 21st day; that is on January 22nd

9) In the year 1990 there was $3500 in a bank account In the year 2000, the account held $5500. If there had been no deposits or withdrawals in that time, what was the rate of interest on the account? In what year will there be double the original amount? (Assume compound interest)

$5500 = 3500(1+r)^{10}$

$\log\frac{5500}{3500} = 10\log(1+r)$

$\frac{0.196}{10} = \log(1+r)$

$10^{0.0196} = 1+r$

$1+r = 1.046$

$r \cong 4.6\%$

10) A radioactive particle has a half life of 90 seconds; that is the amount present will reduce by half every 90 seconds. How much of a 1 kg sample would remain after 5 minutes?

$A = A_0e^{-kt}$

$\frac{A}{A_0} = e^{-90k} = 0.5$

$\ln(e^{-90k}) = \ln 0.5$

$-90k = -0.693$

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$k = 0.0077$

$A = A_0 e^{-0.0077t}$

When $t = 300, A = 1 \times e^{-0.0077 \times 300} \cong 0.099$

Approximately 100 grams will be left

11) The population of a town in the year 2000 was approximately 16,500 and ten years later it was approximately 27,200. Assuming a constant growth rate, what was that rate, and what was the population of the town in 1990?

$27200 = 16500(1 + r)^{10}$

$\frac{27200}{16500} = (1 + r)^{10}$

$(1 + r)^{10} \cong 1.648$

$10 \log(1 + r) = \log 1.648$

$\log(1 + r) = 0.0217$

$1 + r = 10^{0.0217}$

$r = 0.051$

Rate was 5.1%

$P_{1990} \times 1.051^{10} = 16500$

$P_{1990} = \frac{16500}{1.051^{10}} \cong 10036$

The population in 1990 was just over 10000

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Exercise 3

Velocity & Acceleration

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1) The displacement of a car in kilometres from a given point is given by the equation $x = 2t^2 + 10t$

a) What will be the displacement after 3 hours?

When $t = 3, x = 48\ km$

b) What will the velocity and acceleration of the car be after 4 hours?

$$V = \frac{dx}{dt} = 4t + 10$$

When $t = 4$,

$$V = 26\ km\ per\ hr$$

$$A = \frac{d^2x}{dt^2} = 4$$

Acceleration is constant at $4\ km\ hr^{-2}$

c) After how many hours will the car be travelling at the speed limit (30 km per hour)?

$$V = \frac{dx}{dt} = 30\ when$$

$$4t + 10 = 30$$

$$t = 5\ hours$$

2) The velocity of a particle t seconds after it starts moving from the point $x = 0$ is given by the equation $V = 10t + 4$.

a) What is the equation describing the displacement of the particle after t seconds?

$$x = \int V$$

$$= \int 10t = 4 = 5t^2 + 4t$$

b) What is the rate of acceleration of the particle?

$A = 10$ (constant)

3) The acceleration of a particle at time t is described by the equation $a = 10 - 2t$, where t is in seconds, and a is in ms^{-2}. At $t = 0$ the particle was at the origin with a velocity of 5 metres per second

a) What is the equation that describes velocity of the particle at any time?

$$V = \int a$$

$$= \int 10 - 2t = 10t - t^2 + c$$

When $t = 0, V = 5$

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$c = 5$

$V = 10t - t^2 + 5$

b) What is the equation that describes the displacement of the particle at any time?

$x = \int 10t - t^2 + 5$

$= 5t^2 - \frac{t^3}{3} + 5t + c$

At $t = 0, x = 0, so\ c = 0$

$x = 5t^2 - \frac{t^3}{3} + 5t$

c) What will be the displacement and velocity of the particle when the acceleration is zero?

$A = 0\ when\ t = 5$

$x = \frac{325}{3} = 108\frac{1}{3}m$

$V = 30\ ms^{-1}$

4) The velocity of a particle is -3 meters per second. Describe what this means in physical terms.

The particle is slowing down at that rate.

5) Can a particle have positive acceleration and a negative velocity? Explain

No; if a particle is accelerating its velocity must be increasing

6) Can a particle have negative acceleration and a positive velocity? Explain

Yes; a particle can have a positive velocity, but the rate of velocity increase (acceleration) can be getting smaller

7) The velocity of a particle at time t is described by the equation $v = -\frac{3}{2}t^2 + 8$. At $t = 0$, the particle is at position $x = 0$

a) What is the initial velocity of the particle?

When $t = 0, V = 8$

b) Describe the acceleration of the particle at any time t.

$A = -3t^2$

The rate of velocity increase is slowing

c) After how many seconds will the particle return to its original position?

$x = \int -\frac{3}{2}t^2 + 8\ dt$

When $t = 0, x = 0$

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$x = -\frac{1}{2}t^3 + 8t$

$x = 0 \; when \; 8t = \frac{1}{2}t^3$

$t = 0 \; or \; t = 4$

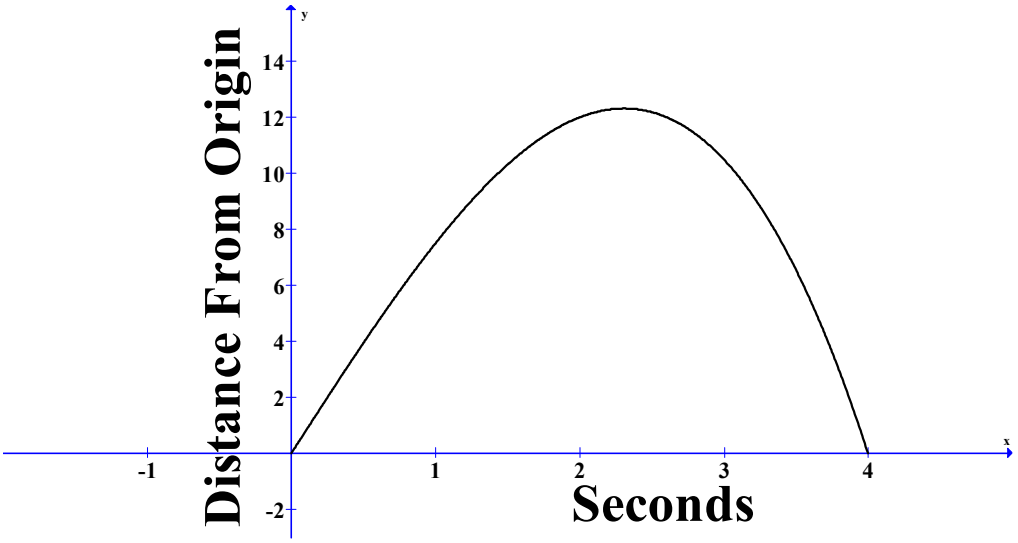
The particle returns to the origin after 4 seconds

d) What will be its velocity at this time?

$t = 4, V = -\frac{3}{2}(4^2) + 8$

$V = -16ms^{-1}$

e) Graph the displacement of the particle as a function of time



8) The acceleration of a particle at any time t (in seconds) is given by the equation $a = 10 - 6t$. At time $t = 0$ the particle is at the position $x = 0$ and has velocity 6 meters per second.

a) What will the velocity be at $t = 4$ seconds?

$V = \int 10 - 6t \; dt$

$V = 10t - 3t^2 + c$

When $t = 0, V = 6$

$c = 6$

$V = 10t - 3t^2 + 6$

When $t = 4, V = -2 \; ms^{-1}$

b) At what time will the particle return to the point $x = 0$?

$x = \int 10t - 3t^2 + 6 \; dt$

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$x = 5t^2 - t^3 + 6t + c$

When $t = 0, x = 0$

$x = 5t^2 - t^3 + 6t$

$x = 0\ when$

$5t^2 - t^3 + 6t = 0$

$t = 0\ or$

$t^2 - 5t - 6 = 0$

$(t - 6)(t + 1) = 0$

$t = 6$

The particle returns to the origin at 6 seconds

c) What will the velocity be when the acceleration is zero?

When $a = 0, 6t = 10$

$t = \frac{10}{6}$

$V = 10 \times \frac{10}{6} - 3 \times \left(\frac{10}{6}\right)^2 + 6$

$V = 86\ ms^{-1}$

9) The distance a particle is from a fixed point is described by the equation $x = 4 - 2\sin 2t$

Find the times when the particle is at rest, when acceleration is zero and when it returns to the fixed point.

$V = -4\cos 2t$

$V = 0\ when\ 2t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots.$

$t = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \ldots.$

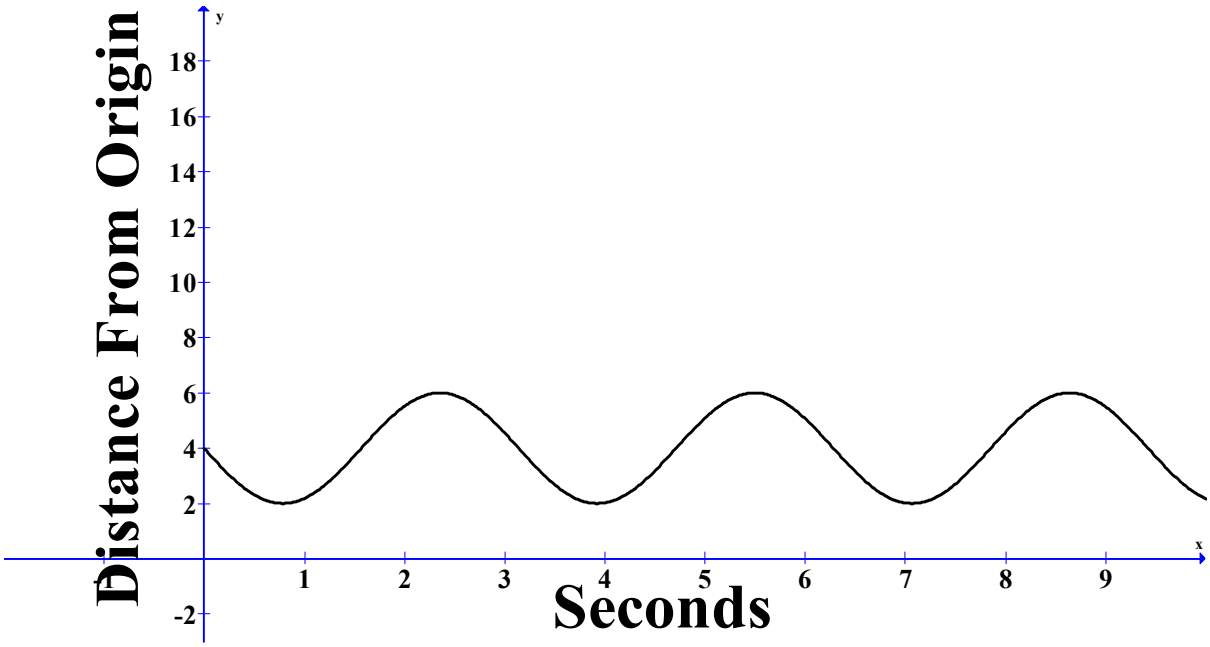
$A = 8\sin 2t$

$A = 0\ when\ 2t = 0, \pi, 2\pi, \ldots..$

$t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \ldots.$

Graph the displacement as a function of t

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Year 12 Mathematics

Exponential & Logarithmic Functions

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Exercise 1

Review of Index Laws

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1) Convert the following to index notation

a) $\sqrt{2}$

$2^{\frac{1}{2}}$

b) $\sqrt{x}$

$x^{\frac{1}{2}}$

c) $\sqrt[3]{3^2}$

$3^{\frac{2}{3}}$

d) $\sqrt[3]{x}$

$x^{\frac{1}{3}}$

e) $\sqrt[4]{x^3}$

$x^{\frac{3}{4}}$

2) Convert the following to surd form

a) $(x)^{\frac{1}{2}}$

$\sqrt{x}$

b) $(x^3)^{\frac{1}{4}}$

$\sqrt[4]{x^3}$

c) $\left(x^{\frac{1}{4}}\right)^3$

$\sqrt[4]{x^3}$

d) $(x)^{\frac{3}{4}}$

$\sqrt[4]{x^3}$

e) $(x)^{\frac{2}{5}}$

$\sqrt[5]{x^2}$

f) $(x)^{\frac{5}{2}}$

$\sqrt{x^5}$

3) Complete the following index laws

a) $a^x a^y = a^{x+y}$

b) $\frac{a^x}{a^y} = a^{x-y}$

c) $a^0 = 1$

d) $(a^x)^y = a^{xy}$

e) $(a \times b)^x = a^x \times b^x$

f) $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$

4) Use index laws to simplify the following

a) $a^2b^3c^2 \times b^2c^2$

$= a^2b^5c^4$

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b) $x^2y^4z \times xz^3$

$= x^3y^4z^4$

c) $x^ay^b \times x^cy^d$

$= x^{a+b} \times y^{b+d}$

d) $\frac{a^4b^3c^2}{a^2c}$

$= a^2b^3c$

e) $\frac{a^2b^2c^2}{a^3bc}$

$= \frac{bc}{a}$

f) $\frac{x^ay^b}{x^cy^d}$

$x^{a-c} \times y^{b-d}$

$= \frac{x^a}{y^b}$

d) $\frac{a^{-2}b^3c^{-1}}{a^2b^{-2}c^3}$

$= \frac{b^5}{a^4c^4}$

e) $a^{-3}b^2c^3 \times a^3b^{-2}c^{-3}$

$= 1$

f) $\frac{a^3b^{-3}c}{a^{-3}c^2}$

$= \frac{a^2}{b^3c}$

g) $\frac{a^{-2}b^3c^2}{a^{-2}b^3c^2}$

$= 1$

5) Use index laws to simplify the following: express your answers with positive indices

a) $a^{-4}b^2c^{-1} \times a^2b^{-5}c^{-2}$

$= \frac{1}{a^2b^3c^3}$

b) $a^{-2}b^{-3} \times a^{-1}b^4c^{-1}$

$= \frac{b}{a^3c}$

c) $x^{-a}y^b \times x^{2a}y^{-2b}$

6) Use index laws to simplify the following leaving your answers in positive index form

a) $(a^2)^3$

$= a^6$

b) $(a^2b^2)^{\frac{1}{2}}$

$= ab$

c) $\left(\frac{a^{\frac{1}{2}}b^{-\frac{1}{2}}}{c^{\frac{1}{2}}}\right)^{-2}$

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$= \frac{bc}{a}$

d) $(a^3b^{-2}c^{-3})^0$

$= 1$

e) $x^4 - x^3$

$= x(x-1)$

7) Simplify the following using index laws

a) $(a^{-3}b^2)^{-2} \div (a^3b^2)^2$

$= a^6b^{-4} \div a^6b^4$

$= \frac{1}{b^8}$

b) $(a^2 \div b^{-2})^{\frac{1}{2}} \times \left(a^{\frac{1}{2}} \div b^{-\frac{1}{2}}\right)^2$

$= (a \div b^{-1}) \times (a \div b^{-1})$

$= \frac{a^2}{b^2}$

c) $(ab)^{-1} \div (a^2b^2)^{-\frac{1}{2}}$

$= (a^{-1}b^{-1}) \div (a^{-1}b^{-1})$

$= 1$

d) $(2a^2)^0 - 2$

$= 1 - 2 = -1$

e) $\frac{4a^0}{(4a)^0} = 1$

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Exercise 2

Logarithms & Exponents

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1) Convert the following to exponential form

a) $\log_{10} 100 = 2$

$10^2 = 100$

b) $\log_{10} x = c$

$10^c = x$

c) $\log_y 10 = x$

$y^x = 10$

d) $\log_2 p = r$

$2^r = p$

e) $\log_x 5 = 12$

$x^{12} = 5$

f) $\log_x y = a$

$x^a = y$

2) Convert the following to logarithmic form

a) $10^2 = 100$

$\log_{10} 100 = 2$

b) $x^3 = 20$

$\log_x 20 = 3$

c) $5^2 = y$

$\log_5 y = 2$

d) $a^2 = r$

$\log_a r = 2$

e) $x^y = 10$

$\log_x 10 = y$

f) $a^b = c$

$\log_a c = b$

3) Prove $\log_a(xy) = \log_a x + \log_a y$

Let $r = \log_a x \ and \ t = \log_a y$

Then $x = a^r, y = a^t$

$xy = a^r \times a^t = a^{r+t}$

$\log_a xy = \log_a a^{r+t} = (r + t) \log_a a$

$\log_a xy = r + t$

$\log_a xy = \log_a x + \log_a y$

4) Prove $\log_a x^b = b \log_a x$

Let $t = \log_a x$

$x = a^t$

$x^b = a^{bt}$

$\log_a x^b = bt = b \log_a x$

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5) Prove that $\log_x 1 = 0$

$\log_x 1 = \log_x \frac{a}{a}$

$= \log_x a - \log_x a = 0$

6) Prove that $\log_x x = 1$

$Let\ \log_x x = t$

$x^t = x$

$t = 1$

7) If $a = 2, b = 3, and\ c = 5$, express the following in terms of a, b and c

a) $\log_{10} 6$

$= \log_{10}(a \times b)$

$\log_{10} a + \log_{10} b$

b) $\log_{10}\left(\frac{1}{25}\right)$

$= \log_{10} 5^{-2}$

$-a \log_{10} c$

c) $\log_{10}\left(\frac{2}{3}\right)$

$= \log_{10}\left(\frac{a}{b}\right)$

$= \log_{10} a - \log_{10} b$

d) $\log_{10}\left(\frac{3}{5}\right)$

$= \log_{10}\left(\frac{b}{c}\right)$

$= \log_{10} b - \log_{10} c$

e) $\log_{10} 16$

$= \log_{10} 2^4$

$= 4 \log_{10} a$

$= a^2 \log_{10} a$

f) $\log_{10} 30$

$= \log_{10}(a \times b \times c)$

$= \log_{10} a + \log_{10} b + \log_{10} c$

g) $\log_{10}\left(\frac{1}{8}\right)$

$= \log_{10} 2^{-3}$

$= -3 \log_{10} 2$

$= -b \log_{10} a$

h) $\log_{10} 10$

$= \log_{10}(2 \times 5)$

$= \log_{10} a + \log_{10} c$

8) Rewrite the following in terms of $\log_{10}$

a) $\log_5 x$

$= \frac{\log_{10} x}{\log_{10} 5}$

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b) $\log_x 5$

$$= \frac{\log_{10} 5}{\log_{10} x}$$

c) $\log_x y$

$$= \frac{\log_{10} y}{\log_{10} x}$$

d) $\log_{100} 10$

$$= \frac{\log_{10} 10}{\log_{10} 100} \left(= \frac{1}{2}\right)$$

e) $\log_x x$

$$= \frac{\log_{10} x}{\log_{10} x} (= 1)$$

f) $\log_1 x$

$$= \frac{\log_{10} x}{\log_{10} 1}$$

Undefined

g) $\log_x 1$

$$= \frac{\log_{10} 1}{\log_{10} x} = 0$$

9) Calculate the following correct to 3 decimal places

a) $\log_3 7$

$$= \frac{\log_{10} 7}{\log_{10} 3} \cong \frac{0.845}{0.477} \cong 1.771$$

b) $\log_2 10$

$$= \frac{\log_{10} 10}{\log_{10} 2} \cong \frac{1}{0.301} \cong 3.22$$

c) $\log_4 64$

$$= \frac{\log_{10} 64}{\log_{10} 4} = \frac{1.806}{0.602} = 3$$

d) $\log_5 5$

$$= 1$$

e) $\log_1 10$

Undefined

f) $\log_{10} 5$

$$= \frac{\log_{10} 5}{\log_{10} 10} \cong 0.699$$

10) Solve the following

a) $3^{-x^2} = \frac{1}{27}$

$$3^{-x^2} = 3^{-3}$$

$$x^2 = 3$$

$$x = \pm\sqrt{3}$$

b) $5(2.5^{2x+1}) = 150$

$$2.5^{2x+1} = 30$$

$$(2x + 1) \log 2.5 = \log 30$$

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$$2x + 1 \cong \frac{1.477}{0.398}$$

$$x \cong 1.356$$

c) $6^{3x} = 32$

$$3x \log 6 = \log 32$$

$$3x \cong \frac{1.505}{0.778}$$

$$x \cong 0.645$$

d) $20 = 10(4)^{x+5}$

$$4^{x+5} = 2$$

$$2^{(2x+10)} = 2^1$$

$$2x + 10 = 1$$

$$x = -\frac{9}{2}$$

e) $1000^{0.15x} = 12{,}000$

$$0.15x \log 1000 = \log 12000$$

$$0.15x = \frac{\log 12000}{3}$$

$$x \cong 9.06$$

11) What is the value of $\ln e^x$?

$$= x$$

12) Solve the following

a) $e^x = 10$

$$\ln e^x = \ln 10$$

$$x \cong 2.3$$

b) $100e^{2x} = 50$

$$e^{2x} = 0.5$$

$$\ln e^{2x} = \ln 0.5$$

$$2x \cong -0.693$$

$$x \cong -0.347$$

c) $25e^{0.5x} = 12$

$$\ln e^{0.5x} = \ln \frac{12}{25}$$

$$0.5x \cong -0.734$$

$$x \cong -1.468$$

d) $\ln x = 4$

$$e^{\ln x} = e^4$$

$$x \cong 54.6$$

e) $4 + 2 \ln x = 14$

$$2 \ln x = 10$$

$$\ln x = 5$$

$$x = e^5$$

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$x \cong 148.41$

f) $\ln(x-4)+\ln(x+3)=\ln 8$

$\ln\big((x-4)(x+3)\big)=\ln 8$

$x^2-x-12=8$

$x^2-x-20=0$

$x=5 \; or \; x=-4$

g) $0.05x=\ln 5$

$x \cong \dfrac{1.609}{0.05}=32.19$

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Exercise 3

Differentiation & Integration

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1) Differentiate the following

a) $y = e^x$

$$\frac{dy}{dx} = e^x$$

b) $y = e^{2x}$

$$\frac{dy}{dx} = 2e^{2x}$$

c) $y = e^{3x-4}$

$$\frac{dy}{dx} = 3e^{3x-4}$$

d) $y = e^{x^2}$

$$\frac{dy}{dx} = 2xe^{x^2}$$

e) $y = e^{x^2+2x-1}$

$$\frac{dy}{dx} = (2x+2)e^{x^2+2x-1}$$

f) $y = e^{\sin x}$

$$\frac{dy}{dx} = \cos x\, e^{\sin x}$$

2) Differentiate the following

a) $y = \ln x$

$$\frac{dy}{dx} = \frac{1}{x}$$

b) $y = \ln(2x)$

$$\frac{dy}{dx} = \frac{2}{x}$$

c) $y = \ln(3x+1)$

$$\frac{dy}{dx} = \frac{3}{3x+1}$$

d) $y = \ln e^{x^2}$

$$\frac{dy}{dx} = 2x$$

e) $y = (\ln x)^2$

$$\frac{dy}{dx} = \frac{2\ln x}{x}$$

f) $y = \ln(x^2 + 2x - 3)$

$$\frac{dy}{dx} = \frac{2x+2}{x^2+2x-3}$$

g) $y = \ln(\sin x)$

$$\frac{dy}{dx} = \frac{\cos x}{\ln(\sin x)}$$

3) Perform the following integrations

a) $\int e^x dx$

$$= e^x + c$$

b) $\int e^{2x}\, dx$

$$\frac{e^{2x}}{2} + c$$

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c) $\int e^{\frac{x}{2}}\,dx$

$= 2e^{\frac{x}{2}} + c$

d) $\int 2e^{2x}dx$

$e^{2x} + c$

e) $\int a^{2x}\,dx$

$\frac{a^{2x}}{2} + c$

4) Perform the following integrations

a) $\int \ln x\,dx$

$= x\ln x - x$

b) $\int \ln 2x\,dx$

Let $u = 2x$

Then $\frac{du}{2} = dx$

$\int = \frac{1}{2}\int \ln u\ du$

From above

$= \frac{1}{2}(2x\ln 2x - 2x) + c$

$= x\ln 2x - x + c$

c) $\int \ln\frac{x}{2}dx$

Let $u = \frac{x}{2}$

$2\ du = dx$

$\int = 2\int u\ du$

$= 2(u\ln u - u) + c$

$= 2\left(\frac{x}{2}\ln\frac{x}{2} - \frac{x}{2}\right) + c$

$= x\ln\frac{x}{2} - x + c$

d) $\int \ln(3x+1)\,dx$

Let $u = 3x + 1$

$dx = \frac{du}{3}$

$\int = \frac{1}{3}\int \ln u\ du$

$= \frac{1}{3}(u\ln u - u) + c$

$= \frac{1}{3}((3x+1)\ln(3x+1) - 3x + 1) + c$

$= \left(x + \frac{1}{3}\right)\ln(3x+1) - x + \frac{1}{3} + c$

e) $\int (\ln x)^2\,dx$

Let $u = \ln x$

$So\ x = e^u$

$x\ du = dx$

$e^u du = dx$

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$$\int = \int u^2 e^u \, du$$

Noting that $e^u = (e^u)$

We can integrate by parts to obtain

$$\int = u^2 \, e^u - \int 2u \, e^u \, du$$

Again noting that $e^u = (e^u)$

And integrating by parts again

$$= u^2 e^u - [2u \, e^u - \int 2e^u \, du]$$

$$= u^2 e^u - 2ue^u + 2e^u$$

(Substituting $x = e^u \; and \; u = \ln x$)

$$= x(\ln x)^2 - 2x \ln x + 2x$$

f) $\int \frac{(\ln x)^3}{x} dx$

Let $u = \ln x$

$x = e^u$

$\frac{du}{dx} = \frac{1}{x}$

$du = \frac{1}{x} \, dx$

$$\int = \int u^3 \, du$$

$$= \frac{u^4}{4} + c$$

$$= \frac{(lnx)^4}{4} + c$$

5) Perform the following integrations

a) $\int \frac{4}{x} dx$

$= 4 \ln x + c$

b) $\int \frac{3}{3x+2} dx$

$= \ln(3x + 2)$+c

c) $2 \int \frac{2x}{x^2-3} dx$

$= 2 \ln(x^2 - 3) + c$

d) $\int \frac{12x}{3x^2+1} dx$

$= 2 \ln(3x^2 + 1) + c$

e) $\int \frac{x-2}{x^2-4x+2} dx$

$= \frac{1}{2} \ln(x^2 - 4x + 2)$+c

f) $\int \cot x \, dx$

$\cot x = \frac{\cos x}{\sin x}$

Let $u = \sin x$

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$$du = \cos x \ dx$$

$$\int \cot x \, dx = \int \frac{\cos x}{\sin x} \, dx$$

$$= \int \frac{1}{u} \, du$$

$$= \ln u + c$$

$$\int \cot x = \ln(\sin x) + c$$

6) Differentiate the following and simplify your answers where possible

a) $y = \ln(x+2) - \ln(x-3)$

$$\frac{dy}{dx} = \frac{1}{x+2} - \frac{1}{x-3}$$

b) $y = \ln(x-4) + \ln(x+1)$

$$\frac{dy}{dx} = \frac{1}{x-4} + \frac{1}{x+1}$$

c) $y = \ln x^2 + \ln(x+1)$

$$\frac{dy}{dx} = \frac{2x}{x^2} + \frac{1}{x+1}$$

d) $y = \ln(x^2+2x+1) - \ln(x+1)$

$$\frac{dy}{dx} = \frac{2x+2}{x^2+2x+1} - \frac{1}{x+1}$$

$$= \frac{2(x+1)}{(x+1)^2} - \frac{1}{x+1}$$

$$= \frac{2}{x+1} - \frac{1}{x+1} = \frac{1}{x+1}$$

e) $y = \ln(\sin x) - \ln(\cos x)$

$$\frac{dy}{dx} = \frac{\cos x}{\sin x} - \frac{-\sin x}{\cos x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\sin x \cos x}$$

$$= \frac{1}{\sin x \cos x}$$

f) $y = \ln(x^2-1) + \ln(x-1)$

$$\frac{dy}{dx} = \frac{2x}{x^2-1} + \frac{1}{x-1}$$

$$= \frac{2x}{(x-1)(x+1)} + \frac{(x+1)}{(x+1)(x-1)}$$

$$= \frac{3x+1}{x^2-1}$$

7) Differentiate the following and simplify your answers where possible

a) $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

$$f(x) = e^x - e^{-x}$$

$$f'(x) = e^x + e^{-x}$$

$$g(x) = e^x + e^{-x}$$

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$g'(x) = e^x - e^{-x}$

$g^2(x) = (e^x + e^{-x})^2$

By the quotient rule

$$\frac{dy}{dx} = \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2}$$

$$= \frac{2}{(e^x + e^{-x})^2}$$

b) $y = \frac{e^x}{e^x - 1}$

$$\frac{dy}{dx} = \frac{(e^x)(e^x - 1) - e^x e^x}{(e^x - 1)^2}$$

$$= \frac{-e^x}{(e^x - 1)^2}$$

c) $y = x(\cos(\ln x) + \sin(\ln x))$

Differentiate each part by the product rule

$$\frac{dy}{dx} = \cos(\ln x) - \frac{x \sin \ln x}{x} + \sin(\ln x) + \frac{x \cos(\ln x)}{x}$$

$$= 2\cos(\ln x)$$

d) $y = \ln(x^2 e^{-x^2})$

Simplify expression using log laws

$$y = \ln x^2 + \ln e^{-x^2} = \ln x^2 - x^2$$

$$\frac{dy}{dx} = \frac{2x}{x^2} - 2x = \frac{2}{x} - 2x$$

e) $y = \ln[(x^2 + 1)(x^3 + 1)^2]$

Simplify

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$$y = \ln(x^2 + 1) + 2\ln(x^3 + 1)$$

$$\frac{dy}{dx} = \frac{2x}{x^2 + 1} + \frac{6x^2}{x^3 + 1}$$

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Year 12 Mathematics

Trigonometry

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Exercise 1

Radian Measurement

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1) Convert the following to exact radians

a) 30°

$\frac{\pi}{6}$

b) 40°

$\frac{2\pi}{9}$

c) 120°

$\frac{2\pi}{3}$

d) 70°

$\frac{7\pi}{18}$

e) 170°

$\frac{17\pi}{18}$

f) 160°

$\frac{8\pi}{9}$

g) 150°

$\frac{5\pi}{6}$

h) 75°

$\frac{5\pi}{12}$

i) 10°

$\frac{\pi}{18}$

j) 130°

$\frac{13\pi}{18}$

k) 165°

$\frac{11\pi}{12}$

l) 60°

$\frac{\pi}{3}$

2) Convert the following radians to degrees

a) $\frac{\pi}{4}$

45°

b) $\frac{4\pi}{9}$

80°

c) $\frac{11\pi}{18}$

110°

d) $\frac{7\pi}{12}$

105°

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e) $\frac{7\pi}{9}$

140°

f) $\frac{\pi}{2}$

90°

g) $\frac{\pi}{9}$

20°

h) π

180°

i) $\frac{5\pi}{18}$

50°

j) $\frac{\pi}{12}$

15°

k) $\frac{3\pi}{4}$

135°

l) $\frac{5\pi}{9}$

100°

3) Convert the following degrees to exact radians

a) 210°

$\frac{7\pi}{6}$

b) 240°

$\frac{4\pi}{3}$

c) 270°

$\frac{3\pi}{2}$

d) 320°

$\frac{16\pi}{9}$

e) 360°

2π

4) Convert the following radians to degrees

a) $\frac{3\pi}{2}$

270°

b) $\frac{5\pi}{4}$

225°

c) $\frac{7\pi}{6}$

210°

d) $\frac{7\pi}{4}$

315°

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e) $\frac{13\pi}{9}$

260°

f) $\frac{17\pi}{15}$

204°

5) Find the length of the following arcs in terms of π

a) Radius 2cm, subtended angle of 30°

$$l = 2\pi \times 2 \times \frac{30}{360}$$

$$= \frac{\pi}{3}$$

b) Radius 10cm, subtended angle of 90°

$$l = 2\pi \times 10 \times \frac{90}{360}$$

$$= 5\pi$$

c) Radius of 12cm, subtended angle of 120°

$$l = 2\pi \times 12 \times \frac{120}{360}$$

$$= 8\pi$$

d) Radius of 1.5m, subtended angle of 70°

$$l = 2\pi \times 1.5 \times \frac{70}{360}$$

$$= \frac{7\pi}{12}$$

e) Radius of 33cm, subtended angle of 100°

$$l = 2\pi \times 33 \times \frac{100}{360}$$

$$= \frac{55\pi}{3}$$

f) Radius of 2.25cm, subtended angle of 135°

$$l = 2\pi \times 2.25 \times \frac{135}{360}$$

$$= \frac{27\pi}{16}$$

6) Calculate the areas of the following sectors in terms of π

a) Radius 10cm, subtended angle of 220°

$$A = \pi 10^2 \times \frac{220}{360}$$

$$= \frac{550\pi}{9}$$

b) Radius 4cm, subtended angle of 10°

$$A = \pi 4^2 \times \frac{10}{360}$$

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$= \frac{4\pi}{9}$

c) Radius 15cm, subtended angle of 135°

$A = \pi 15^2 \times \frac{135}{360}$

$= \frac{675\pi}{8}$

d) Radius 8cm, subtended angle of 110°

$A = \pi 8^2 \times \frac{110}{360}$

$= \frac{176\pi}{9}$

e) Radius 1cm, subtended angle of 180°

$A = \pi 1^2 \times \frac{180}{360}$

$= \frac{\pi}{2}$

f) Radius 10cm, subtended angle of 360°

$A = \pi 10^2 \times \frac{360}{360}$

$= 100\pi$

7) Solve the following equations for x in the interval stated

a) $2\sin x + 1 = 0, [0, 2\pi]$

$\sin x = -\frac{1}{2}$

$x = \frac{11\pi}{12}$

b) $\sqrt{3}\cos x - 1 = 0, [0, 2\pi]$

$\cos x = \frac{1}{\sqrt{3}}$

$x = \frac{\pi}{3}, \frac{5\pi}{3}$

c) $\sin 2x = \frac{1}{\sqrt{2}}, [0, \pi]$

$2x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

$x = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$

d) $\tan 4x - 1 = 0, [\pi, 2\pi]$

$\tan 4x = 1$

$4x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}, \ldots$

$x = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16}, \ldots$

In the interval:

$x = \frac{17\pi}{16}, \frac{21\pi}{16}, \frac{25\pi}{16}, \frac{29\pi}{16}$

e) $4\cos 4x = 2, [0, \pi]$

$\cos 4x = \frac{1}{2}$

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$$4x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

$$x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}$$

8) Solve the following in the interval stated

a) $2\cos 3x = 1, [0,2\pi)$

$$\cos 3x = \frac{1}{2}$$

$$3x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \ldots$$

$$x = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}, \frac{15\pi}{9}$$

b) $\sin 2x = \sin x, \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$$x = 0$$

c) $\sin x - \cos x = 1, [0, 2\pi]$

$$\tan x = 1$$

$$x = \frac{\pi}{4}, \frac{5\pi}{4}$$

d) $2\sin 2x \cos 2x = \cos 2x, [-\pi, \pi]$

$$2\sin 2x = 1$$

$$2x = \frac{\pi}{6}, \frac{5\pi}{6}, -\frac{7\pi}{6}$$

$$x = \frac{\pi}{12}, \frac{5\pi}{12}, -\frac{7\pi}{12}$$

e) $\sin^2 x - \sin x = 2, (-2\pi, 2\pi)$

$$(\sin x - 2)(\sin x + 1) = 0$$

$$\sin x = -1$$

$$x = -\pi, \frac{3\pi}{2}$$

f) $\sin x + \cos x = 1, [0, 2\pi]$

$$x = 0, \frac{\pi}{2}, 2\pi$$

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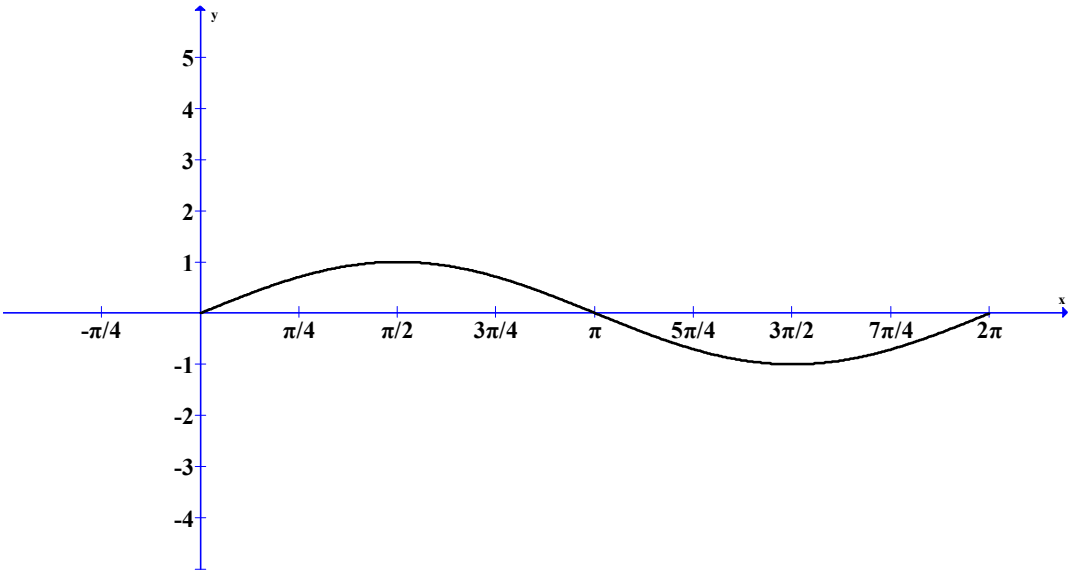
Exercise 2

Graphing Trigonometric Functions

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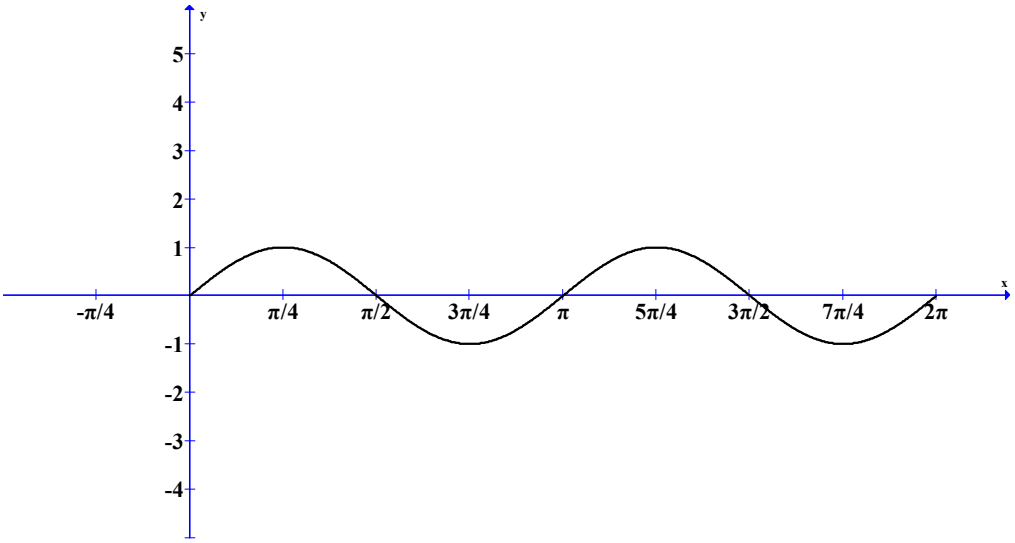
1) Graph the following in the range 0 to 2π, and state the domain for each

a) $\sin x$



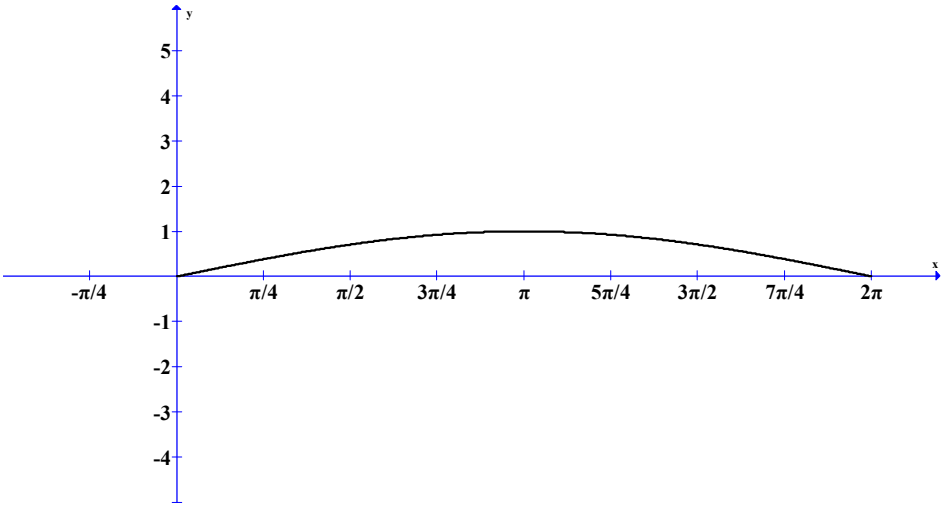
-1 to 1

b) $\sin 2x$

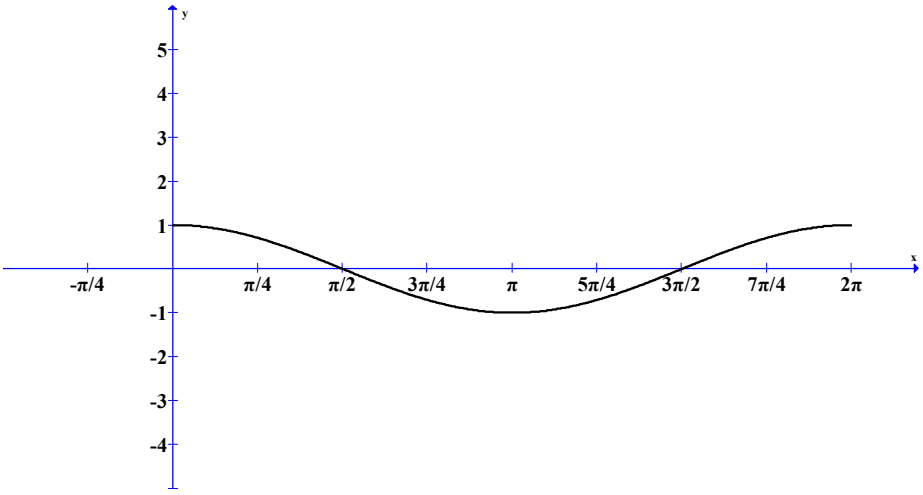


c) $\sin \frac{1}{2}x$

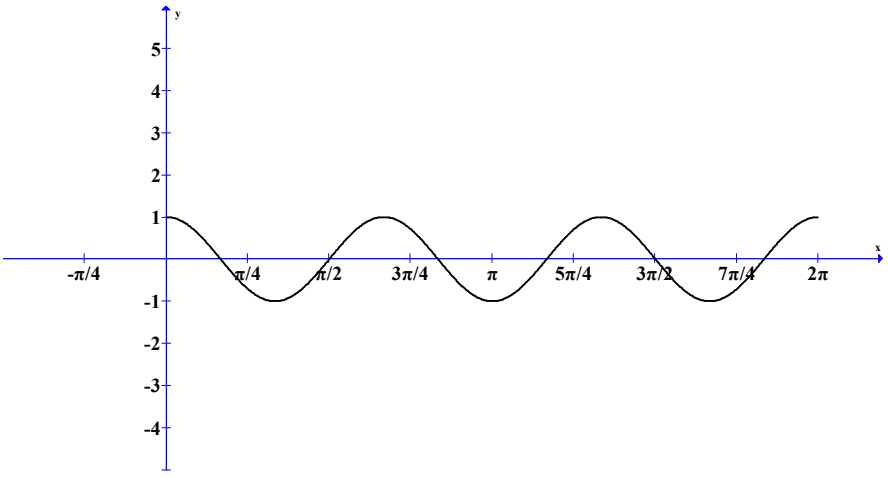
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d) $\cos x$

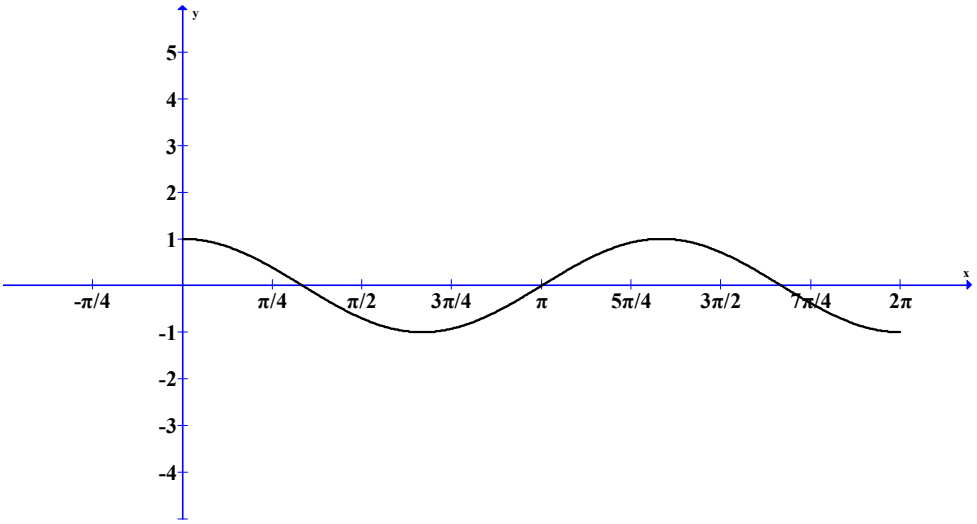


e) $\cos 3x$



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f) $\cos\frac{3}{2}x$



2) From your graphs in question 1, what effect on the range is produced by the varying of B in the general equations

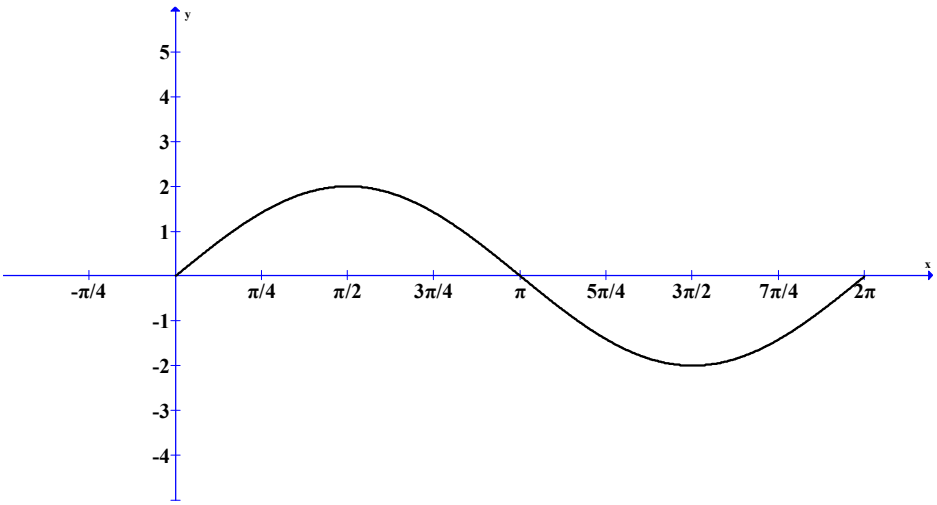
$y = \sin Bx$, and $y = \cos Bx$

What general effect does the value of "B" have on such graphs?

The period of the curve increases or decreases with the value of B

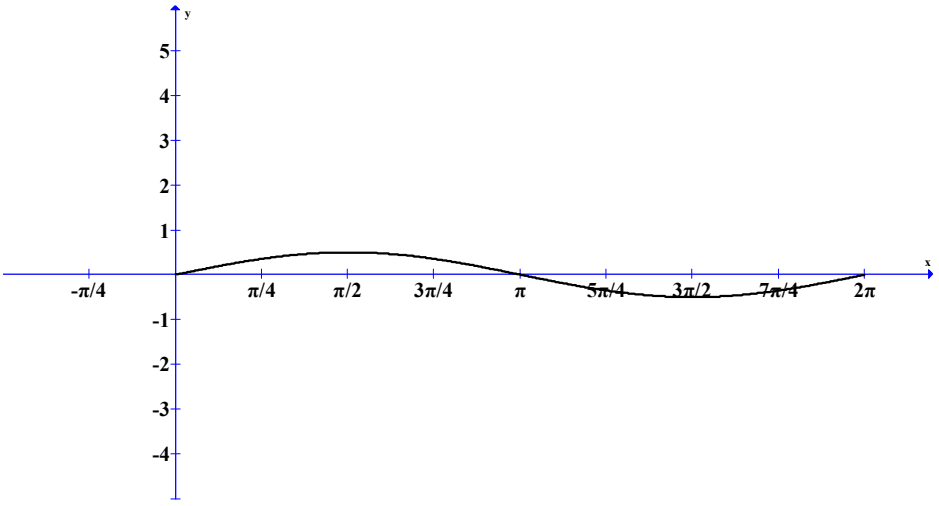
3) Graph the following in the range 0 to 2π, and state the domain for each

a) $2\sin x$

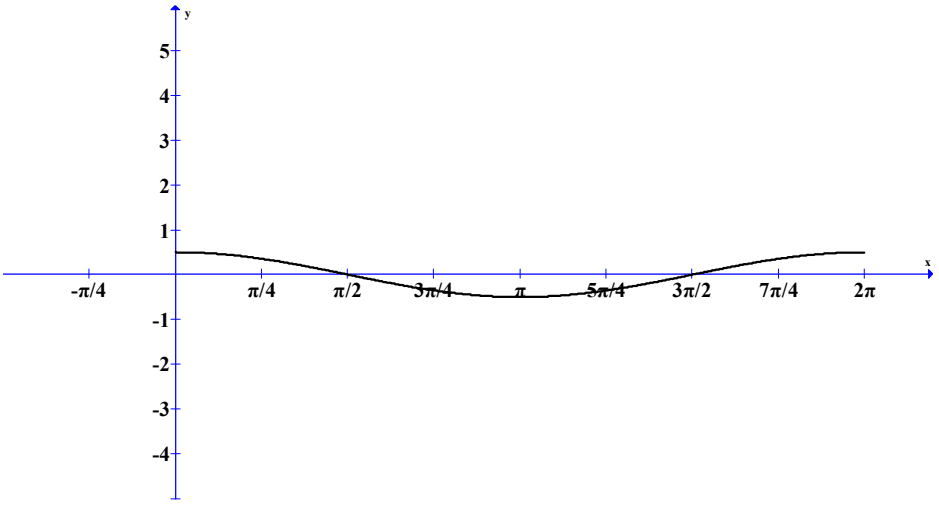


b) $\frac{1}{2}\sin x$

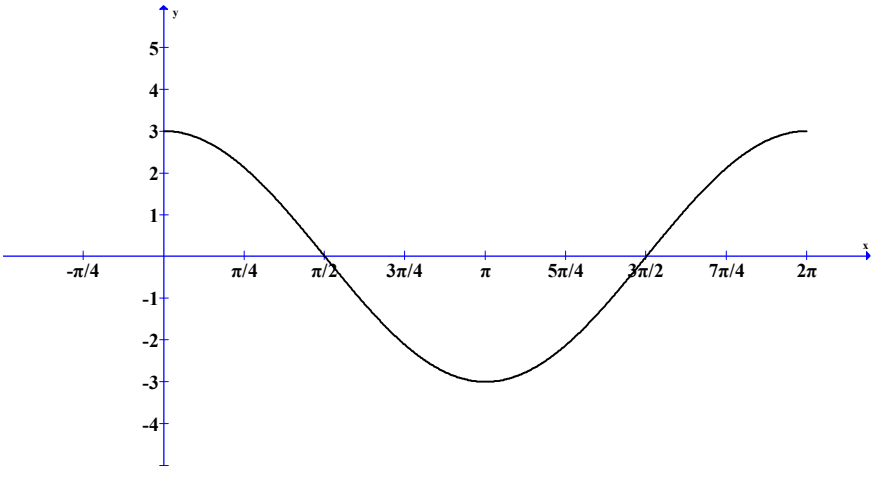
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c) $\frac{1}{2}\cos x$



d) $3\cos x$



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4) From your graphs in question 3, what effect on the range is produced by the varying of A in the general equations

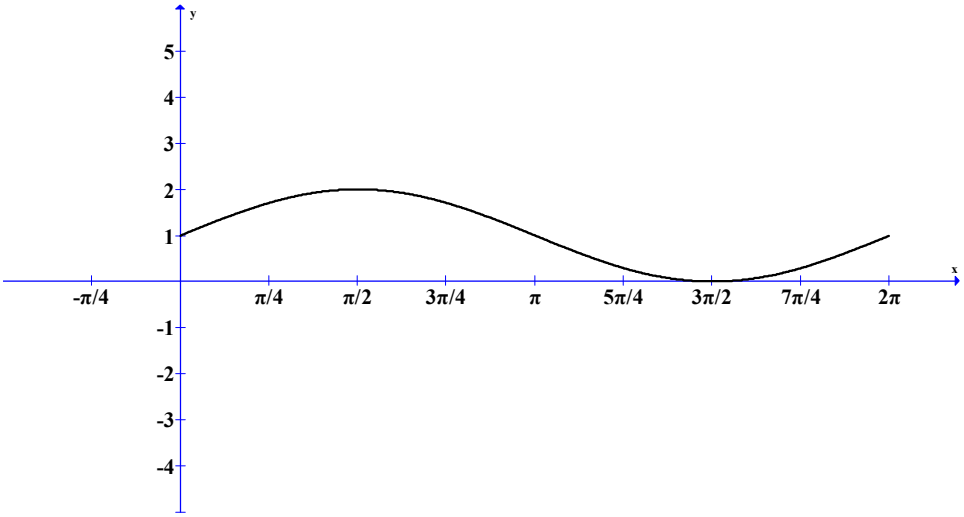
$y = A\sin x$, and $y = A\cos x$

What general effect does the value of "A" have on such graphs?

The amplitude of the curve changes with the value of A

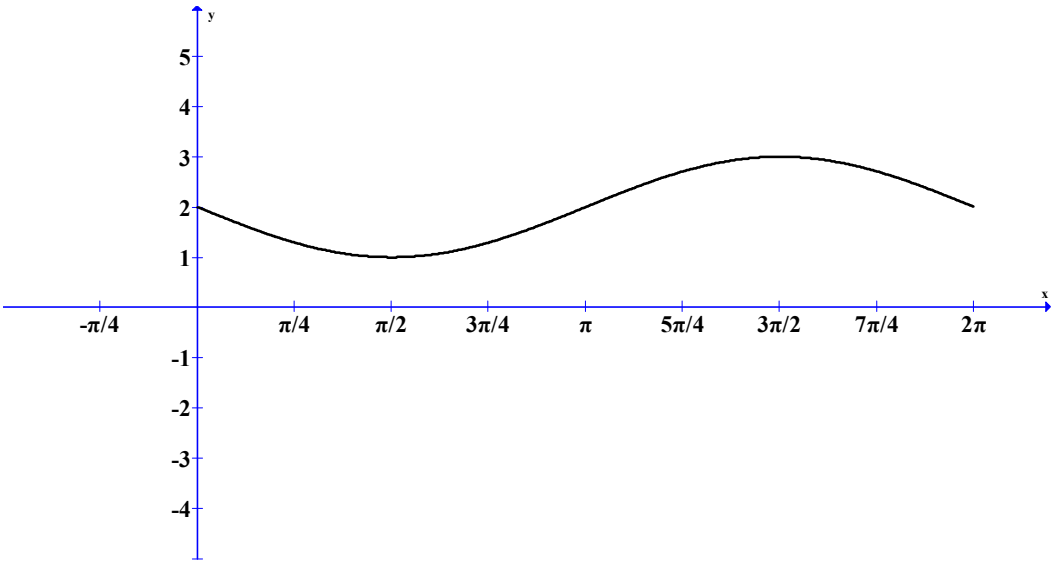
5) Graph the following in the range 0 to 2π, and state the domain for each

a) $y = 1 + \sin x$

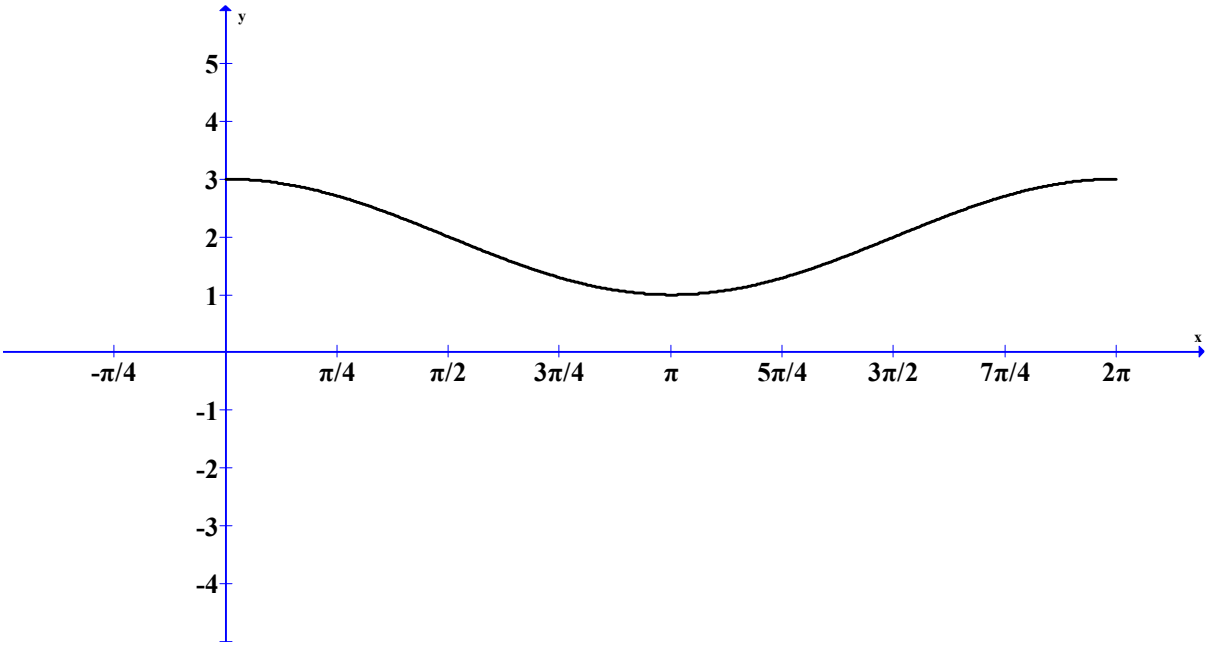


b) $y = 2 - \sin x$

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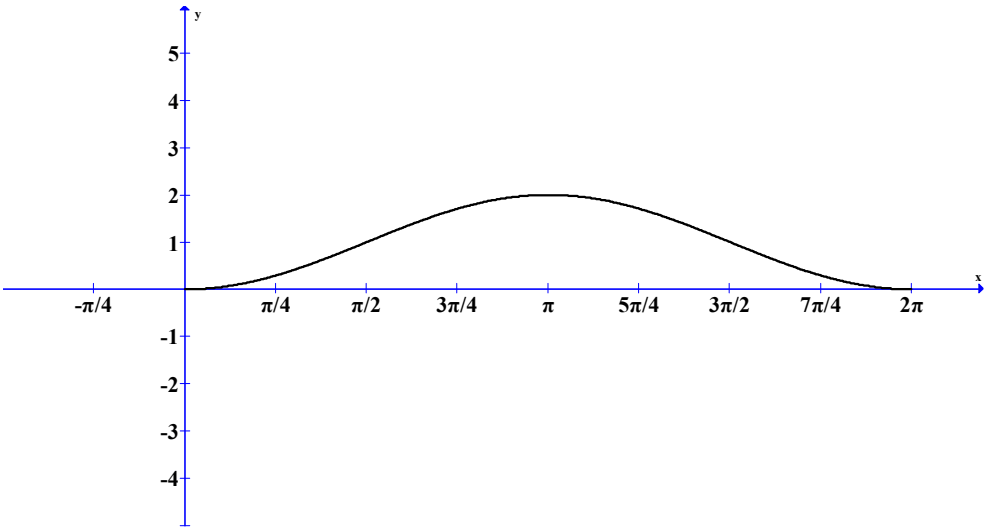


c) $y = 2 + \cos x$



d) $y = 1 - \cos x$

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6) From your graphs in question 5, what effect is produced by the varying of "C" in the general equations

$y = C \pm \sin x$, and $y = C \pm \cos x$

The curve moves up or down the y axis C units

7) Solve the following by drawing graphs of the functions

a) $\sin x = x$

$x = 0$

b) $\cos 2x = x + 1$

$x = 0$

c) $\sin 2x = \frac{x}{2}$

$x = 0, \frac{2\pi}{5}$

d) $2\cos x = x - 1$

$x = \frac{7\pi}{16}$

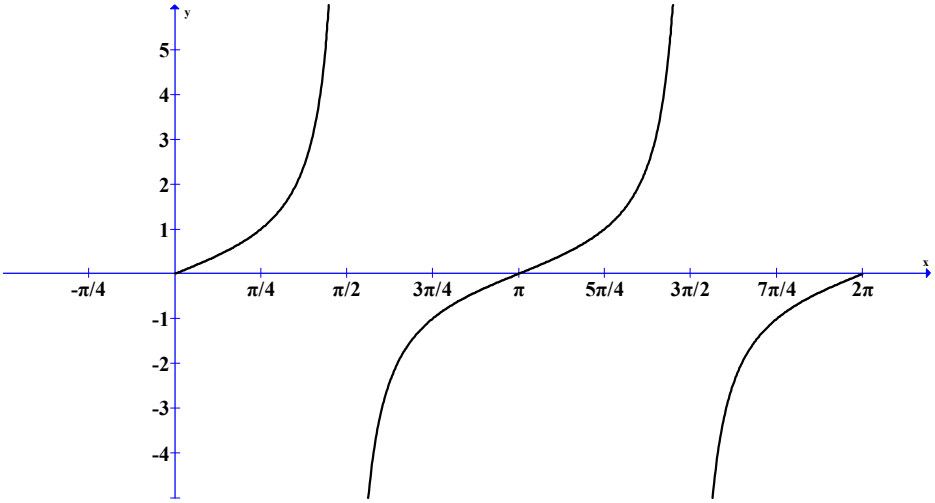
e) $1 - \sin x = \cos x$

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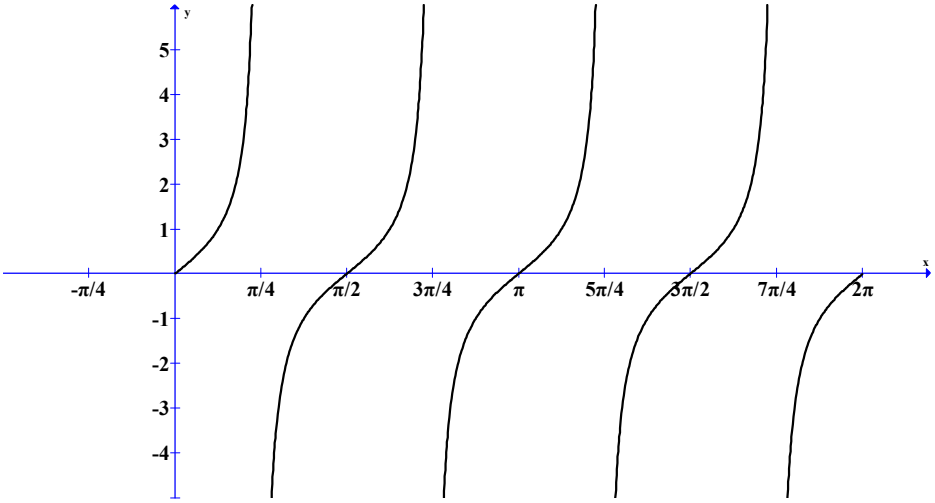
$x = 0, \frac{\pi}{2}, 2\pi$

8) Graph the following in the range 0 to 2π, state the domain, and any values in the range for which the function is undefined

a) $y = \tan x$

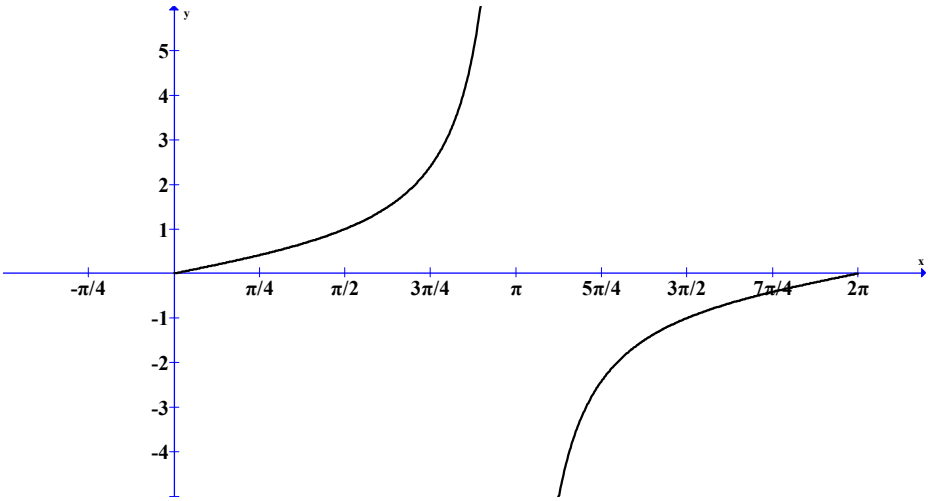


b) $y = \tan 2x$

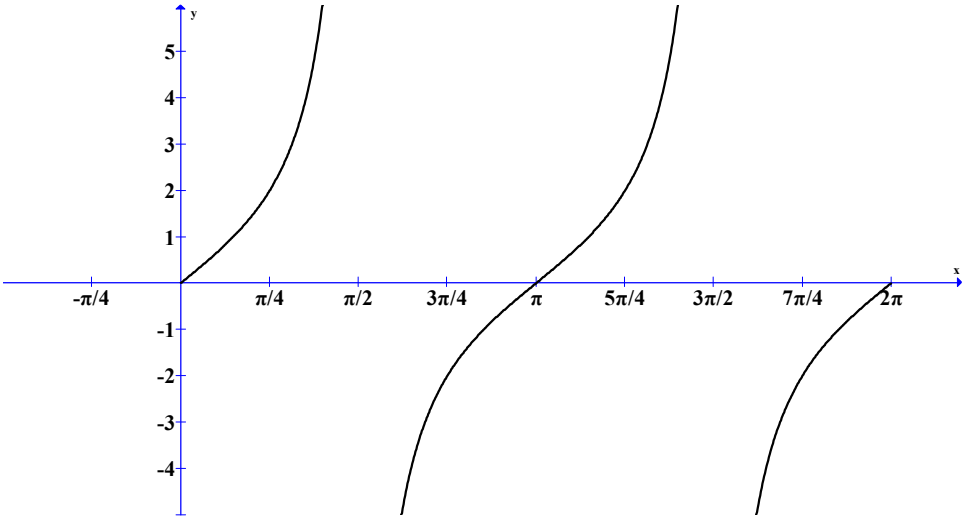


c) $y = \tan \frac{1}{2}x$

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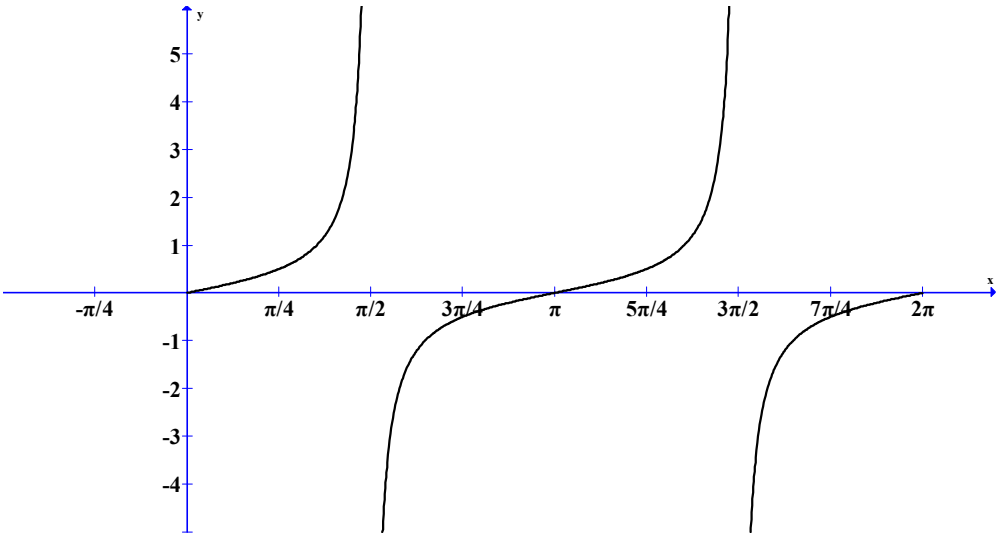


d) $y = 2\tan x$



e) $y = \frac{1}{2}\tan x$

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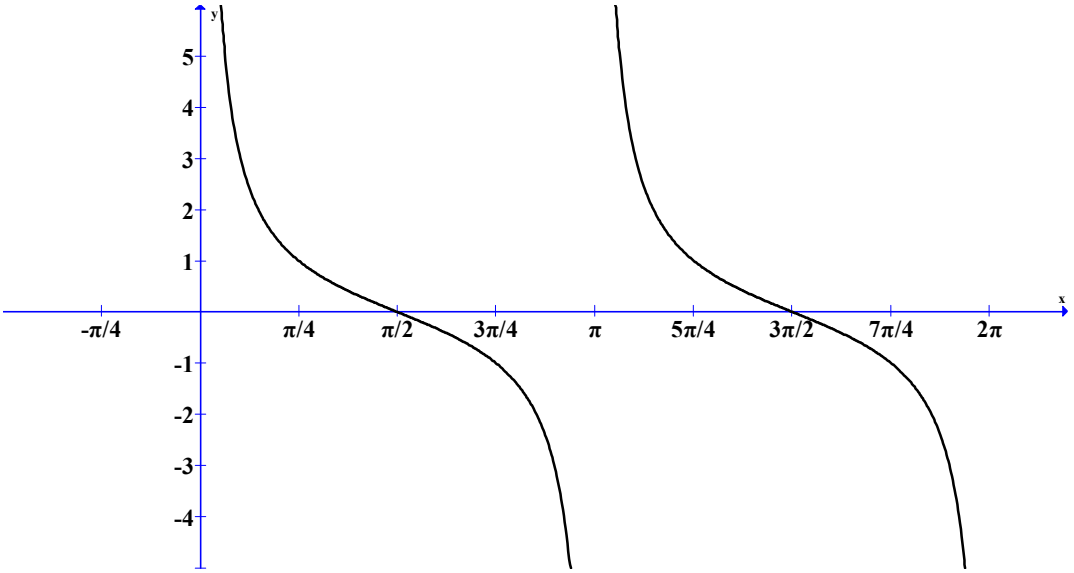


9) What effect does varying the values of A and B have on the graph and the range and domain of functions of the type

$y = A\tan Bx$

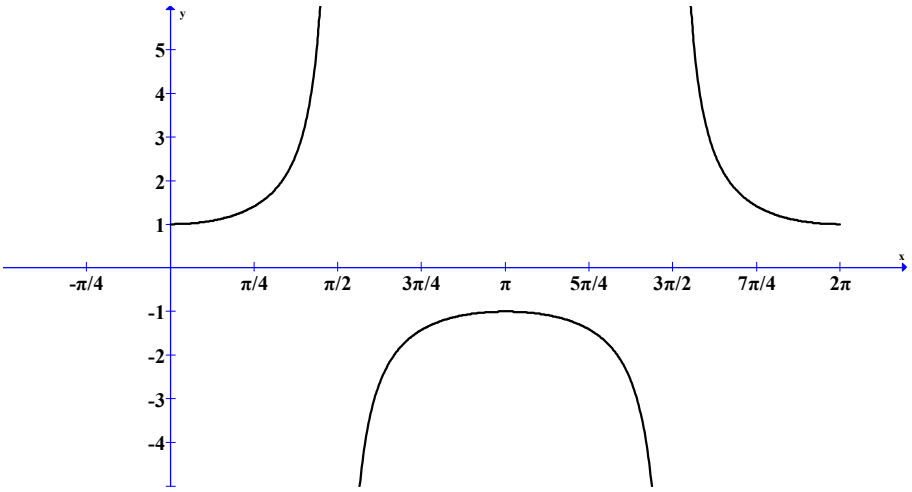
10) Graph the following in the range 0 to 2π

a) $y = \cot x$

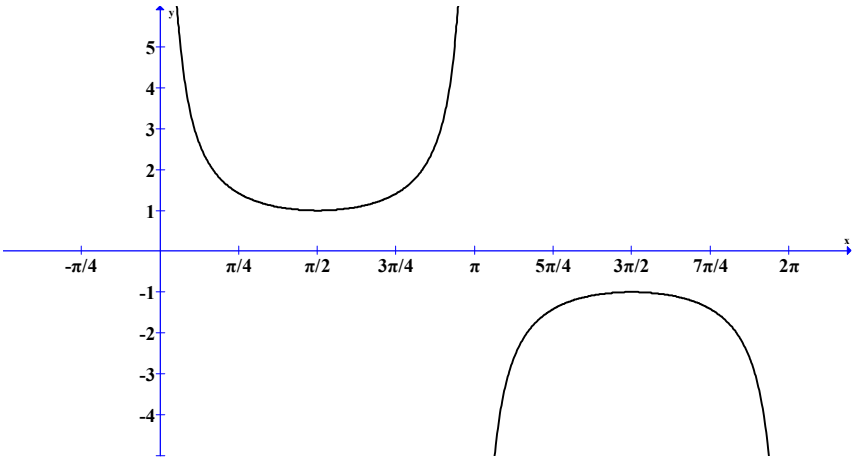


b) $y = \sec x$

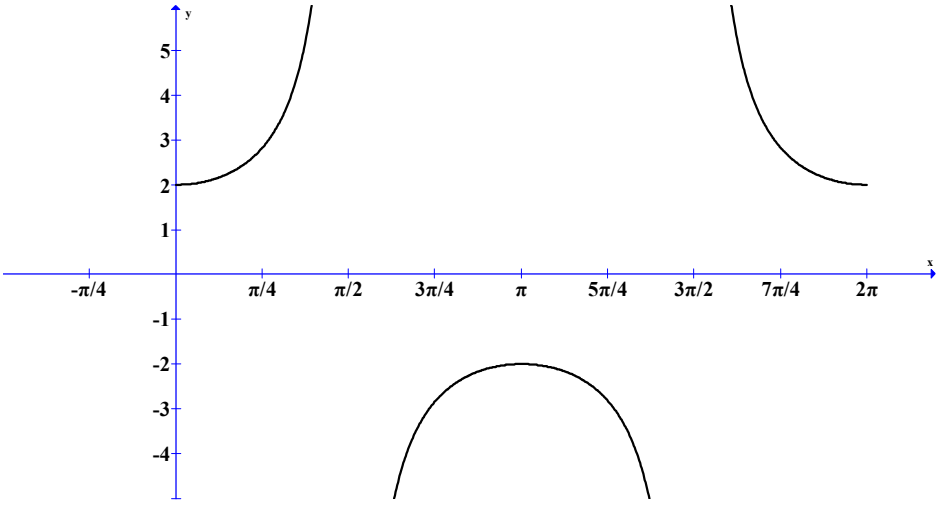
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c) $y = \csc x$

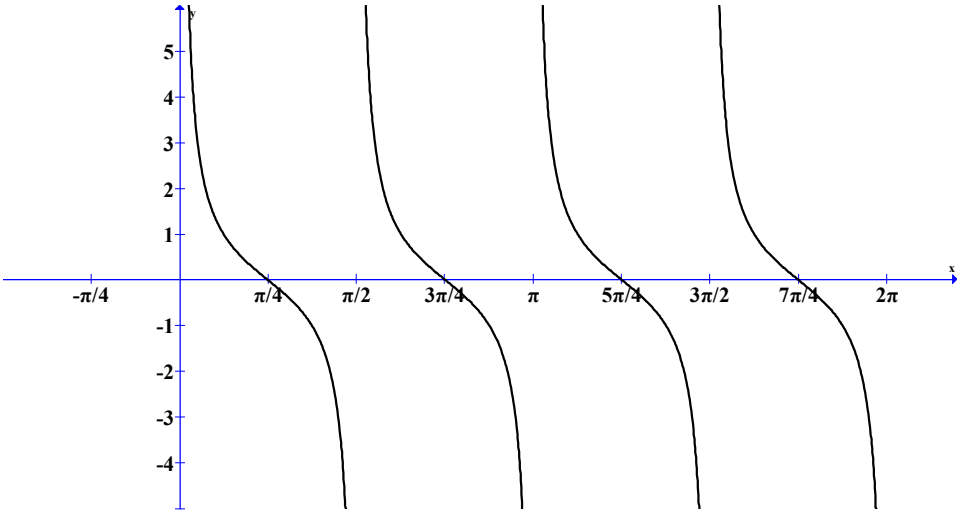


d) $y = 2\sec x$

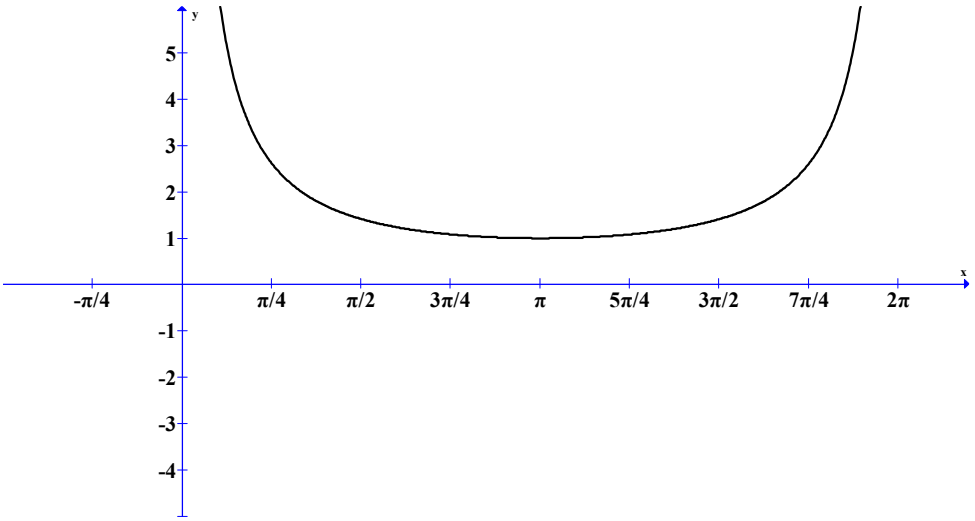


e) $y = \cot 2x$

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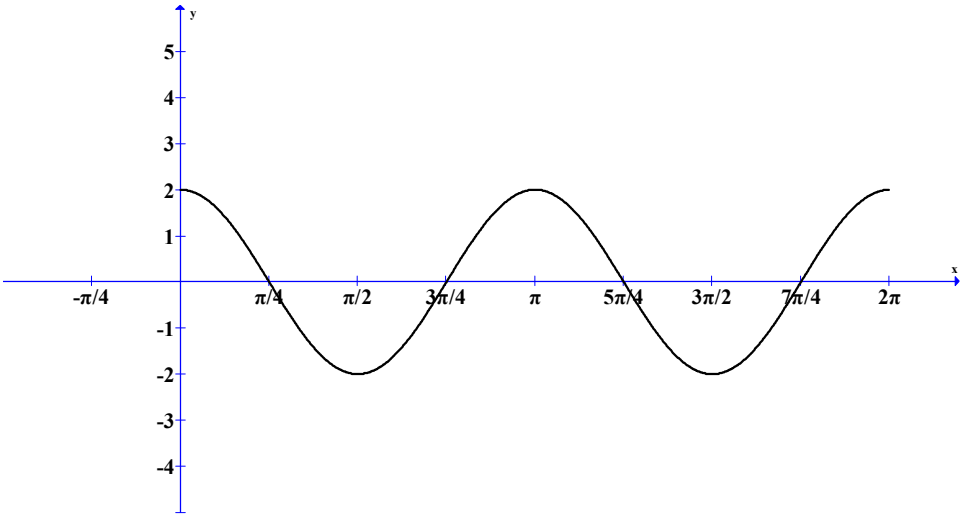
f) $y = \csc\frac{1}{2}x$



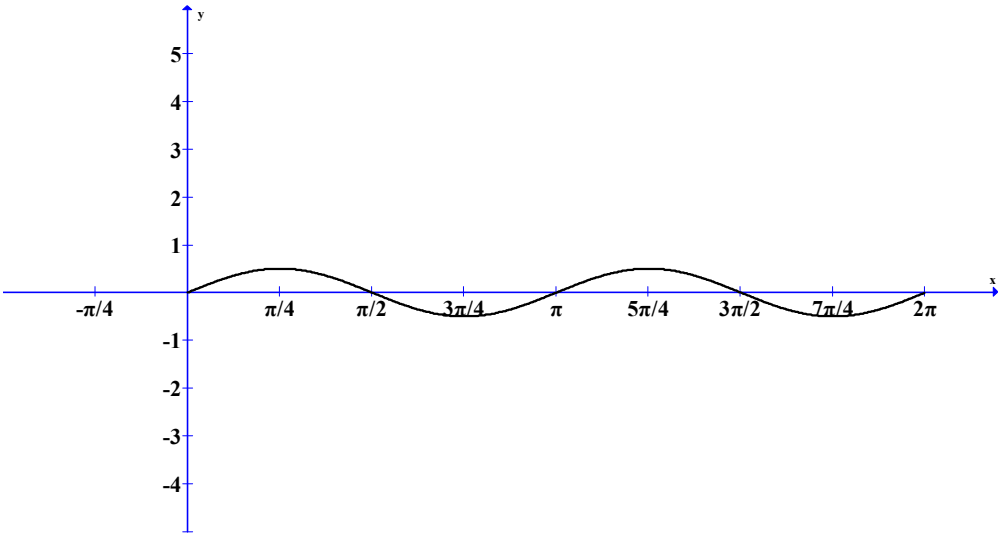
11) Draw a rough sketch of the following in the range 0 to 2π, and then graph formally to check your sketch

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a) $y = 2\cos 2x$

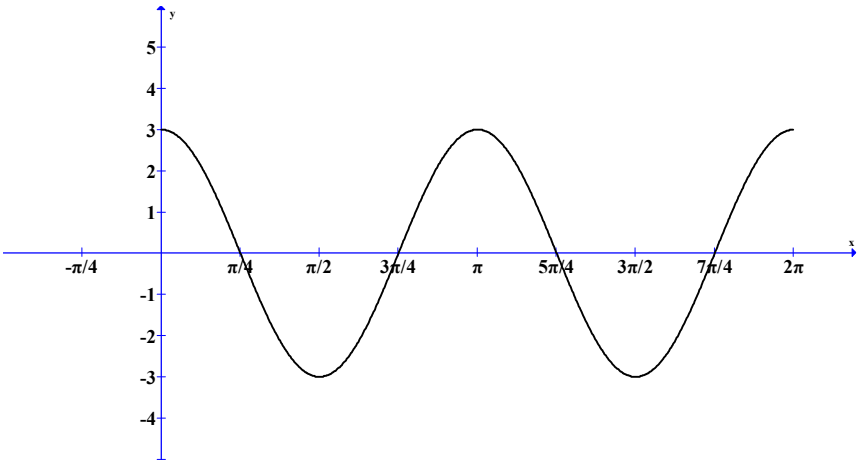


b) $y = \frac{1}{2}\sin 2x$

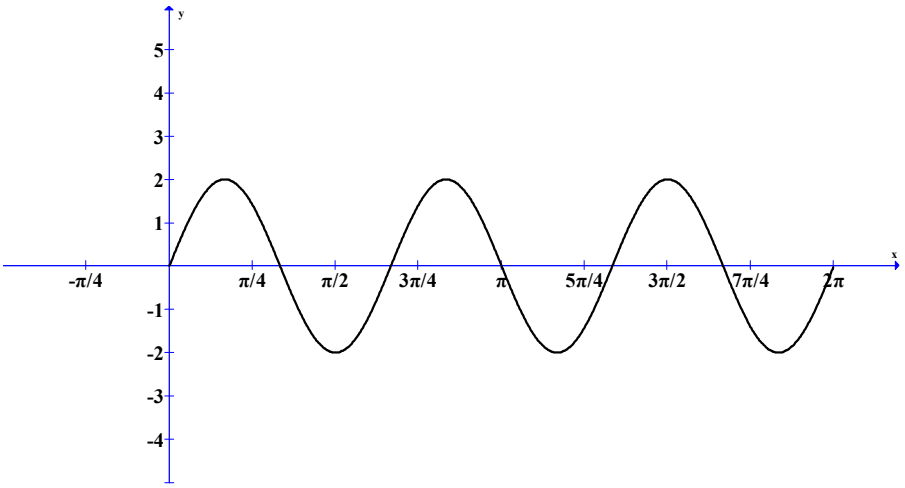


c) $y = 3\cos 2x$

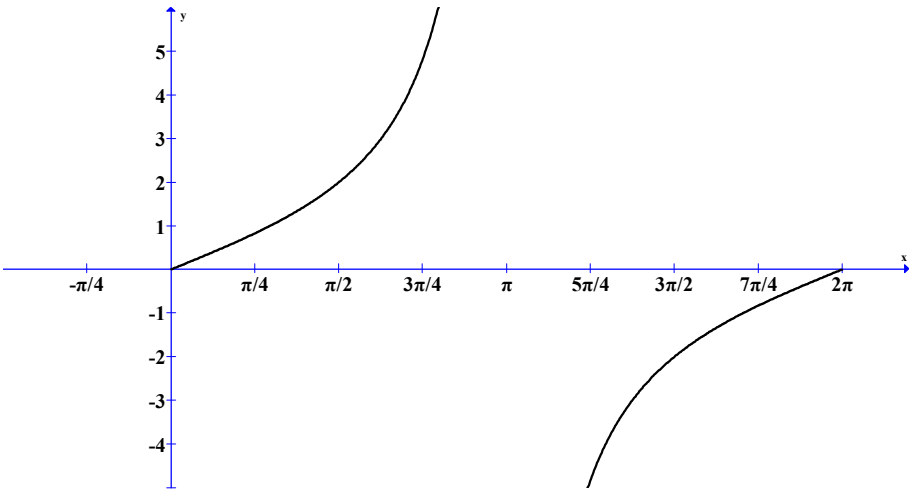
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d) $y = 2\sin 3x$

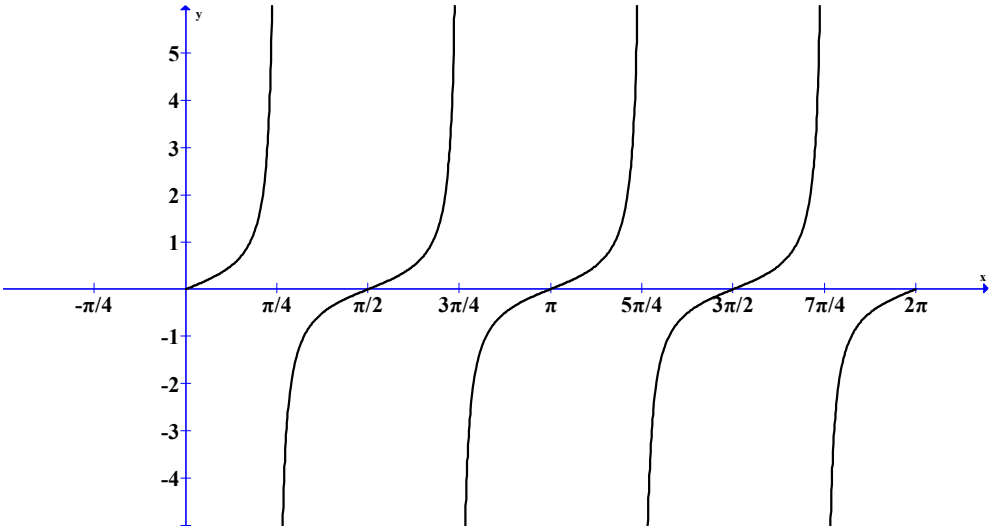


e) $y = 2\tan\frac{1}{2}x$



f) $y = \frac{1}{2}\tan 2x$

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Exercise 3

Differentiation & Integration

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1) Find the derivatives of the following

a) $y = \sin x$

$$\frac{dy}{dx} = \cos x$$

b) $y = \sin 2x$

$$\frac{dy}{dx} = 2\cos 2x$$

c) $y = \sin \frac{1}{4}x$

$$\frac{dy}{dx} = \frac{1}{4}\cos \frac{1}{4}x$$

d) $y = \sin 3x$

$$\frac{dy}{dx} = 3\cos 3x$$

e) $y = \sin(x + 2)$

$$\frac{dy}{dx} = \cos(x + 2)$$

f) $y = \sin(x + 1)$

$$\frac{dy}{dx} = \cos(x + 1)$$

g) $y = -\sin(x - 3)$

$$\frac{dy}{dx} = -\cos(x - 3)$$

2) Find the derivatives of the following

a) $y = \cos x$

$$\frac{dy}{dx} = -\sin x$$

b) $y = -\cos 2x$

$$\frac{dy}{dx} = 2\sin 2x$$

c) $y = \cos \frac{1}{2}x$

$$\frac{dy}{dx} = -\frac{1}{2}\sin \frac{1}{2}x$$

d) $y = \tan x$

$$\frac{dy}{dx} = \sec^2 x$$

e) $y = \tan 2x$

$$\frac{dy}{dx} = 2\sec^2 x$$

f) $y = \sin(2x - 1)$

$$\frac{dy}{dx} = 2\cos(2x - 1)$$

g) $y = \cos(3x + 2)$

$$\frac{dy}{dx} = -3\sin(3x + 2)$$

h) $y = \sin\left(-\frac{1}{2}x - 1\right)$

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$$\frac{dy}{dx} = -\frac{1}{2}\cos\left(-\frac{1}{2}x - 1\right)$$

i) $y = \cos\left(2x - \frac{1}{2}\right)$

$$\frac{dy}{dx} = -2\sin\left(2x - \frac{1}{2}\right)$$

3) Find derivatives of the following

a) $y = 2\sin x$

$$\frac{dy}{dx} = 2\cos x$$

b) $y = 2\sin\frac{1}{2}x$

$$\frac{dy}{dx} = \cos\frac{1}{2}x$$

c) $y = 4\cos 2x$

$$\frac{dy}{dx} = -8\sin 2x$$

d) $y = 3\cos\frac{1}{2}x$

$$\frac{dy}{dx} = -\frac{3}{2}\sin\frac{1}{2}x$$

e) $y = 2\sin(2x + 1)$

$$\frac{dy}{dx} = 4\cos(2x + 1)$$

f) $y = 3\cos(x - 1)$

$$\frac{dy}{dx} = -3\sin(x - 1)$$

g) $y = -2\tan\left(x + \frac{1}{2}\right)$

$$\frac{dy}{dx} = -2\sec^2\left(x + \frac{1}{2}\right)$$

h) $y = \frac{1}{2}\tan\frac{1}{2}x$

$$\frac{dy}{dx} = \frac{1}{4}\sec^2\frac{1}{2}x$$

4) Perform the following integrations

a) $\int \sin x\, dx$

$$y = -\cos x + c$$

b) $\int \cos x\, dx$

$$y = \sin x + c$$

c) $\int \sin(-2x)\, dx$

$$y = \frac{1}{2}\cos(-2x) + c$$

d) $\int \frac{\cos 2x}{2}\, dx$

$$y = \frac{\sin 2x}{4} + c$$

e) $2\int \frac{sin2x}{2}\, dx$

$$y = -\frac{\cos 2x}{4} + c$$

5) Perform the following integrations

a) $\frac{1}{2}\int \cos\left(\frac{1}{2}x + 1\right) dx$

$$y = \sin\left(\frac{1}{2}x + 1\right) + c$$

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b) $2\int \sin(2x-3)\,dx$

$y = -\cos(2x-3) + c$

c) $\int \cos 4x\,dx$

$y = \frac{1}{4}\sin 4x + c$

d) $\int 2 - \cos 2x\,dx$

$y = 2x - \frac{1}{2}\sin 2x + c$

e) $\int \left(\sin\frac{1}{2}x\right) + 1\,dx$

$y = -2\cos\frac{1}{2}x + x + c$

6) Integrate the following

a) $\int \sec^2 x\,dx$

$y = \tan x$

b) $\int \frac{\sec^2 x}{2}\,dx$

$y = \frac{1}{2}\tan x$

c) $\int (3\sin x - 2\sec^2 x)\,dx$

$y = -3\cos x - 2\tan x + c$

d) $\int \left(2\cos\frac{1}{2}x - \sin 2x + \sec^2 x\right)dx$

$y = 4\sin\frac{1}{2}x + \frac{1}{2}\cos 2x + \tan x + c$

e) $\int (\cos x\, e^{\sin x})\,dx$

Let $y = e^{\sin x}$

$\frac{dy}{dx} = \cos x\, e^{\sin x}$

$\int = \int 1\,dy = y + c$

$= e^{\sin x} + c$

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